

PROJECT APPRAISAL

Module 6

Shadow Pricing Marketed Inputs and Outputs



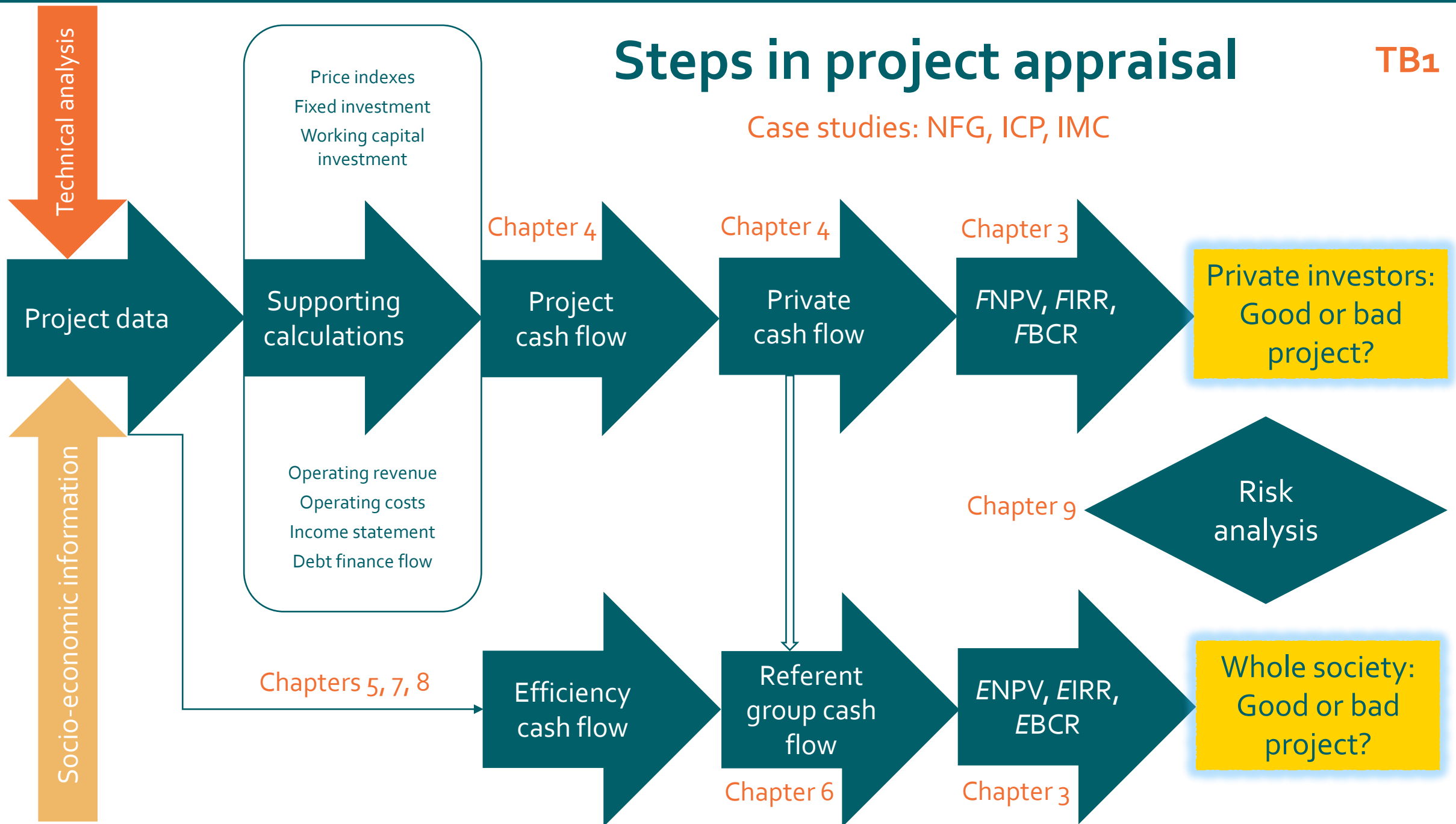
Outline – Module 6

- Net social benefits
- Impacts of a project
- Shadow pricing approach: An introduction
- Shadow pricing marketed inputs: A small-scale project
- Shadow pricing marketed outputs: A small-scale project
- Shadow pricing marketed inputs: A large-scale project
- Shadow pricing marketed outputs: A large-scale project
- The efficiency pricing rules
- Conversion factors of project inputs

Steps in project appraisal

TB1

Case studies: NFG, ICP, IMC

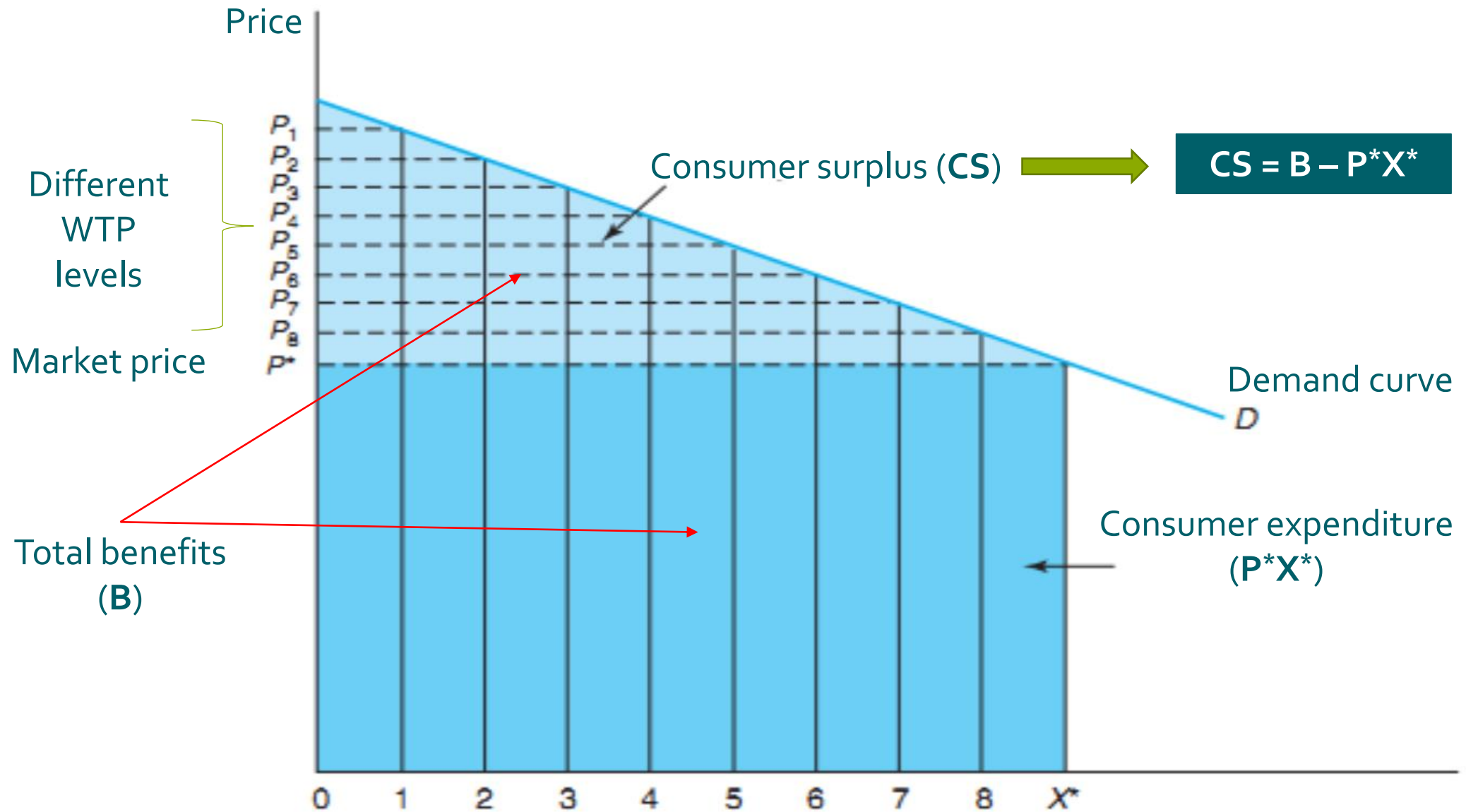


Net social benefits

Microeconomic foundations

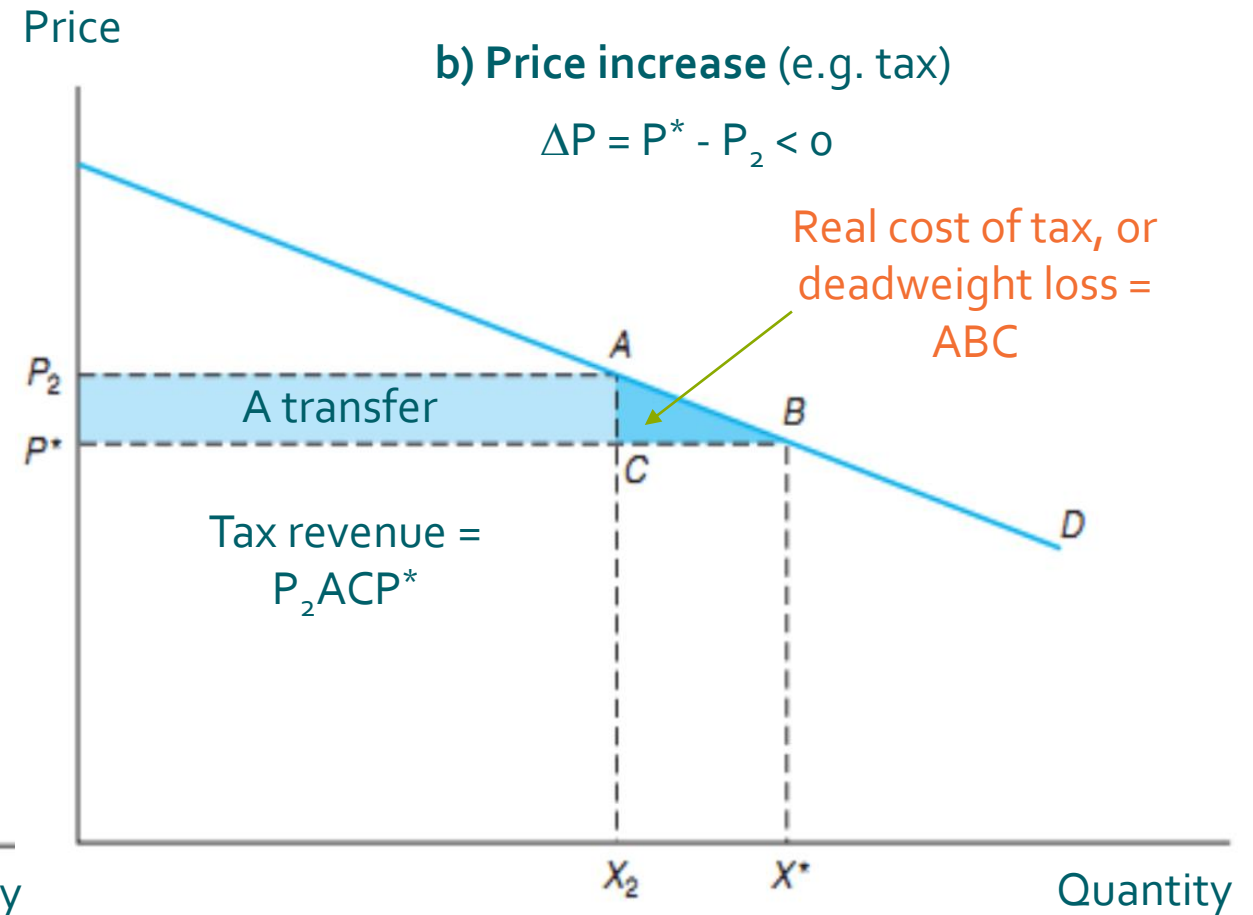
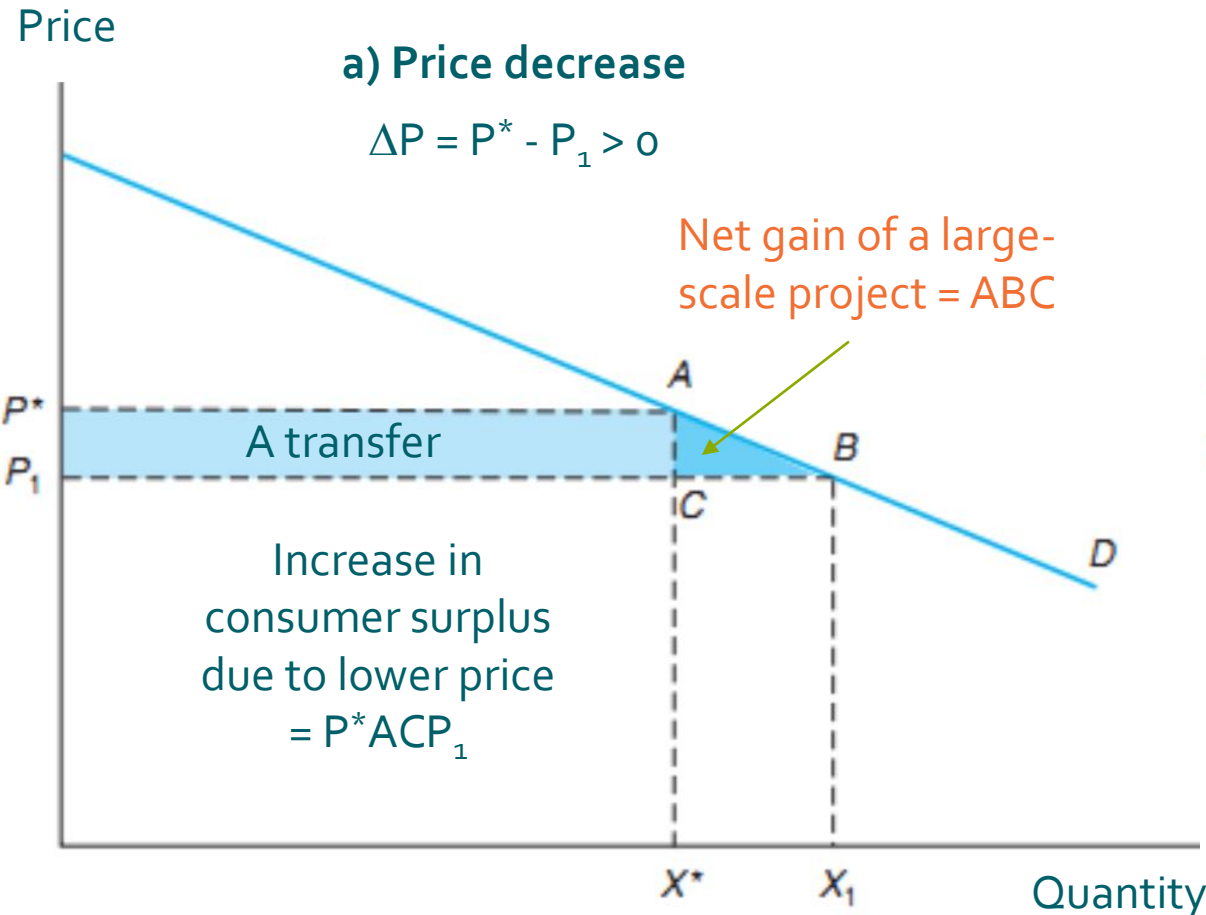


Demand curve, consumer surplus

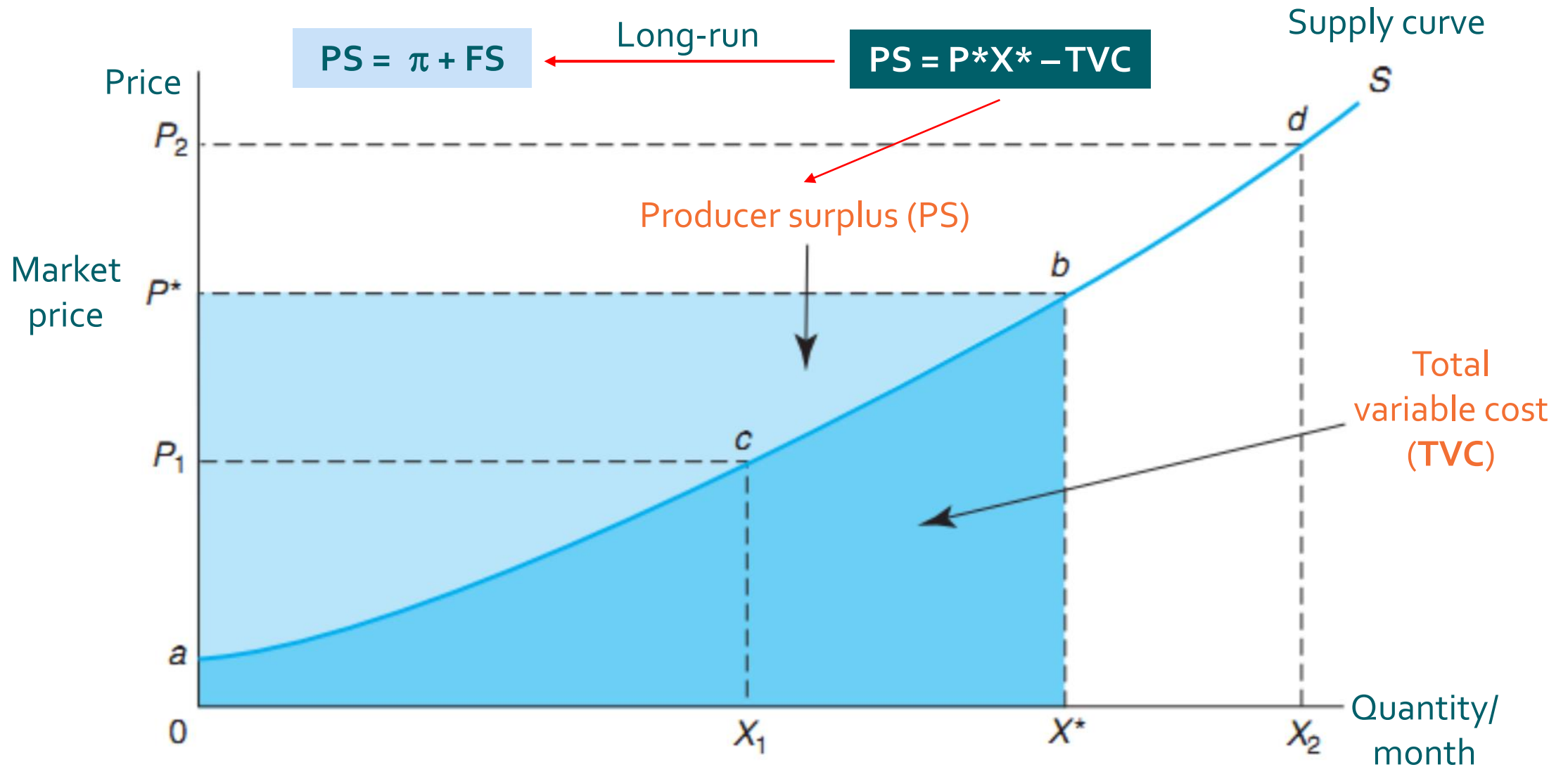


Demand curve, consumer surplus

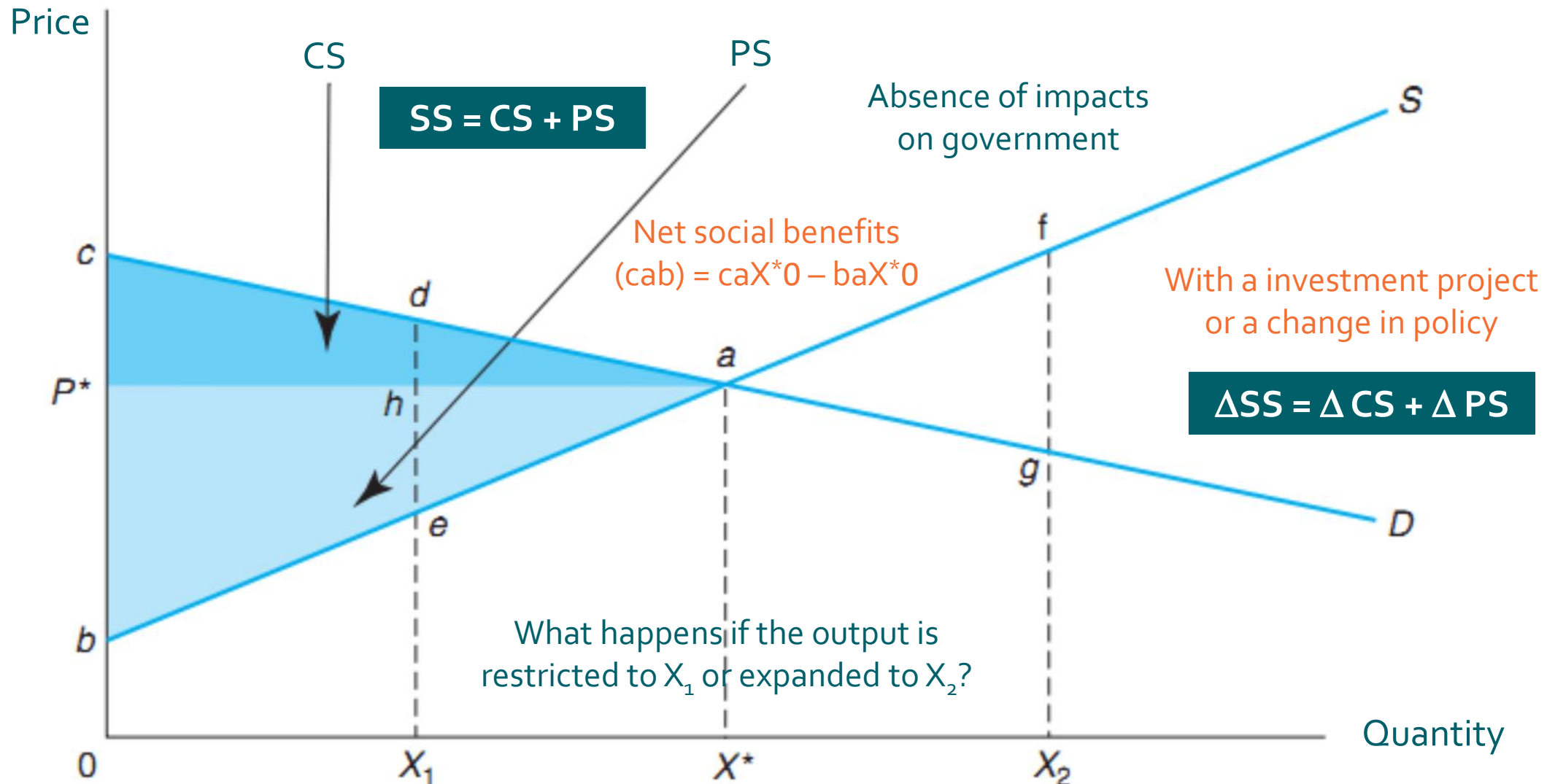
$$\Delta CS = -(\Delta P)(X^*) - \frac{1}{2}(\Delta X)(\Delta P)$$



Supply curve, producer surplus



Net social benefits (ΔSS)



Net social benefits (ΔSS)

- From a practical standpoint, it is useful to treat government as a distinct actor or sector in society. The analyst should include financial inflows to or outflows from government (i.e. the net budget impacts on government, **GS**) :

$$SS = CS + PS + GS$$

- The incremental net social benefit of an investment project or a change in policy is given by:

$$\Delta SS = \Delta CS + \Delta PS + \Delta GS$$

Net social benefits (ΔSS)

- For non-tradable outputs and inputs of a project:
 - $\text{Economic benefit} = \text{Financial benefit} + \Delta SS$
 - $\text{Economic cost} = \text{Financial cost} + \Delta SS$
- For tradable outputs and inputs of a project:
 - Economic benefits and costs are measured by the border prices (**FOB**: exported outputs, or exportable inputs; **CIF**: imported inputs, or import-replacing outputs), and shadow exchange rate.

Net social benefits (ΔSS)

- Let's denote ΔEV as economic benefit or cost of a project's output or input, and ΔQ as quantity of a project's output or input.
- Shadow prices or efficiency prices (P_e) can be defined as the following formula:

$$P_{e_i} = \frac{\Delta EV_i}{\Delta Q_i}$$

Impacts of a project

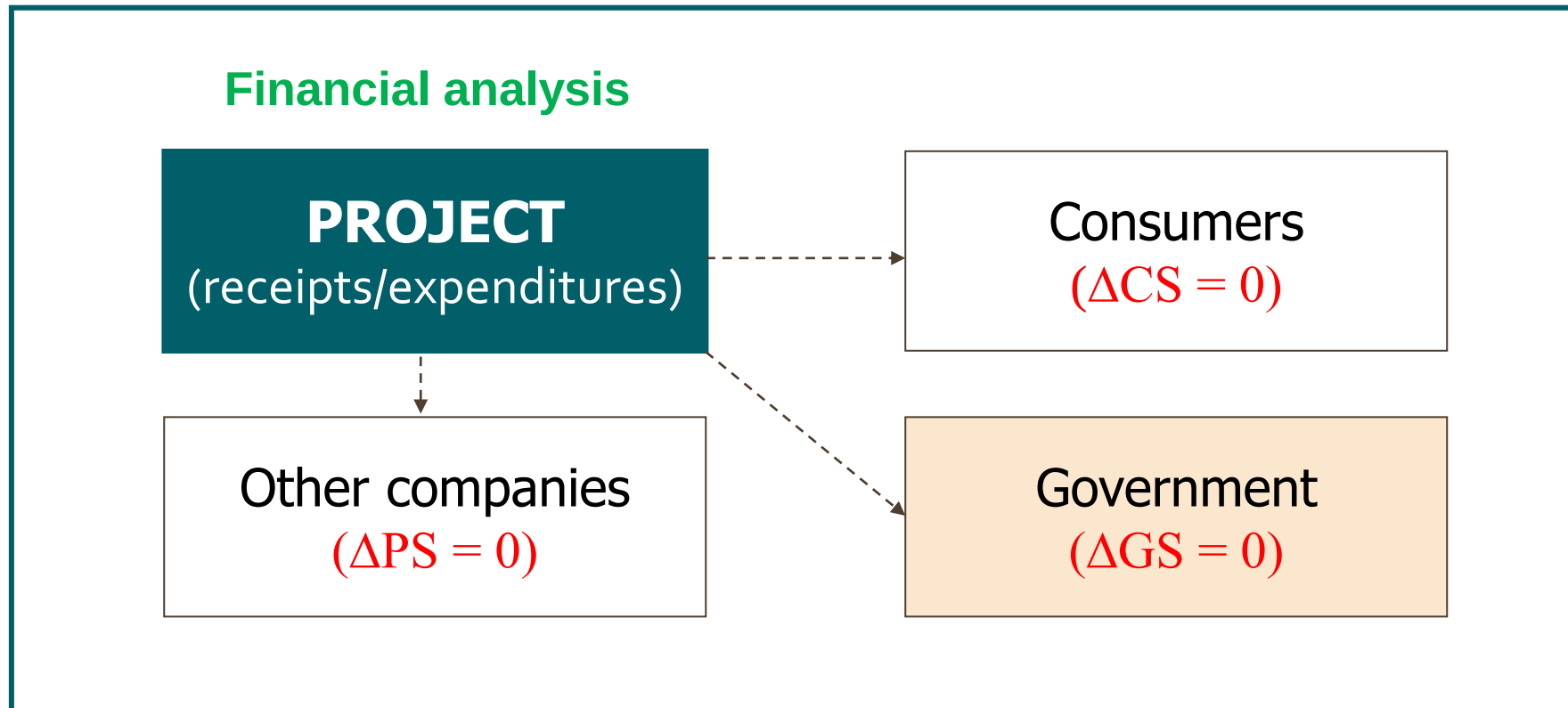
On other actors of the economy



Impacts of a project

If we lived in the perfectly competitive economy of the textbooks, with no external effects; **all projects were relatively small**, and we were concerned solely with economic efficiency, we could have terminated the discussion at the market analysis.

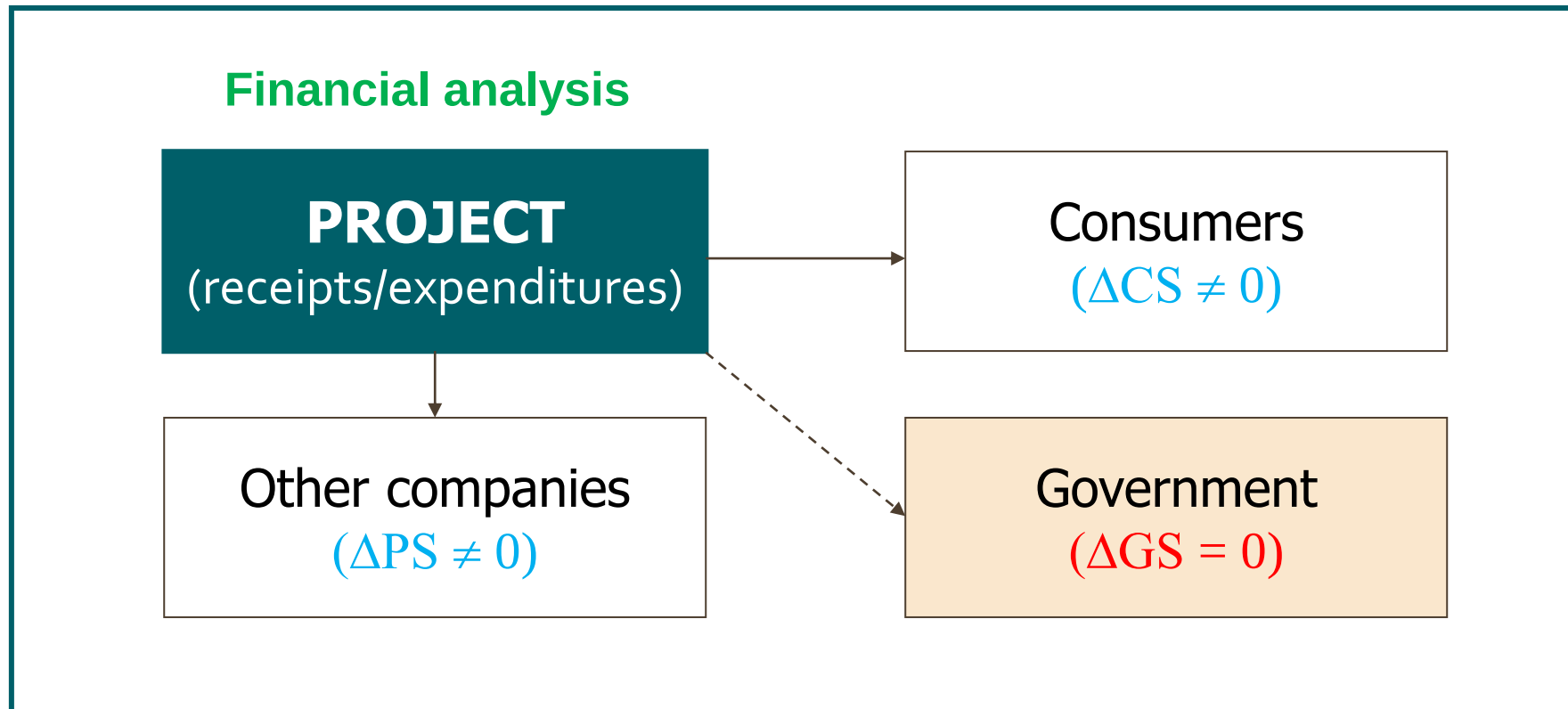
Economic analysis



Impacts of a project

If we lived in the perfectly competitive economy of the textbooks, with no external effects; and were concerned solely with economic efficiency of **large-scale projects**, we could have adjusted the market prices to account for changes in consumer and producer surpluses.

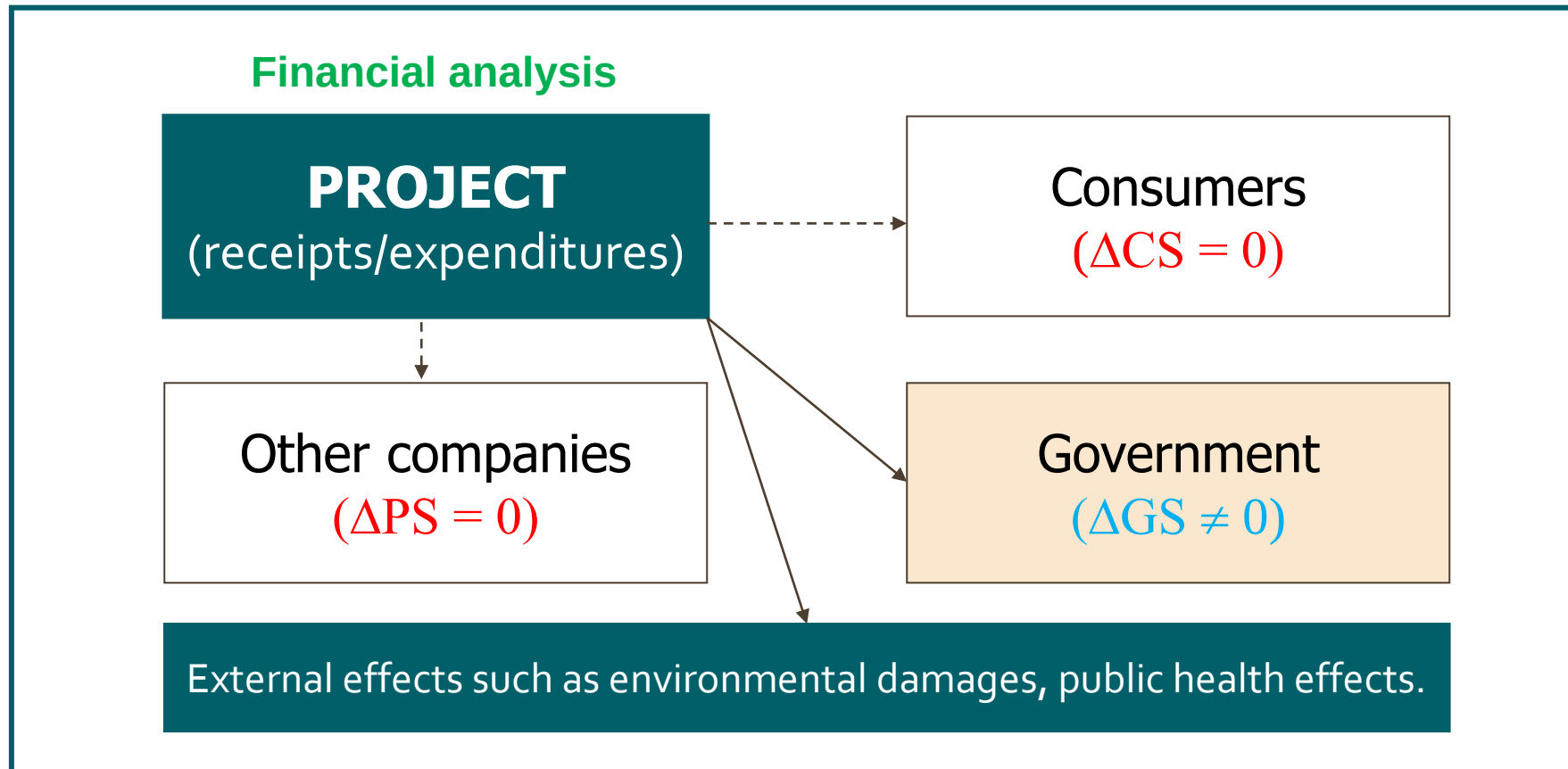
Economic analysis



Impacts of a project

All real-world economies depart from a perfectly competitive market economy to a greater or lesser extent: some markets have a small number of buyers or sellers; taxes are levied on outputs and inputs; regulations such as rent control or minimum wages are in place; and some goods and services which affect economic welfare may not be marketed. If we are concerned solely with economic efficiency of a **small-scale project**, we should adjust the market prices to account for the distortions and add external effects.

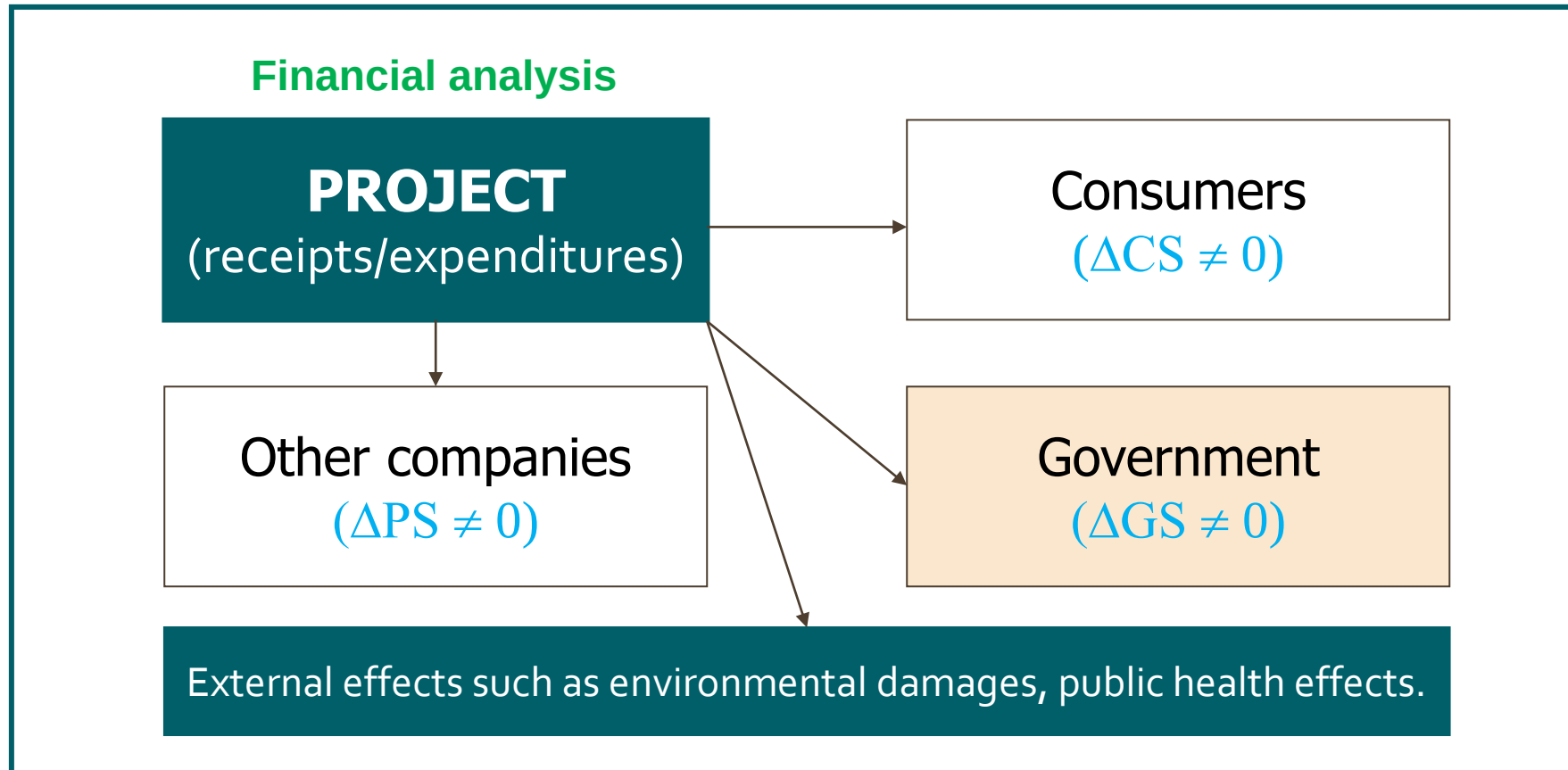
Economic analysis



Impacts of a project

All real-world economies depart from a perfectly competitive market economy to a greater or lesser extent: some markets have a small number of buyers or sellers; taxes are levied on outputs and inputs; regulations such as rent control or minimum wages are in place; and some goods and services which affect economic welfare may not be marketed. If we are concerned solely with economic efficiency of a **large-scale project**, we should adjust the market prices and add external effects.

Economic analysis



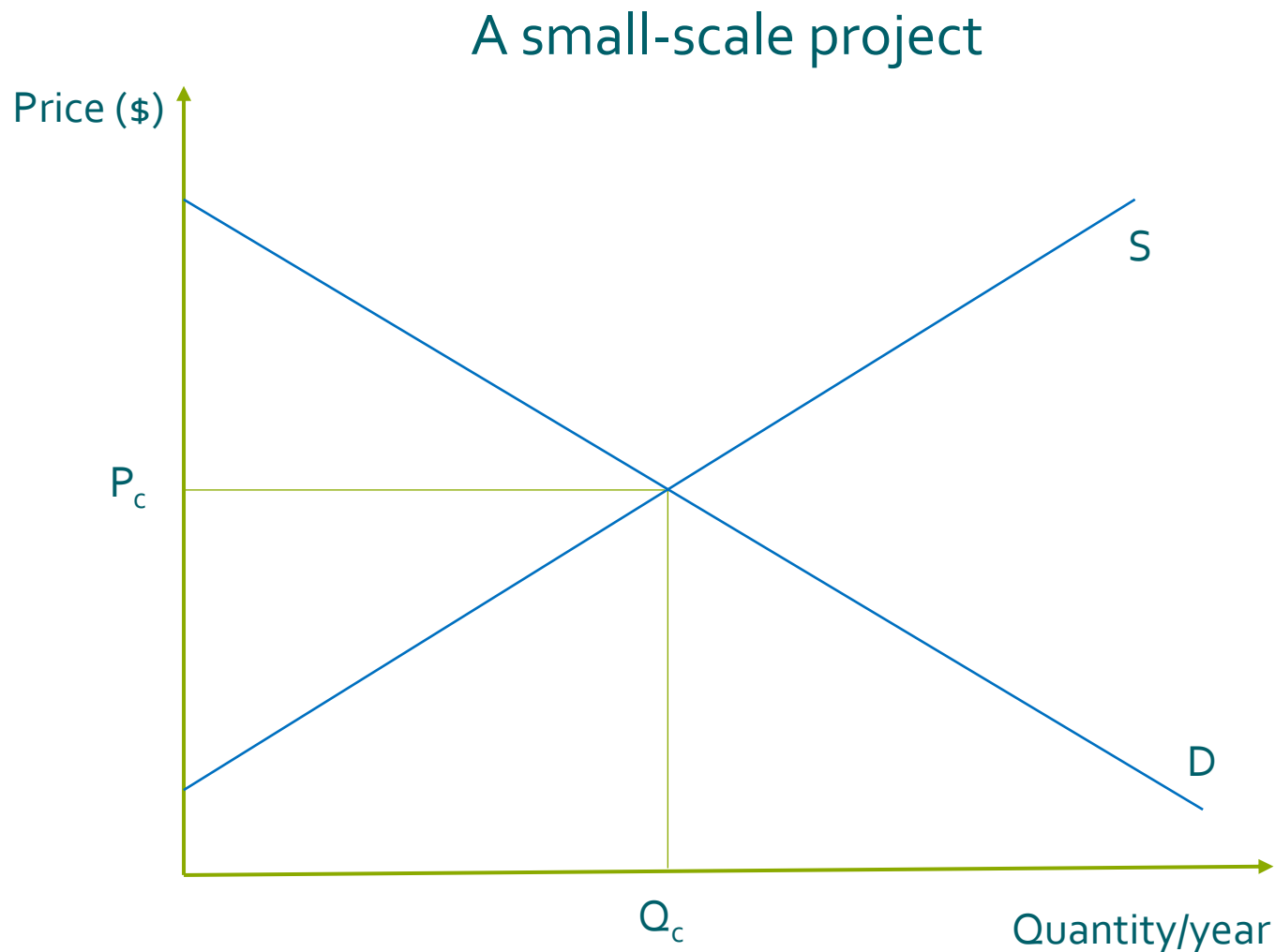
Shadow pricing

An introduction



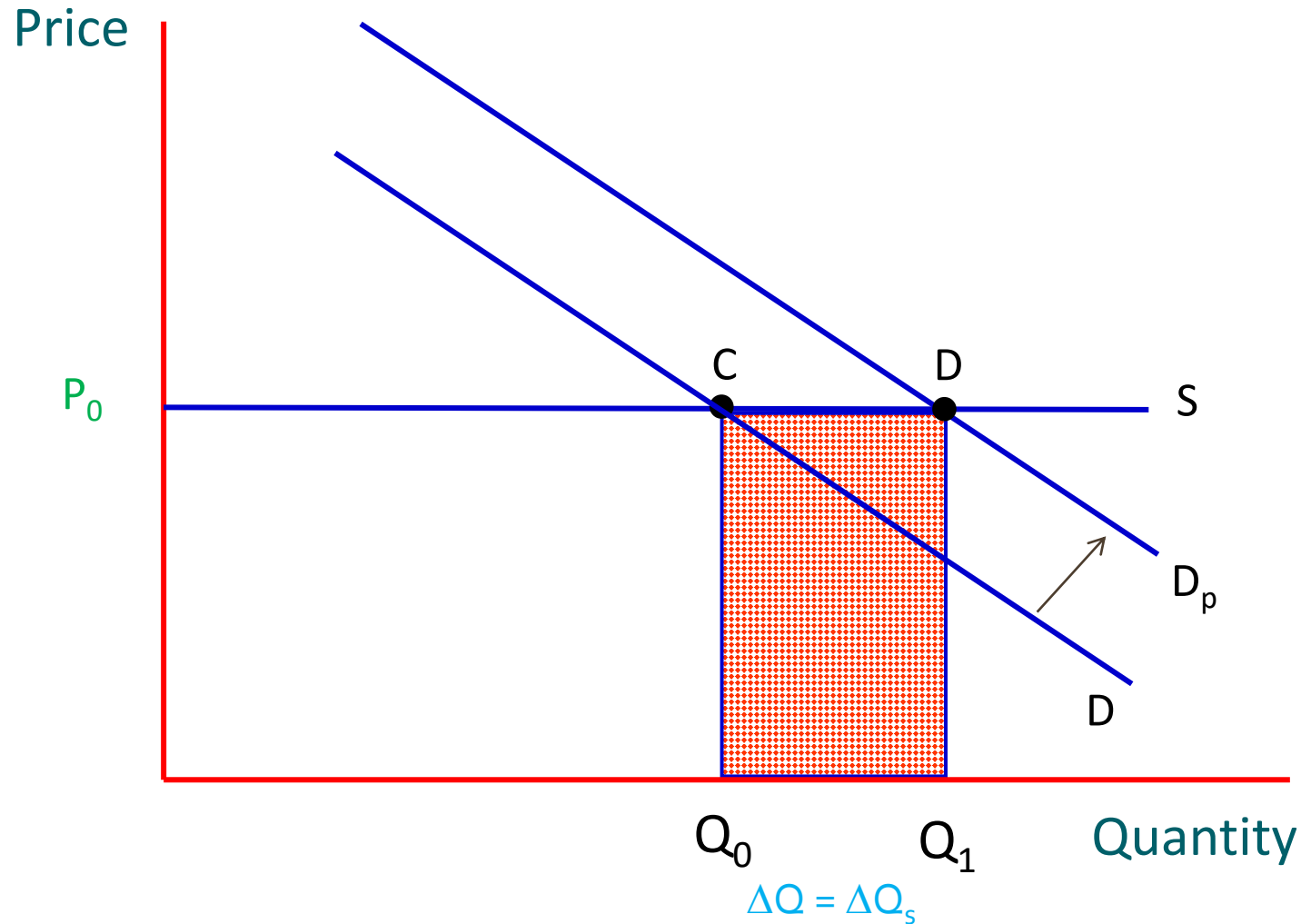
A perfectly competitive market

- The demand curve, D , tells us, at each level of a good or service, how much someone is willing to pay for one extra unit of it.
- The supply curve, S , tells us, at each level of a good or service, what an extra unit of a good or service will cost to produce. This is the opportunity cost of producing an additional unit of a good or service.
- At the equilibrium point, the demand price P_d equals the supply price P_s ; and market price P_m also equals shadow price $P_e (= P_c)$. **Shadow-pricing rule is not necessary here.**



A perfectly competitive market

(A small-scale project's input)



Financial cost:

$$\begin{aligned}\Delta TC &= -P_0(Q_1 - Q_0) = -P_0\Delta Q \\ &= -Q_0CDQ_1\end{aligned}$$

Economic cost:

$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P_0\Delta Q + 0 + 0 + 0 \\ &= -Q_0CDQ_1\end{aligned}$$

=> Economic cost = Financial cost

$$\Rightarrow P_e = P_0$$

A perfectly competitive market

(A small-scale project's output)

Financial benefit:

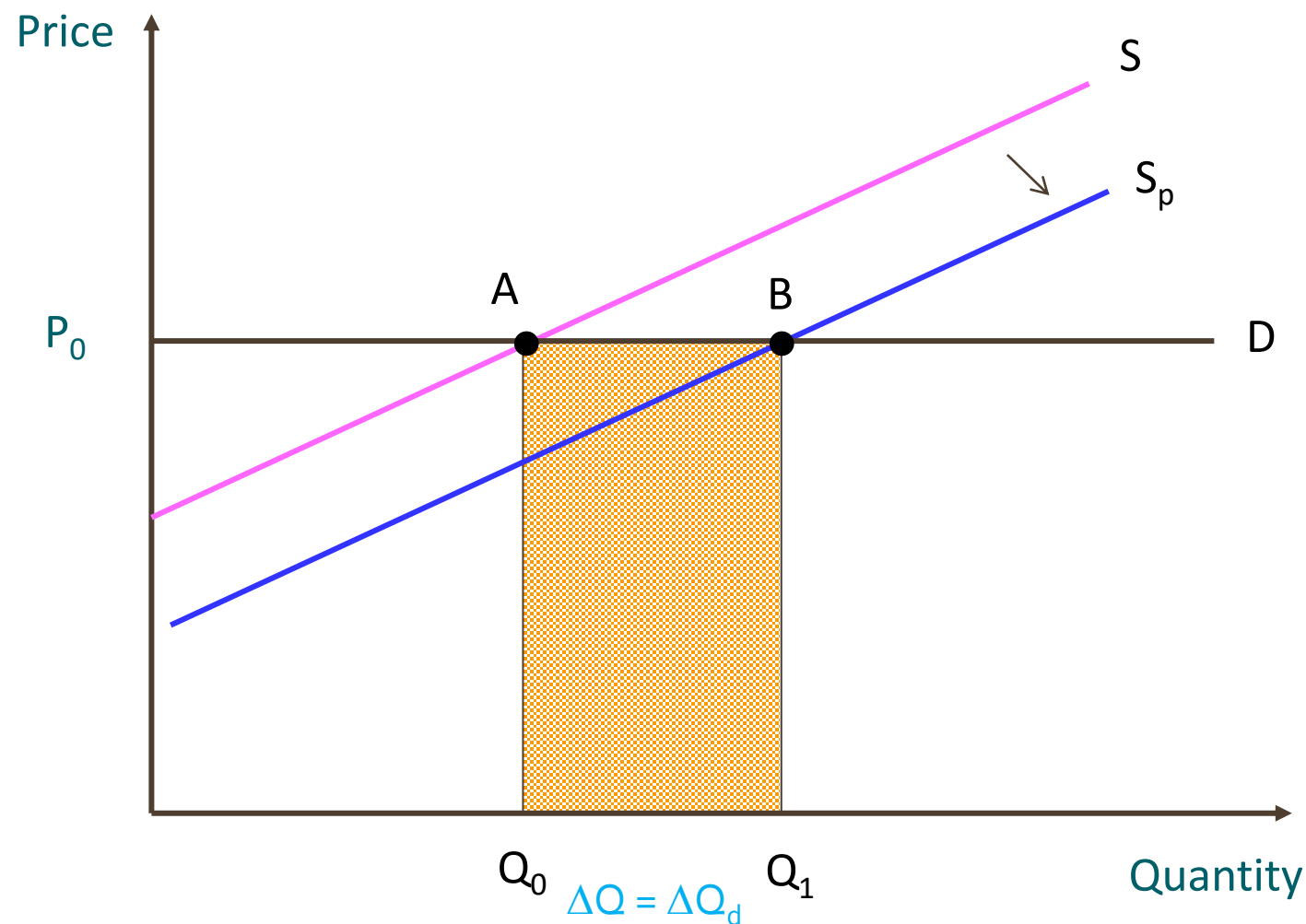
$$\begin{aligned}\Delta TR &= P_0(Q_1 - Q_0) = P_0\Delta Q \\ &= Q_0ABQ_1\end{aligned}$$

Economic benefit:

$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P_0\Delta Q + 0 + 0 + 0 \\ &= Q_0ABQ_1\end{aligned}$$

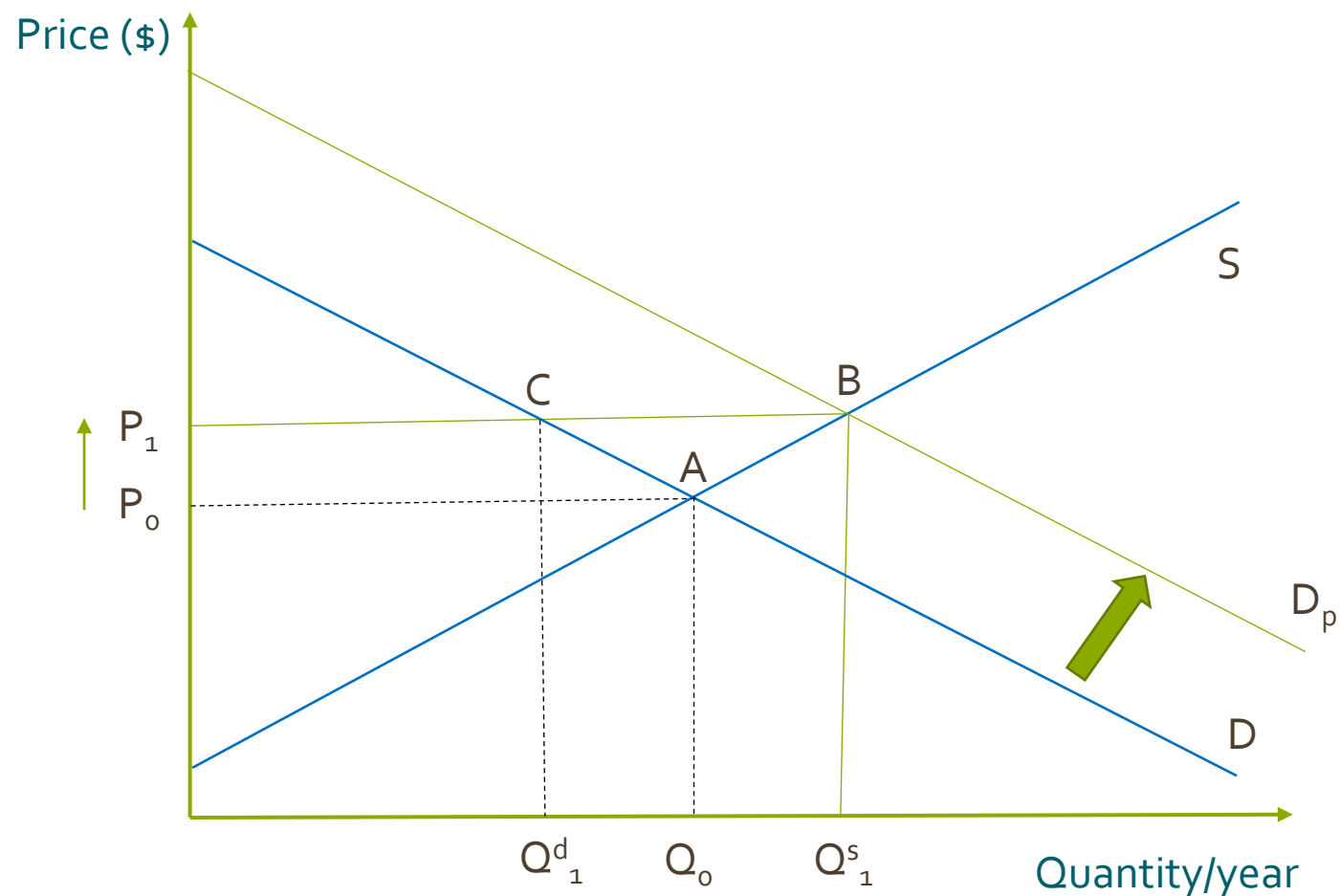
=> Economic benefit = Financial benefit

$$\Rightarrow P_e = P_0$$



A perfectly competitive market

- A large-scale project's input may cause the market price to rise from P_0 to P_1 ; and a higher market price may lead to an increase in market supply ($Q_0Q_1^s$) and a decrease in demand of other buyers ($Q_1^dQ_0$).
- Economic cost of the incremental quantity supplied is measured by the area below the supply curve ($Q_0ABQ_1^s$) and economic cost of the replaced quantity demanded is measured by the area below the demand curve ($Q_1^dCAQ_0$).
- The area **ABC** is the increase in social surplus.
- The shadow price will be the average of the with and without project's prices [i.e. $P_e = (P_0 + P_1)/2$].



A perfectly competitive market

(A large-scale project's input)

Financial cost:

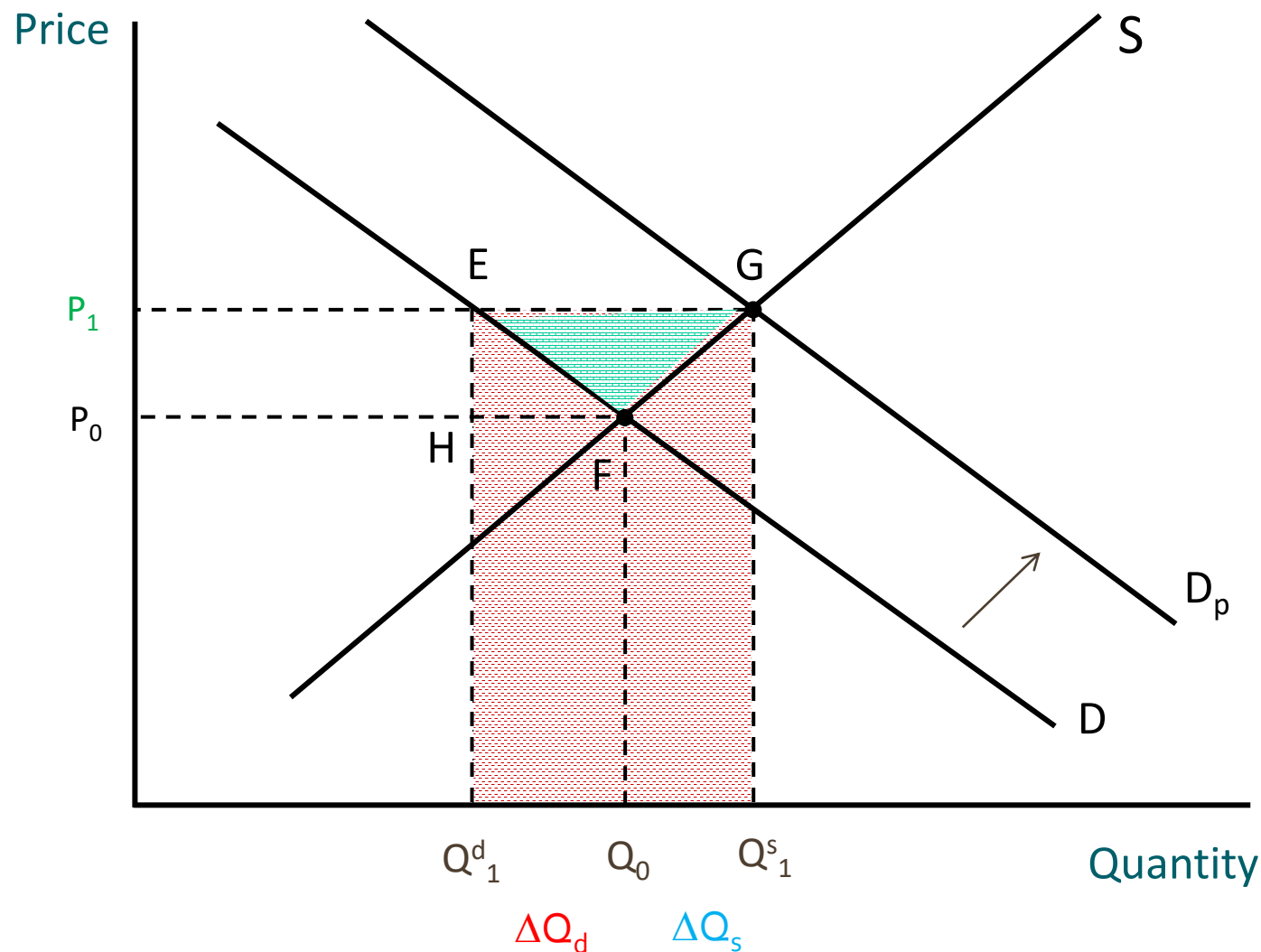
$$\begin{aligned}\Delta TC &= -P_1(Q_1^s - Q_1^d) = -P_1\Delta Q \\ &= -Q_1^d EGO_1^s\end{aligned}$$

Economic cost:

$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P_1\Delta Q - P_0FEP_1 + P_0FGP_1 + 0 \\ &= -Q_1^d EFGO_1^s\end{aligned}$$

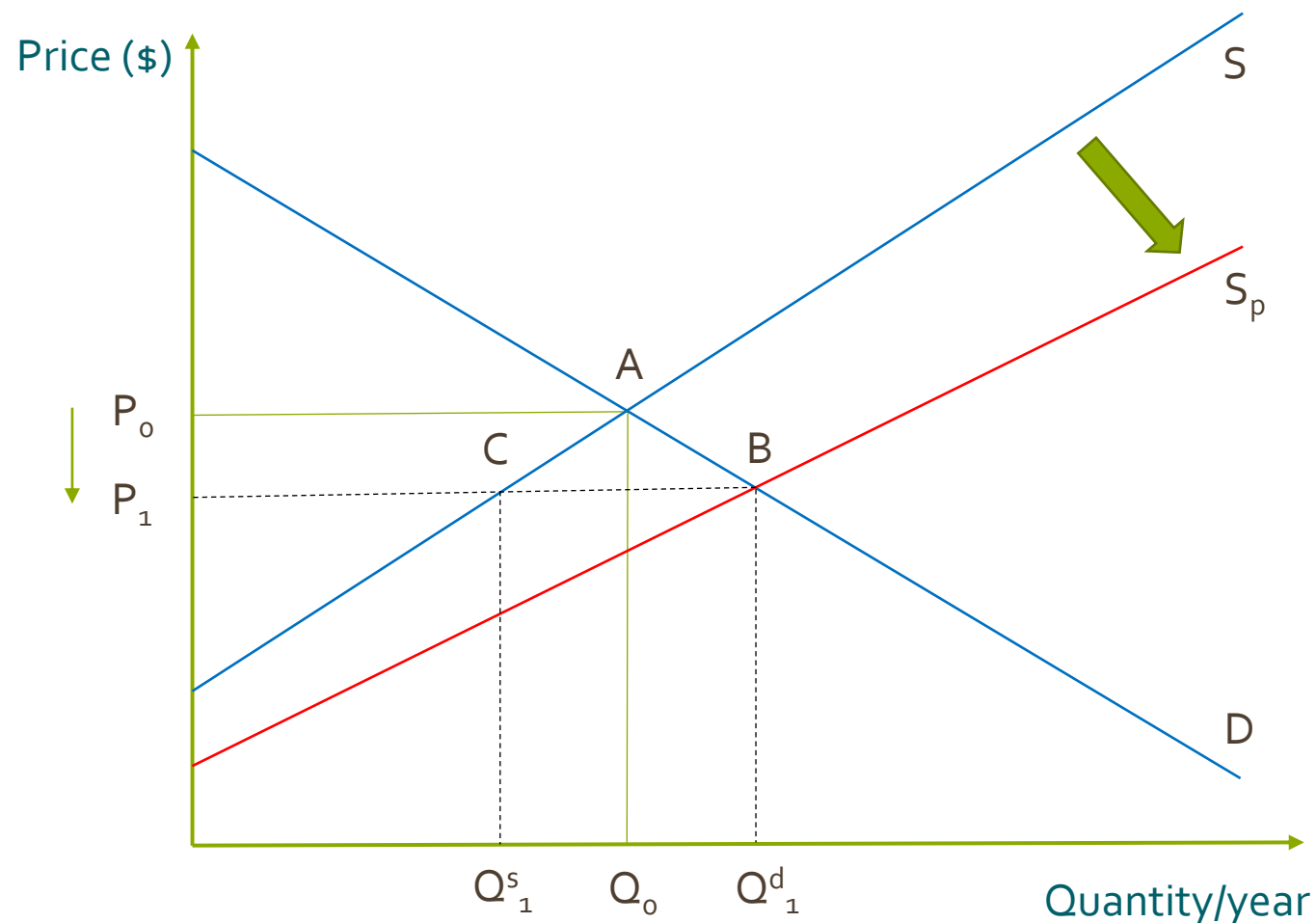
=> Economic cost < Financial cost

$$\Rightarrow P_e = (P_0 + P_1)/2 < P_1$$



A perfectly competitive market

- A large-scale project's output may cause the market price to fall from P_0 to P_1 ; and a lower market price may lead to an increase in market demand ($Q_0 Q_1^d$) and a decrease in supply of other producers ($Q_1^s Q_0$).
- Economic benefit of the incremental quantity demanded is measured by the area below the demand curve ($Q_0 A B Q_1^d$) and economic benefit of the replaced quantity supplied is measured by the area below the supply curve ($Q_1^s C A Q_0$).
- The area **ABC** is the increase in social surplus.
- The shadow price will be the average of the with and without project's prices [i.e. $P_e = (P_0 + P_1)/2$].



A perfectly competitive market

(A large-scale project's output)

Financial benefit:

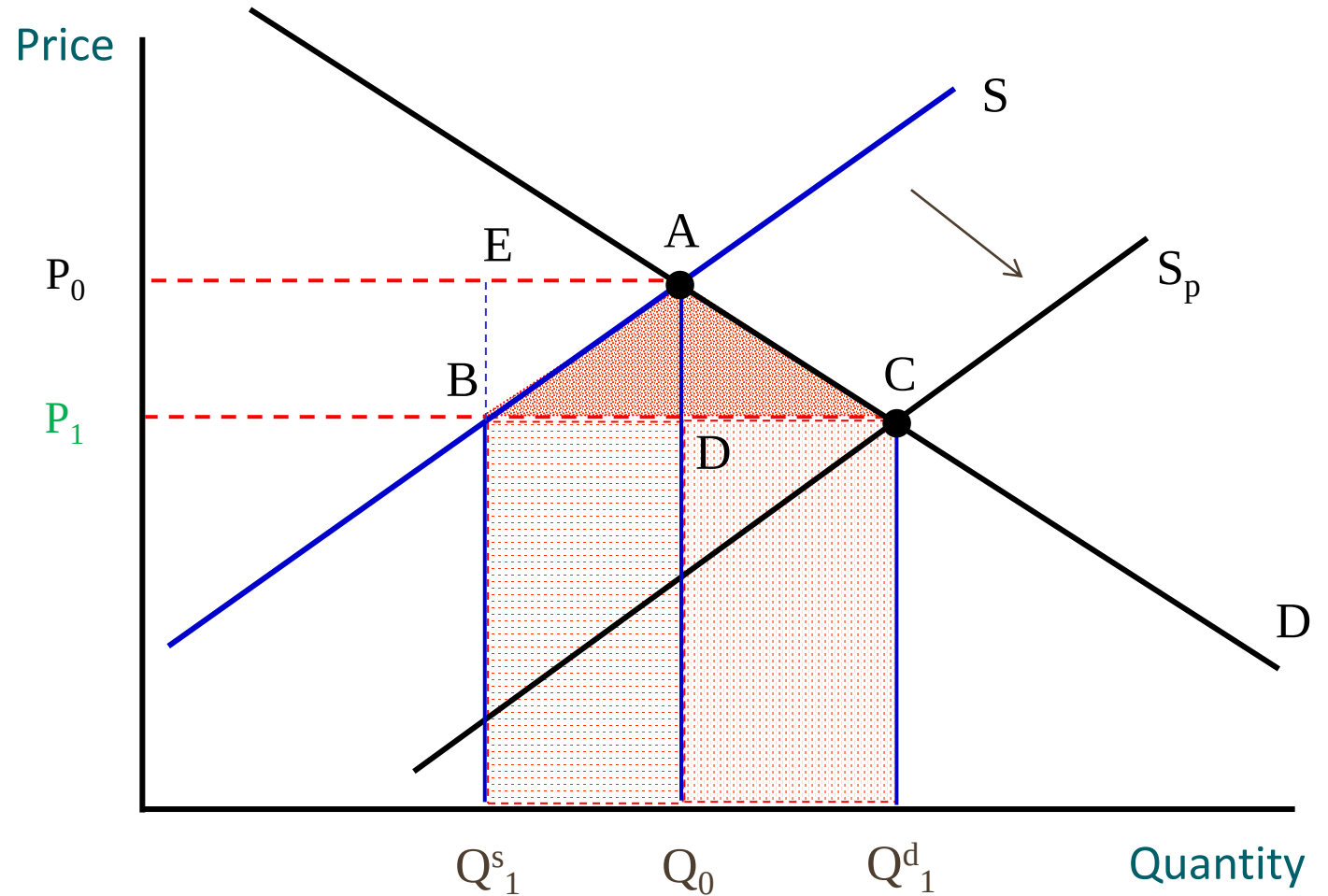
$$\begin{aligned}\Delta TR &= P_1(Q_d - Q_s) = P_1\Delta Q \\ &= Q_s^s B C Q_d^d\end{aligned}$$

Economic benefit:

$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P_1\Delta Q + P_0ACP_1 - P_0ABP_1 + 0 \\ &= Q_s^s B A C Q_d^d\end{aligned}$$

=> Economic benefit > Financial benefit

$$\Rightarrow P_e = (P_0 + P_1)/2 > P_1$$



A perfectly competitive market

- The observed demand price may not represent the marginal value of a good or service, and the observed supply price may not represent marginal cost.
- In such cases, the approach we adopt is known as **shadow-pricing**: where appropriate, we adjust the observed imperfect market price so that it measures marginal benefit or marginal cost, and we use the adjusted price – the shadow price – in the Efficiency Analysis.
- Shadow prices are sometimes also referred to as efficiency prices.
- It must be stressed again that these prices are used for the purposes of project appraisal only, and are not prices at which project inputs or outputs are actually traded.

Shadow pricing marketed inputs

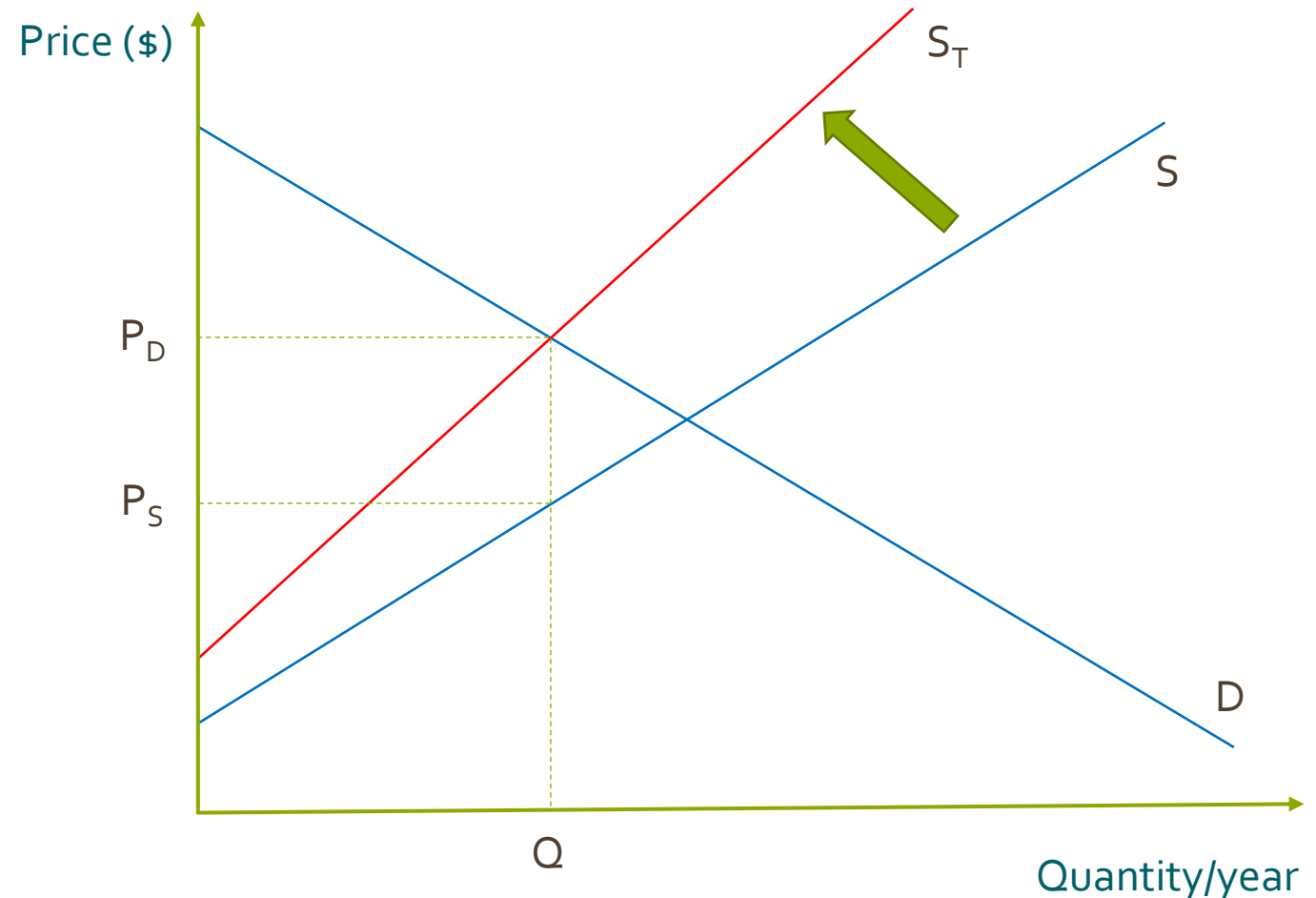
Distorted markets, small-scale projects



Indirect taxes on a project's input

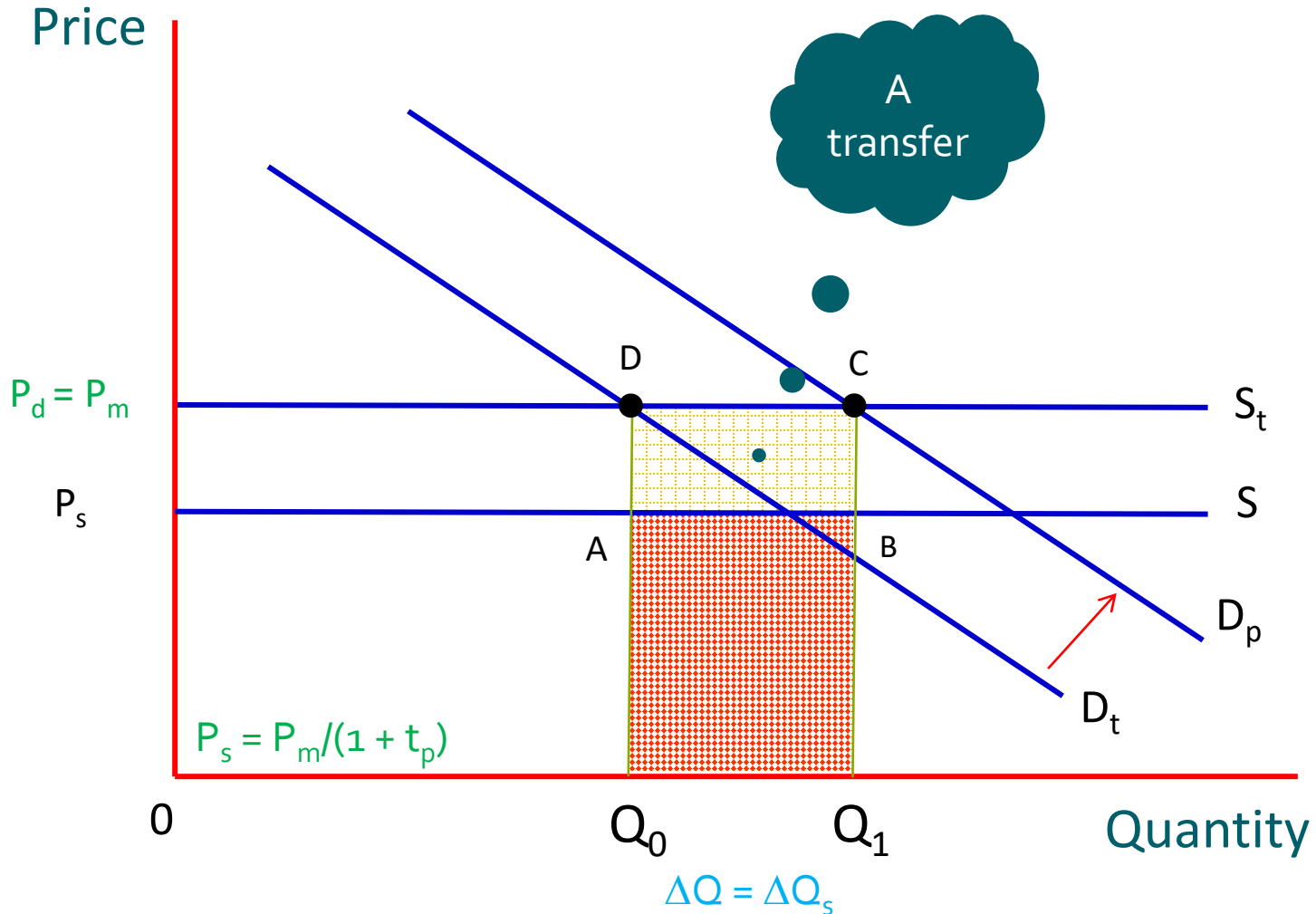
- Suppose an indirect tax t_p on the producers, then the market price of the input is $P_m = P_d > P_s = P_m/(1 + t_p)$.
- If a small number of additional units of the input are produced to meet the demand of the project, the shadow price is measured by P_s .
- If additional units cannot be produced in the time available, they will have to be diverted to the project from existing users who value them at P_d . Hence, the shadow is measured by P_d (?).

Percentage tax levied at producers level



Indirect taxes on a project's input

(A small-scale project's input)



Financial cost:

$$\Delta TC = -P_d(Q_1 - Q_0) = -P_d \Delta Q$$

$$= -Q_0 D C Q_1$$

Economic cost:

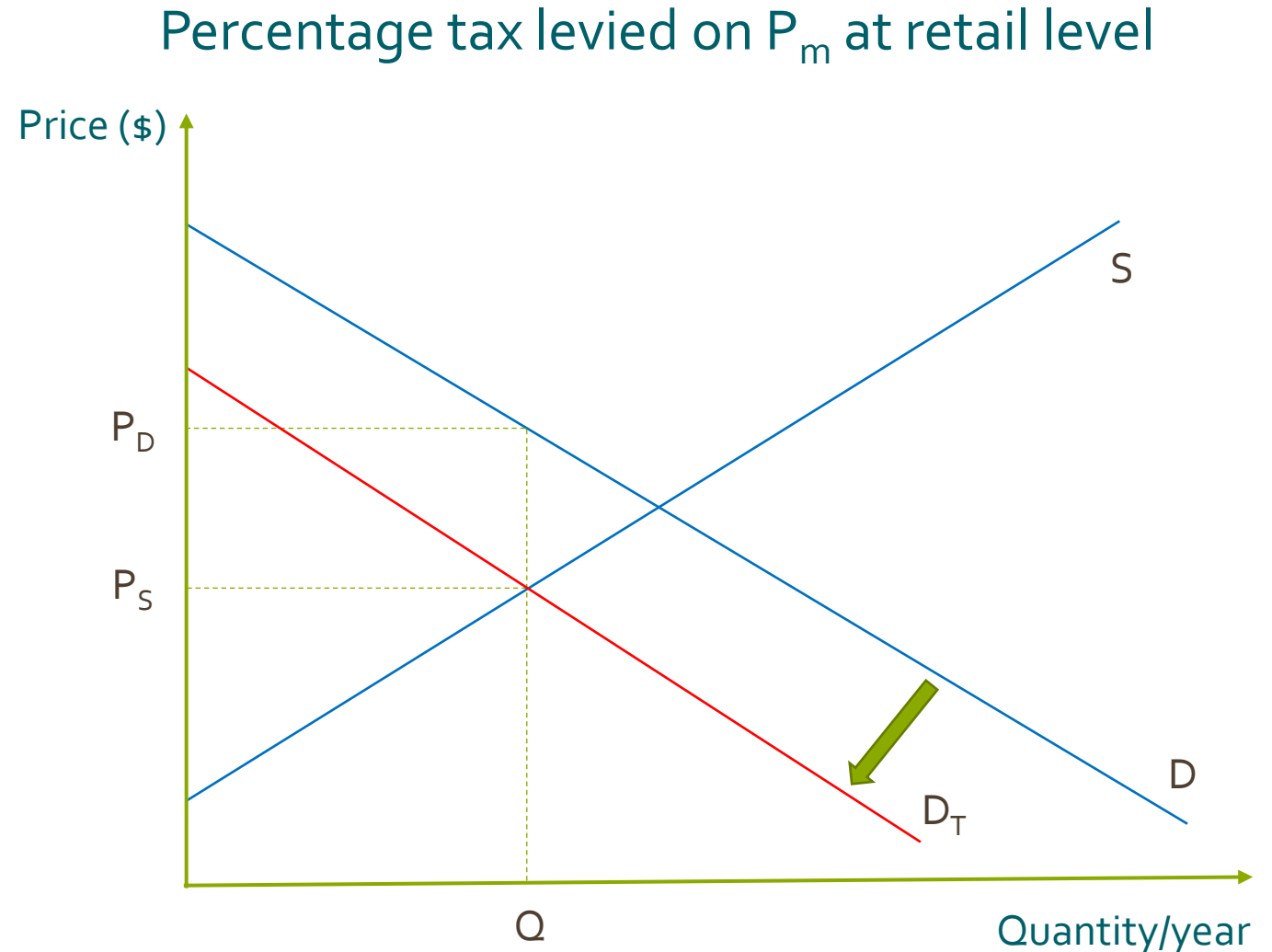
$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P_d \Delta Q + 0 + 0 + ABCD \\ &= -Q_0 ABQ_1\end{aligned}$$

=> Economic cost < Financial cost

$$\Rightarrow P_e = P_s < P_d$$

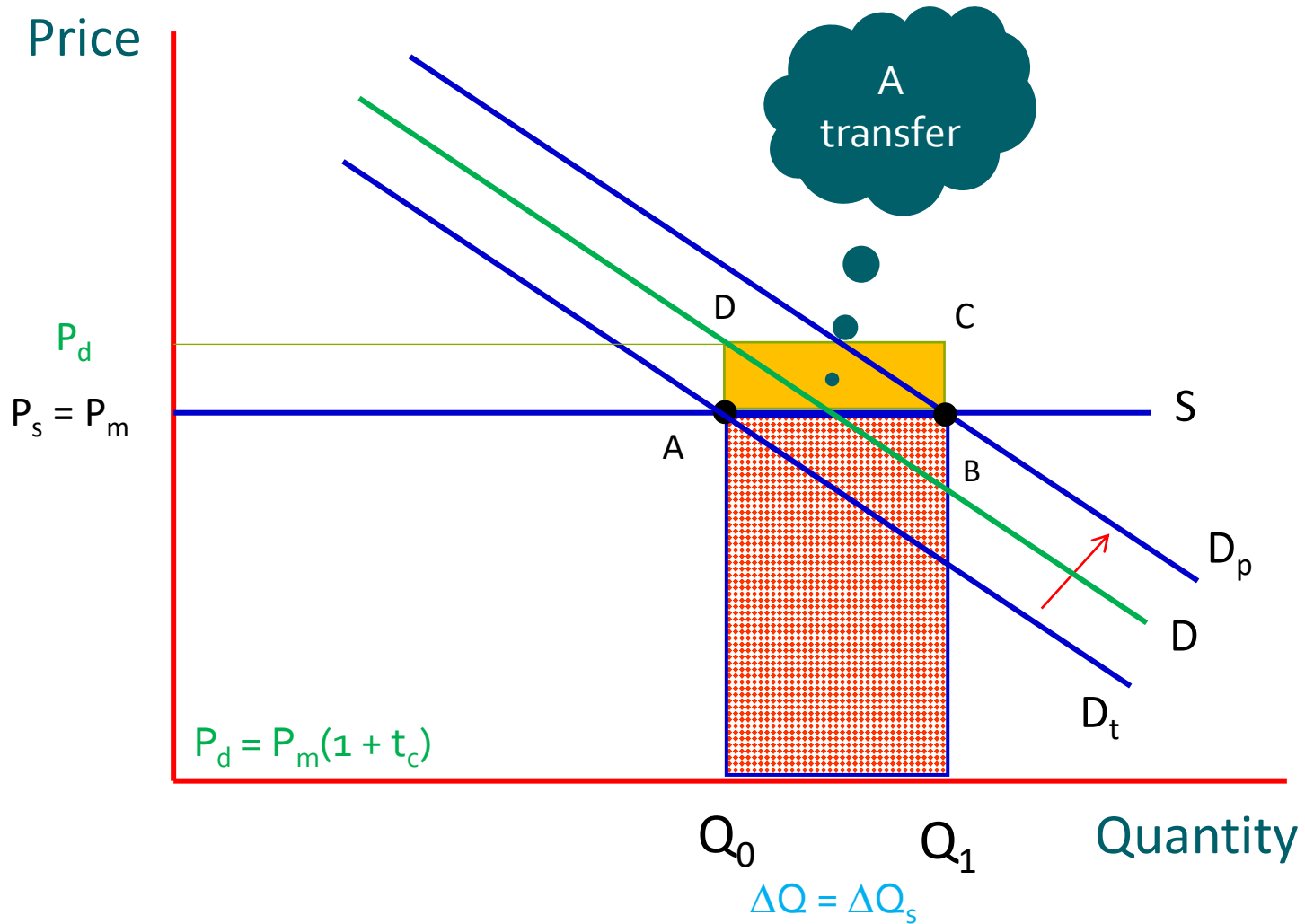
Indirect taxes on a project's input

- Suppose an indirect tax t_c on the consumers, then the market price of the input is $P_m = P_s < P_d = P_m(1 + t_c)$.
- If a small number of additional units of the input are produced to meet the demand of the project, the shadow price is measured by P_s .
- If additional units cannot be produced in the time available, they will have to be diverted to the project from existing users who value them at P_d . Hence, the shadow is measured by P_d (?).



Indirect taxes on a project's input

(A small-scale project's input)



Financial cost:

$$\Delta TC = -P_d(Q_1 - Q_o) = -P_d \Delta Q$$

$$= -Q_o DC Q_1$$

Economic cost:

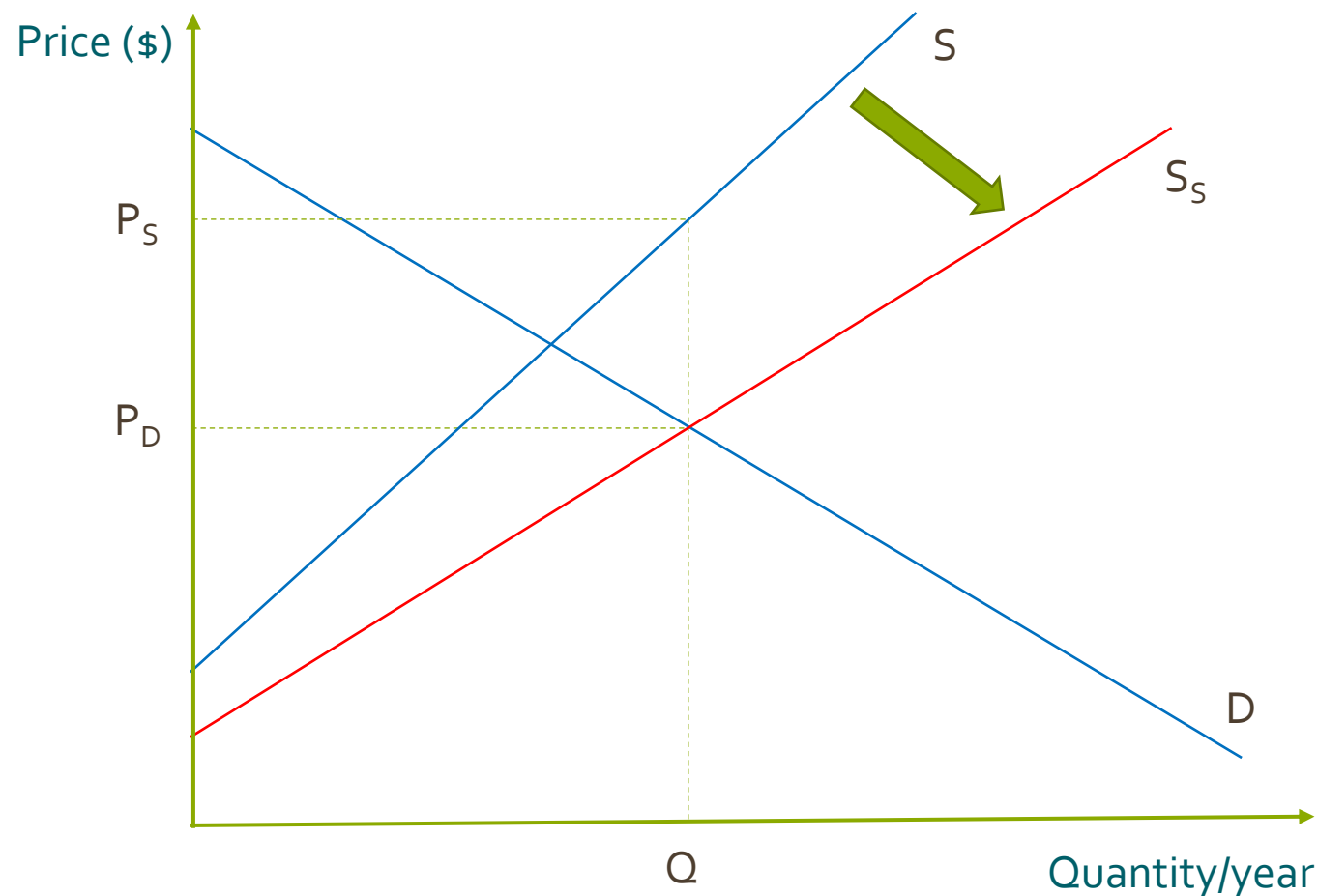
$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P_d \Delta Q + 0 + 0 + ABCD \\ &= -Q_0 ABQ_1\end{aligned}$$

=> Economic cost < Financial cost

$$\Rightarrow P_e = P_s < P_d$$

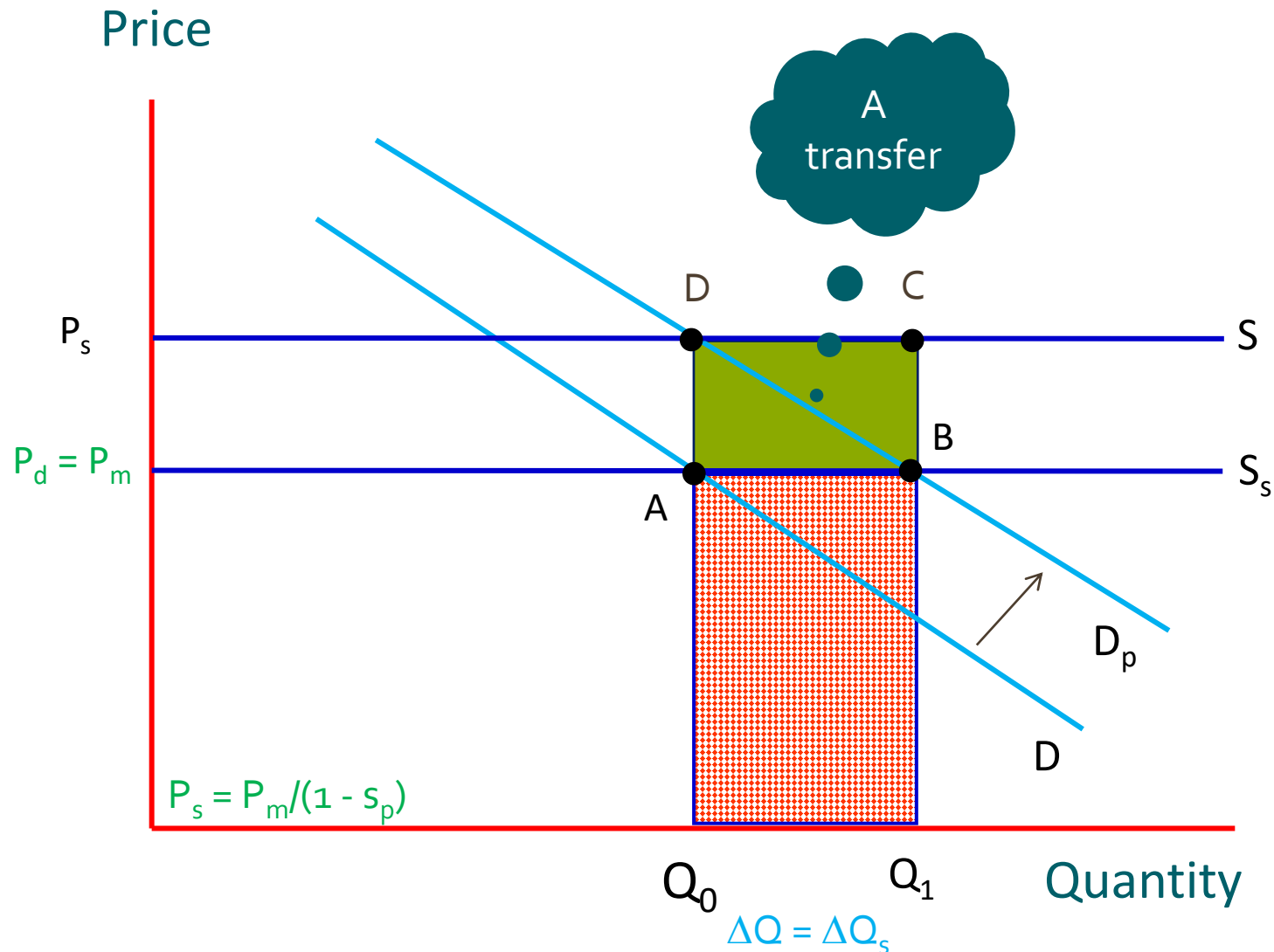
A project's input subject to a subsidy

- Suppose an input of a project, such as sugar) is subject to an *ad valorem* subsidy s_p , the market price of this input is $P_m = P_d < P_s = P_m/(1 + s_p)$.
- If extra units of this input is produced to meet the project's requirements, the opportunity cost of production is the supply price (i.e. shadow price), P_s .
- If there is a quota on total production, the input required for the project will have to be diverted from an alternative use valued at the market price, $P_m (= P_d)$. This will be the price used in the efficiency analysis.



A project's input subject to a subsidy

(A small-scale project's input)



Financial cost:

$$\begin{aligned}\Delta TC &= -P_d(Q_1 - Q_0) = -P_d\Delta Q \\ &= -Q_0ABQ_1\end{aligned}$$

Economic cost:

$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P_d\Delta Q + 0 + 0 - ABCD \\ &= -Q_0DCQ_1\end{aligned}$$

=> Economic cost > Financial cost

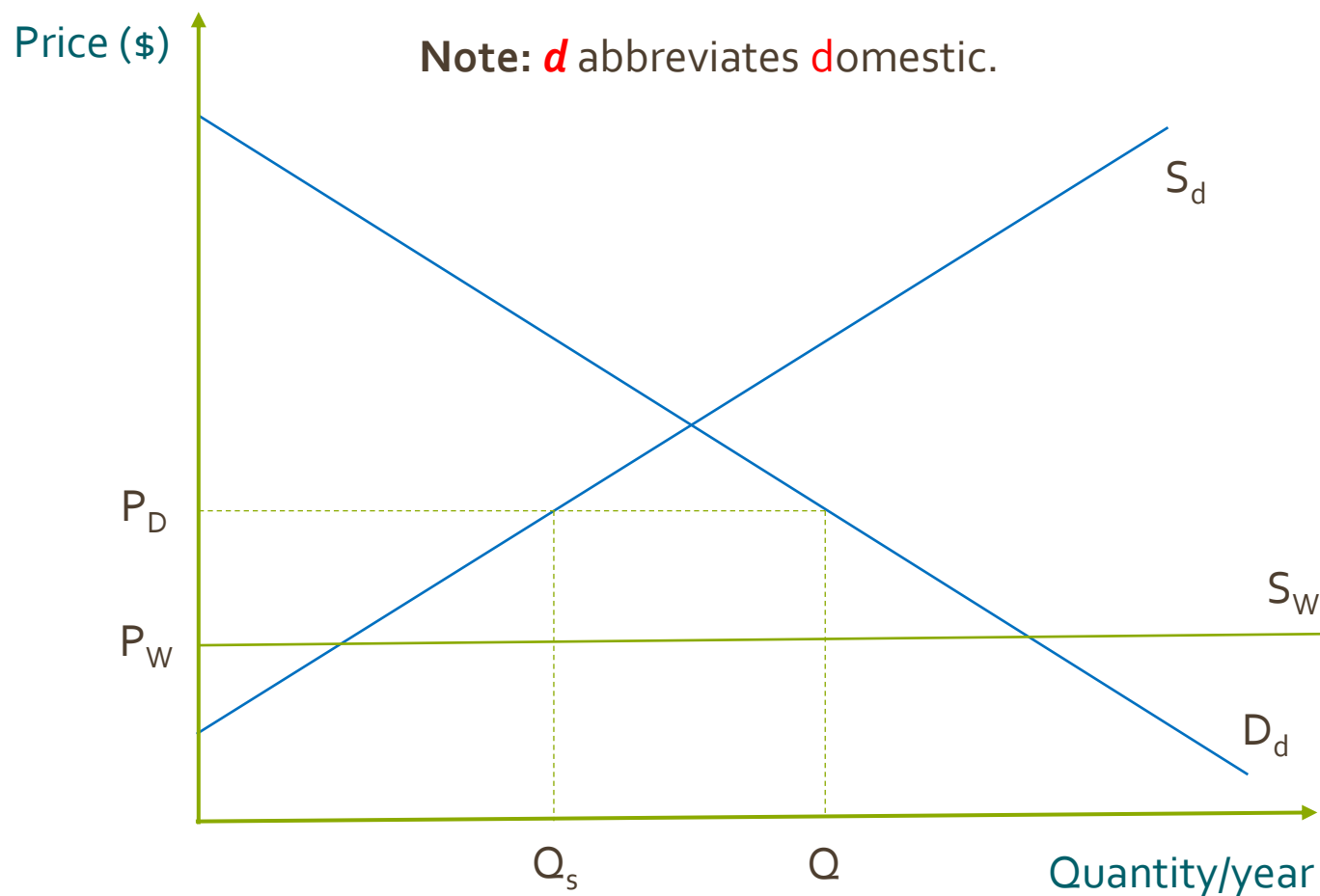
=> $P_e = P_s > P_d$

Gross versus net price concepts

- When we talk of the *gross of tax* price, or the *after-tax* price, or the *tax-inclusive* price of a good or service, we mean its price including the tax – P_d .
- The *net of tax*, or *before-tax*, or *tax-exclusive* price is its price not including the tax – P_s .
- Similarly, when we talk of a *gross of subsidy*, or *before-subsidy*, or *unsubsidized* price, we mean the price before the subsidy is applied – P_s .
- The *net of subsidy*, or *after-subsidy*, or *subsidized* price is the price once the subsidy has been applied - P_d .

An imported input subject to a tariff

- If domestic production of an input is protected by a tariff t_m , then P_d is the cost of an extra unit of domestic production; P_w is the cost of an extra unit of imports.
- Assuming the absence of a quota on the volume of imports, the small increase in input demand contributed by the project will be met by imports. While the project will pay the market price, $P_m = P_d = P_w(1 + t_m)$ for the input, the cost to the economy of an additional unit of imports is P_w (i.e. shadow price).
- If further imports are prevented by an import quota, the quantity of the input used by the project will be supplied by additional domestic production, diverted from other uses. The opportunity cost of project input will be measured by either P_d .



An imported input subject to a tariff

(A small-scale project's input)

Financial cost:

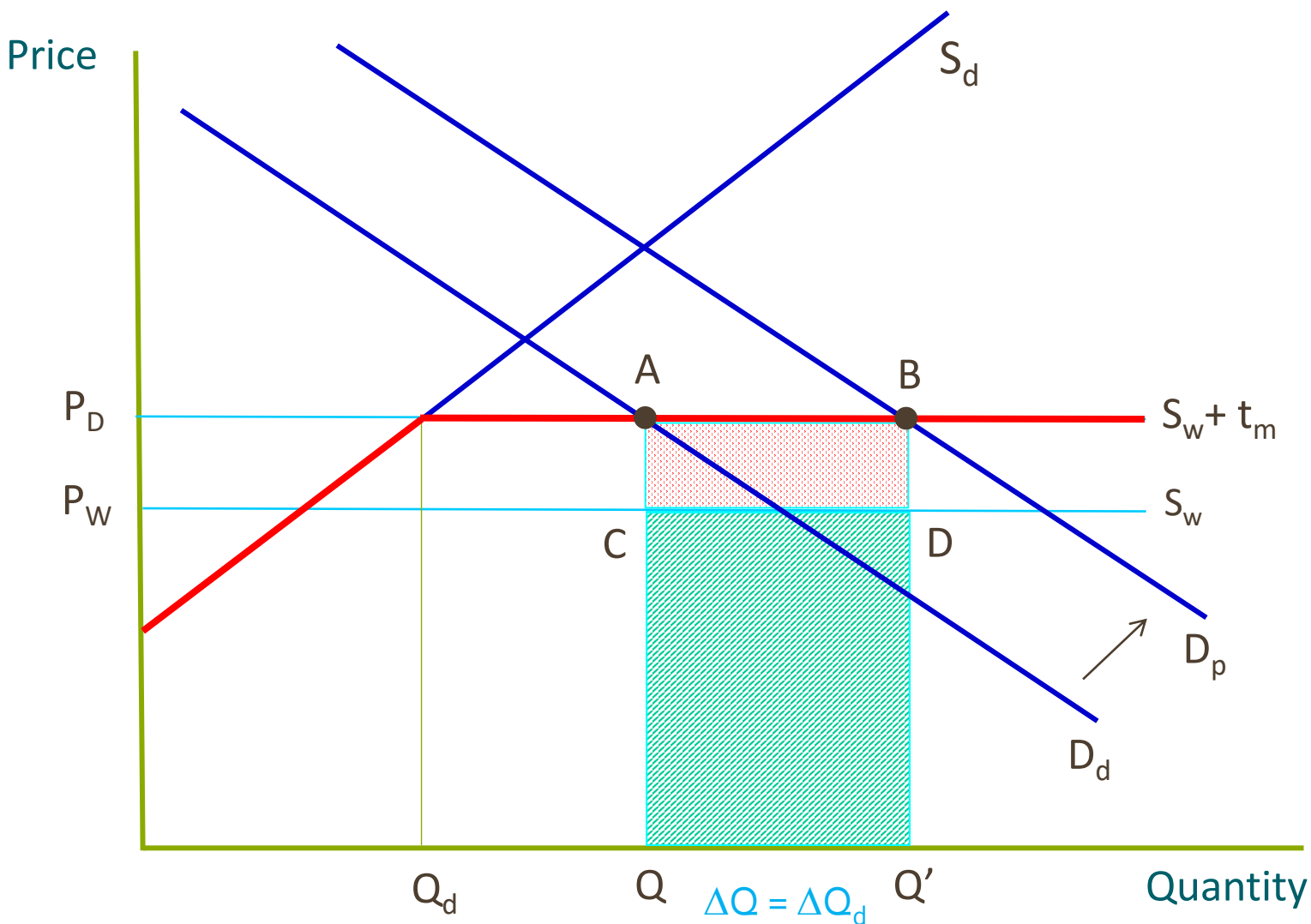
$$\begin{aligned}\Delta TC &= -P_d(Q' - Q) = -P_d\Delta Q \\ &= -QABQ'\end{aligned}$$

Economic cost:

$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P_d\Delta Q + 0 + 0 + ABCD \\ &= QCDQ'\end{aligned}$$

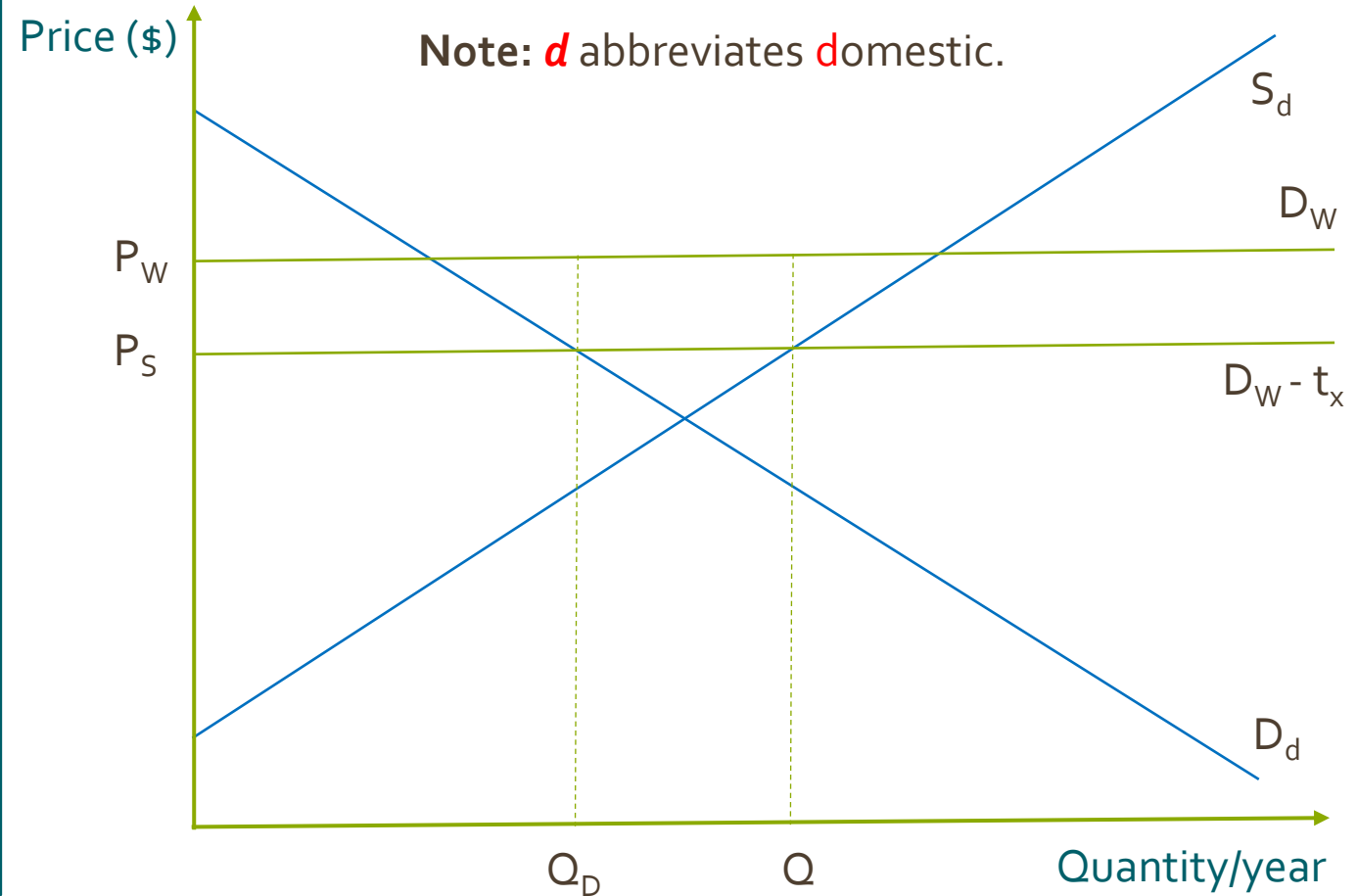
=> Economic cost < Financial cost

$$\Rightarrow P_e = P_w < P_d$$



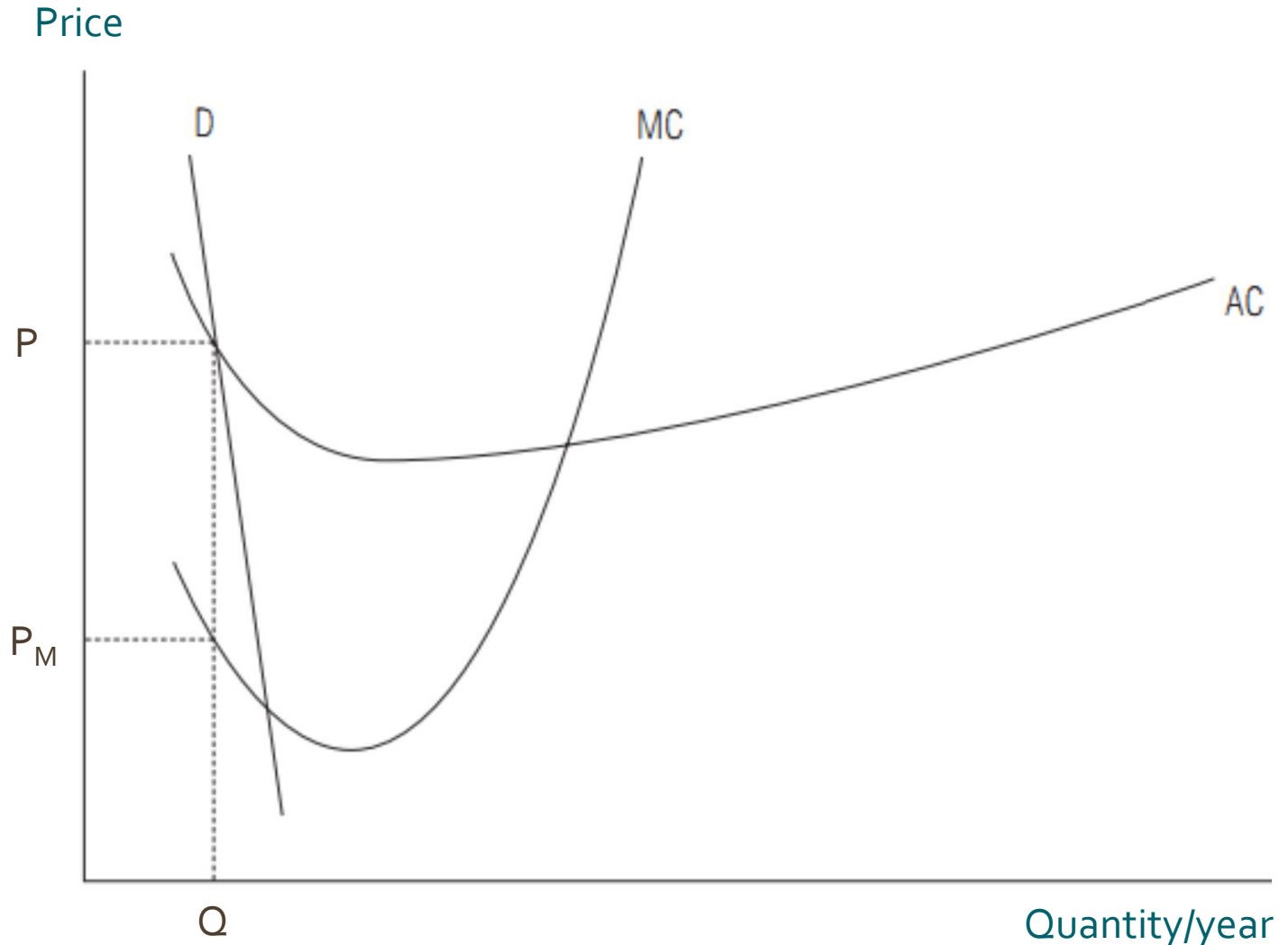
An exportable input

- If a product such as coffee is levied an export tax t_x , then $P_s (= P_w - t_x)$ is the cost of an extra unit of domestic production; P_w is the benefit of an extra unit of exports.
- If a small number of additional units of the input are produced to meet the demand of the project, the shadow price is measured by P_s .
- If additional units cannot be produced in the time available, they will have to be diverted to the project from an exportable source. Hence, the shadow is measured by P_w .



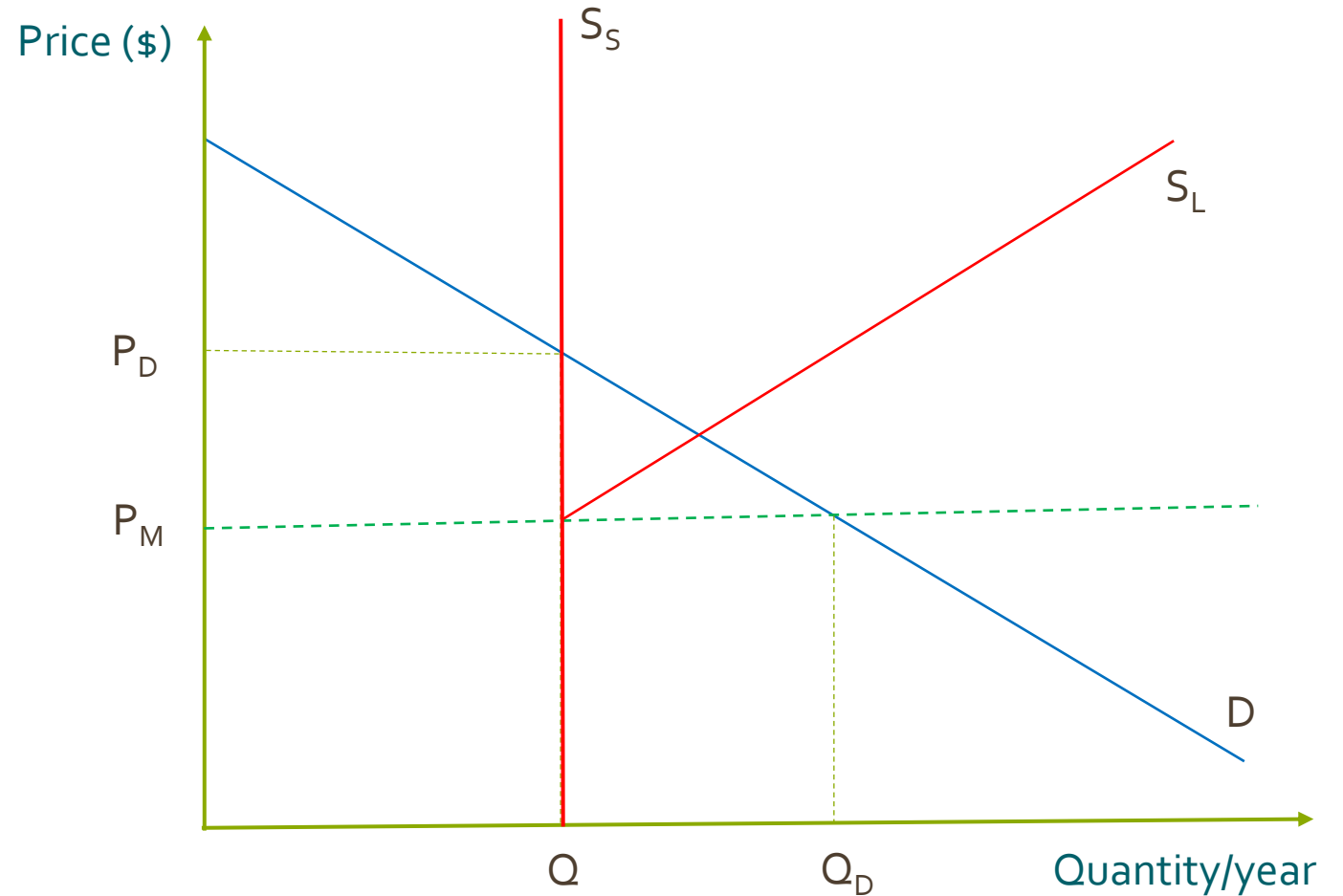
An input under conditions of decreasing unit cost

- One of the features of electricity production is the economies of scale; this means that the average cost of production continues to fall as the chosen level of output increases until that level becomes quite large.
- Since the power utility is a monopoly, the government regulates the price of electricity at a level sufficient to cover the utility's costs – at the regulated price P .
- If the small quantity of power used in the project will be in addition to the total supply, it is valued at its supply price, or marginal cost – P_M (shadow price).
- If the utility is operating at capacity, the power used in the project will be diverted from some other use, and it is valued at the market price, P .



A project's input subject to a maximum price

- The market for rental units with rent control at P_M . The short-run supply curve of rental units, S_S , is vertical at the current quantity supplied, Q . The long-run supply curve, S_L , offers the possibility of adding to the current stock.
- If an apartment occupied by one of the project experts is not available for rent to a local resident who placed a value of P_d . Hence, P_d is the relevant shadow price of the input in an Efficiency Analysis.



A project's input subject to a maximum price

(A small- or large-scale project's input)

Financial cost:

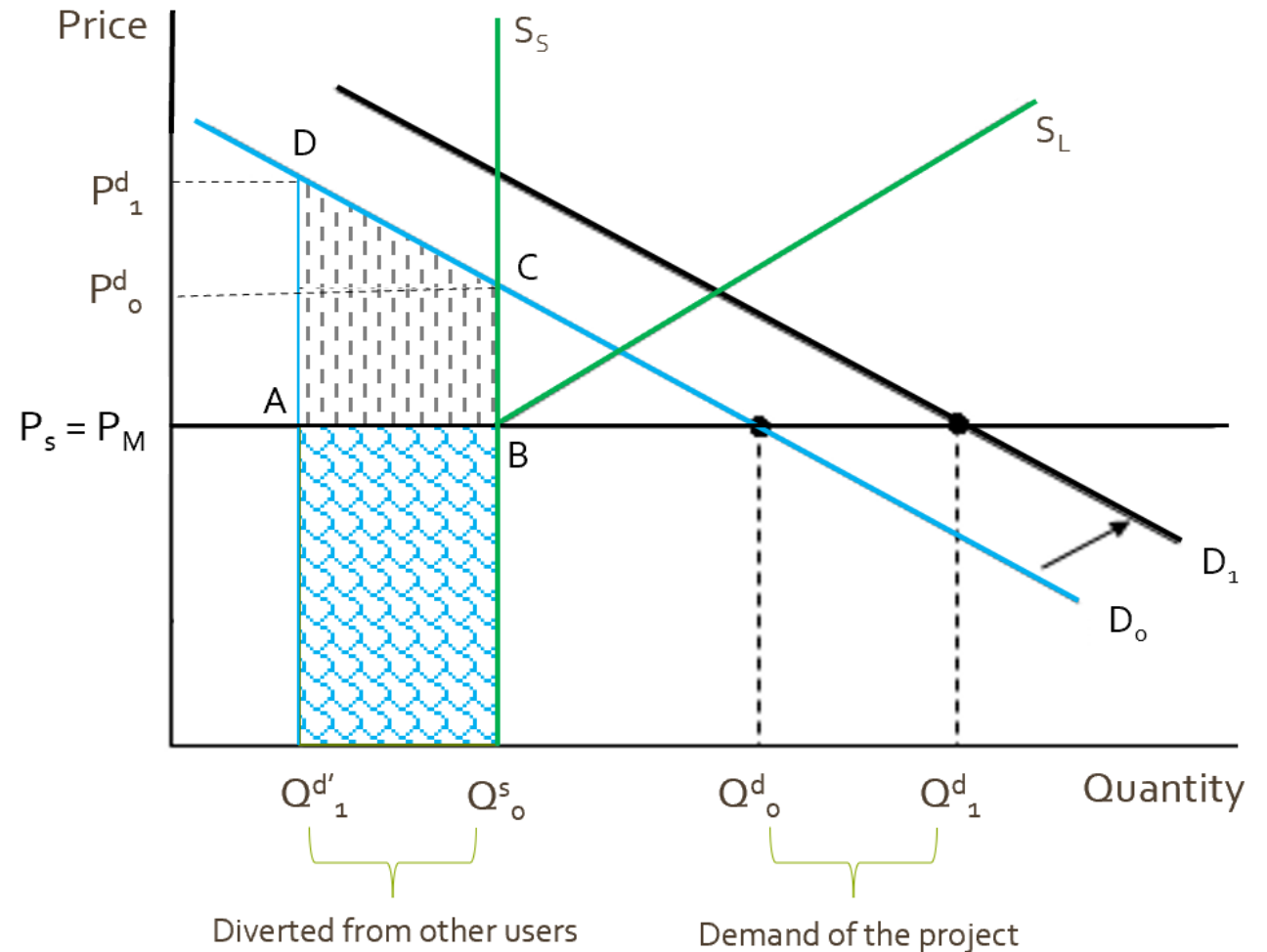
$$\begin{aligned}\Delta TC &= -P_s(Q_o^s - Q_1^{d'}) = -P_s\Delta Q \\ &= -Q_1^{d'}ABQ_o^s\end{aligned}$$

Economic cost:

$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P_s\Delta Q - ABCD + 0 + 0 \\ &= -Q_1^{d'}DCQ_o^s\end{aligned}$$

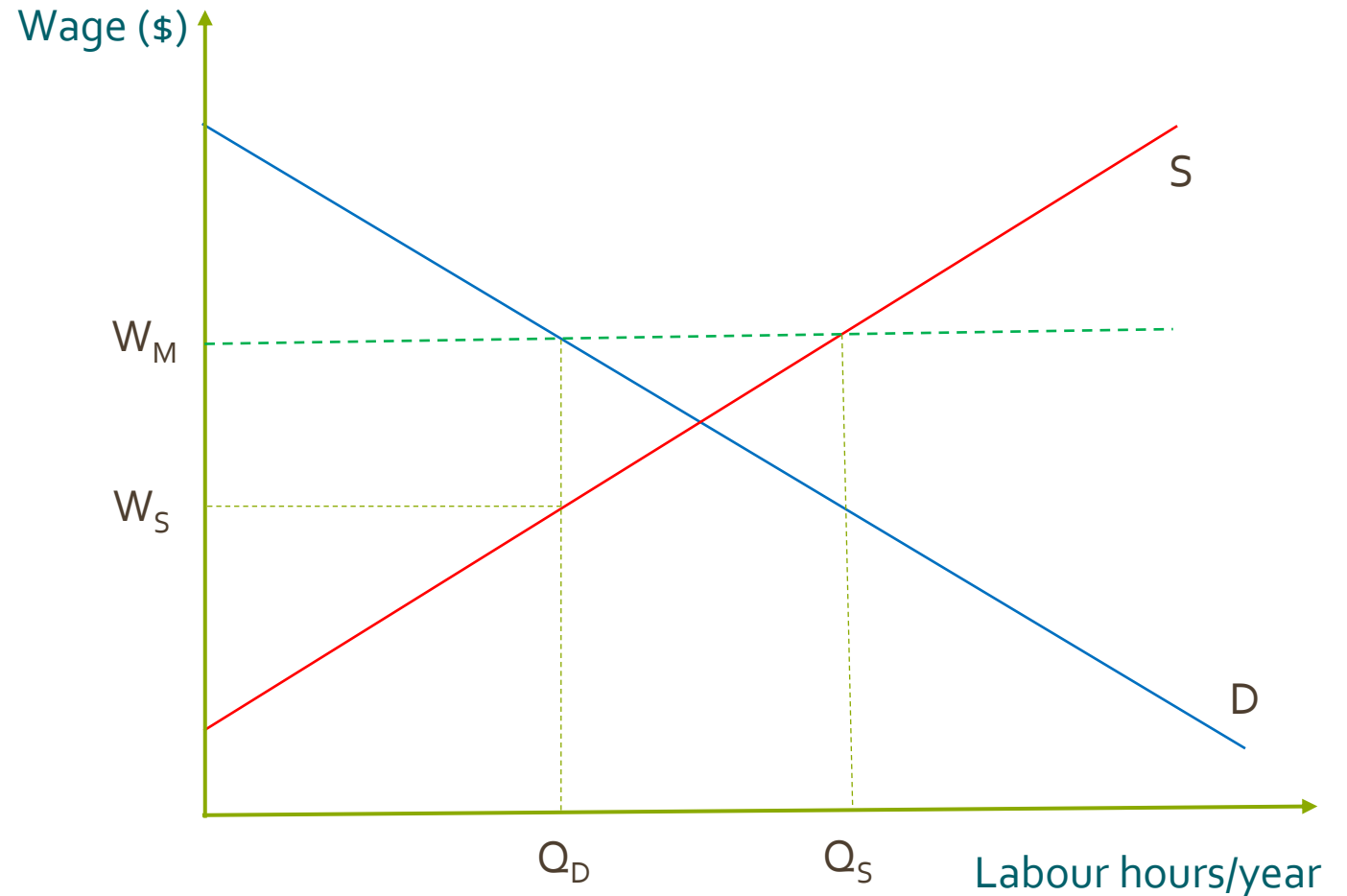
=> **Economic cost > Financial**

$$\Rightarrow P_e = P_d = (P_o^d + P_1^d)/2 > P_s$$



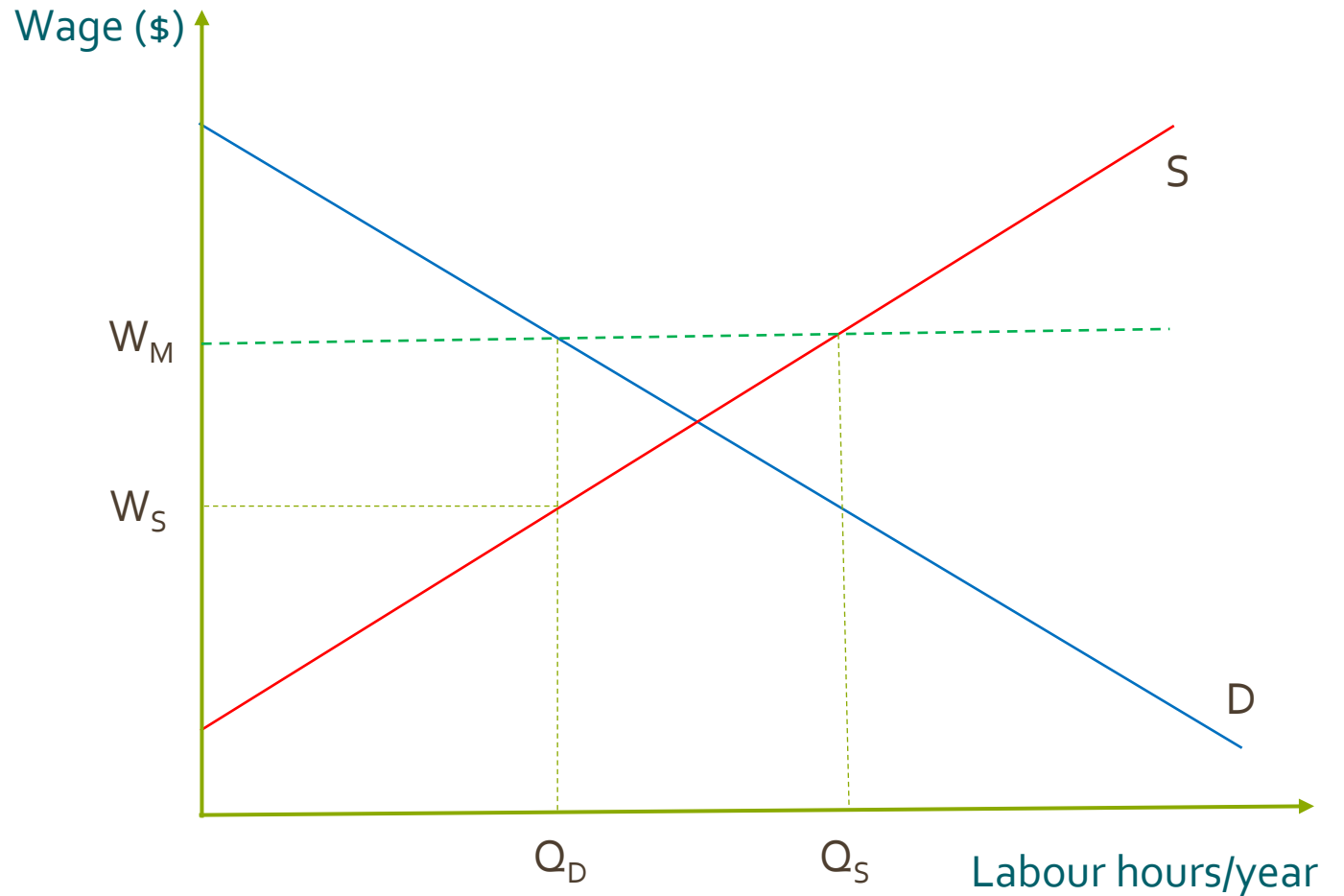
Unskilled labour subject to a minimum wage

- By the *wage* we mean the unit cost of labour to the project consisting, perhaps, of take-home pay, income tax withheld, superannuation contributions by the employer and payroll tax.
- There are two prices of labour at the market equilibrium, Q_D – the supply price W_s (the wage which a marginal unit of labour would be willing to accept) and the market wage, W_M (the cost to the employer of a marginal unit of labour).



Unskilled labour subject to a minimum wage

- If the worker hired for the (public) project would otherwise be unemployed, it might be tempting to say that the opportunity cost is **zero** (sometimes).
- If the labour supply elasticity exceeds zero, an extra unit of labour has an opportunity cost measured by W_s – that represents the value to the economy of the marginal hour of unemployed labour in its alternative occupation – work in the informal economy, such as gardening, or just plain leisure.



Unskilled labour subject to a minimum wage

(A small- or large-scale project's input)

Financial cost:

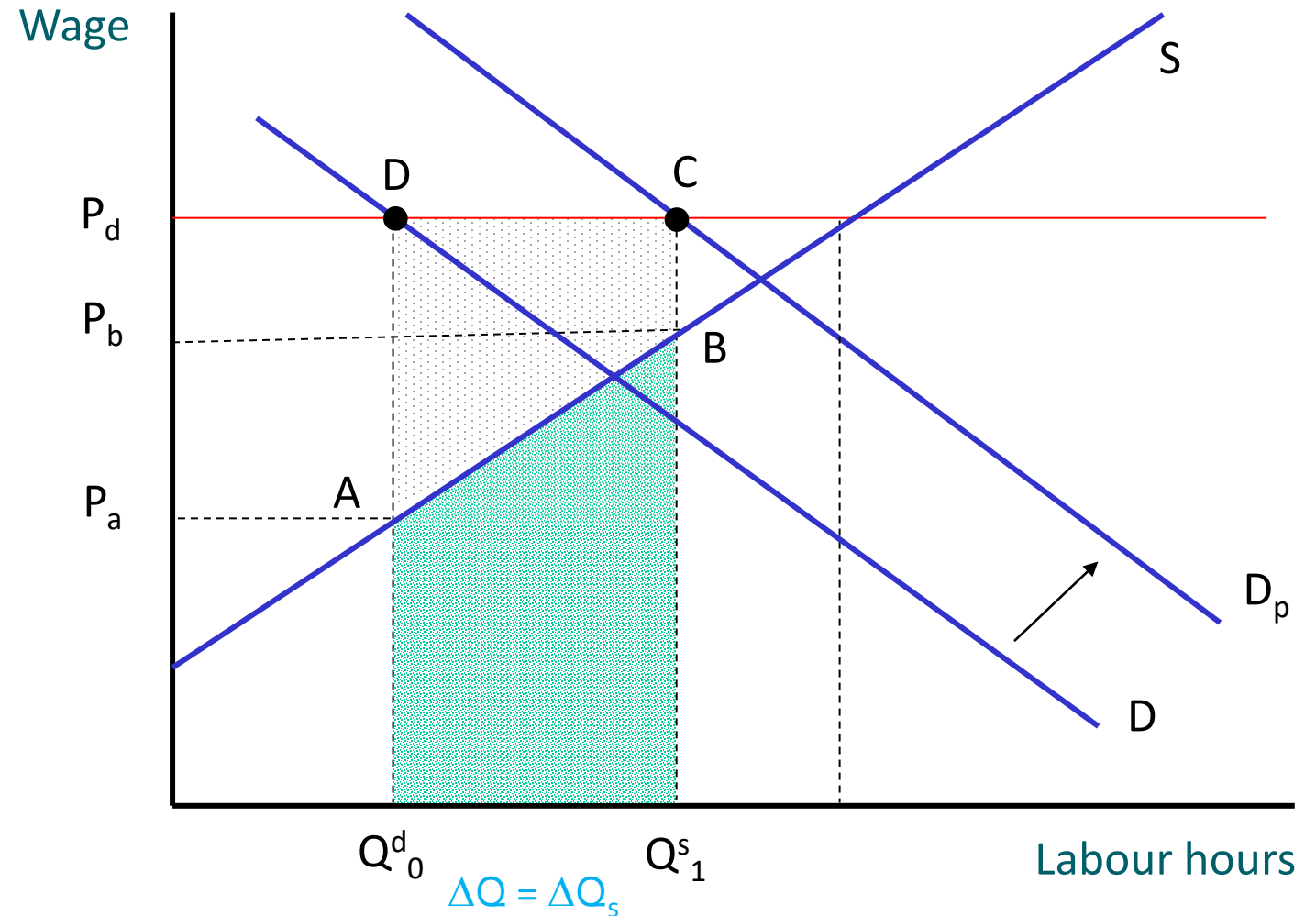
$$\begin{aligned}\Delta TC &= -P_d(Q_{1d} - Q_{0d}) = -P_d\Delta Q \\ &= -Q_0^d DCQ_1^s\end{aligned}$$

Economic cost:

$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P_d\Delta Q + 0 + ADCB + 0 \\ &= -Q_0^d ABQ_1^s\end{aligned}$$

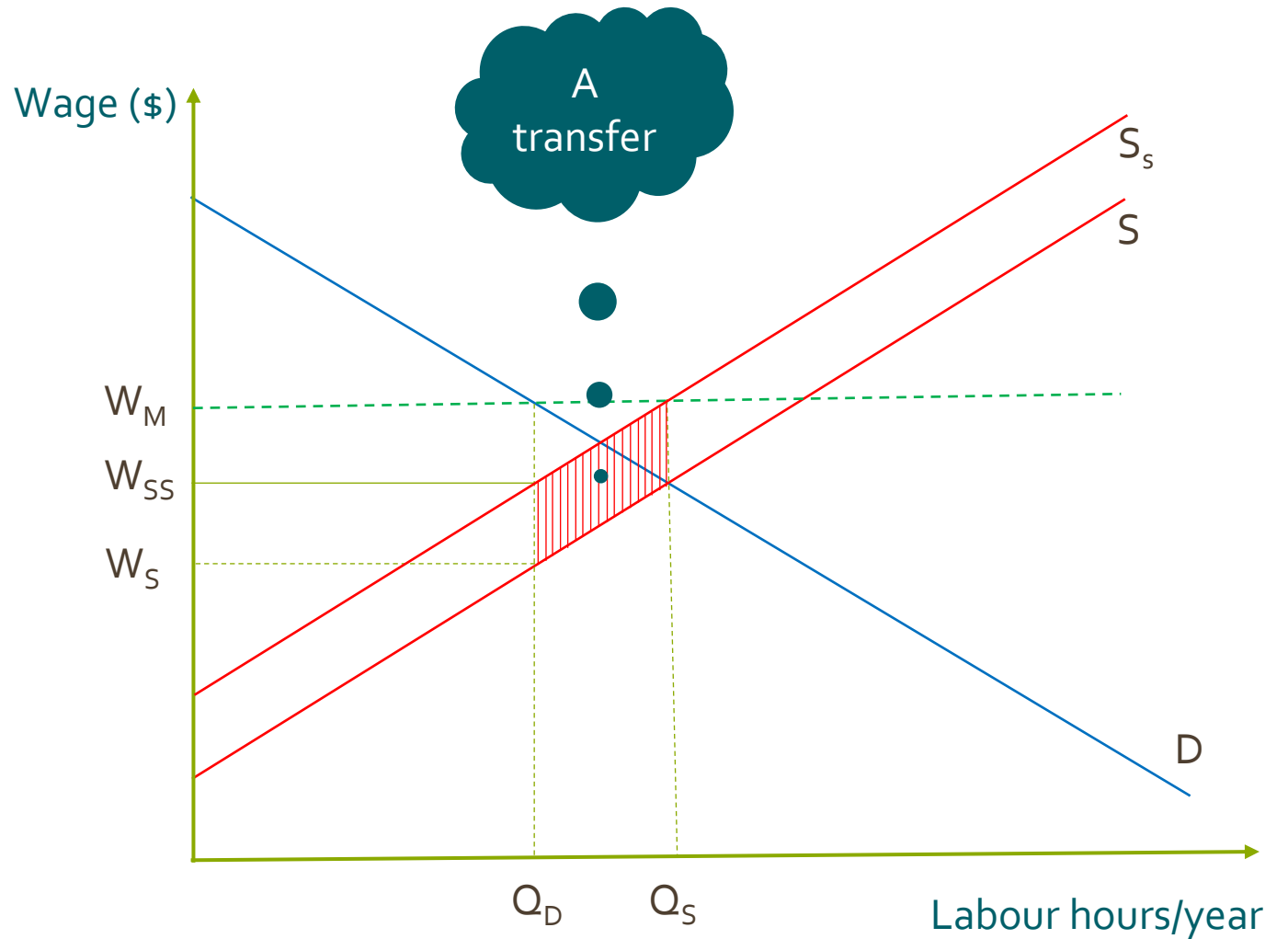
=> **Economic cost < Financial cost**

$$\Rightarrow P_e = P_s = (P_0^s + P_1^s)/2 < P_d$$



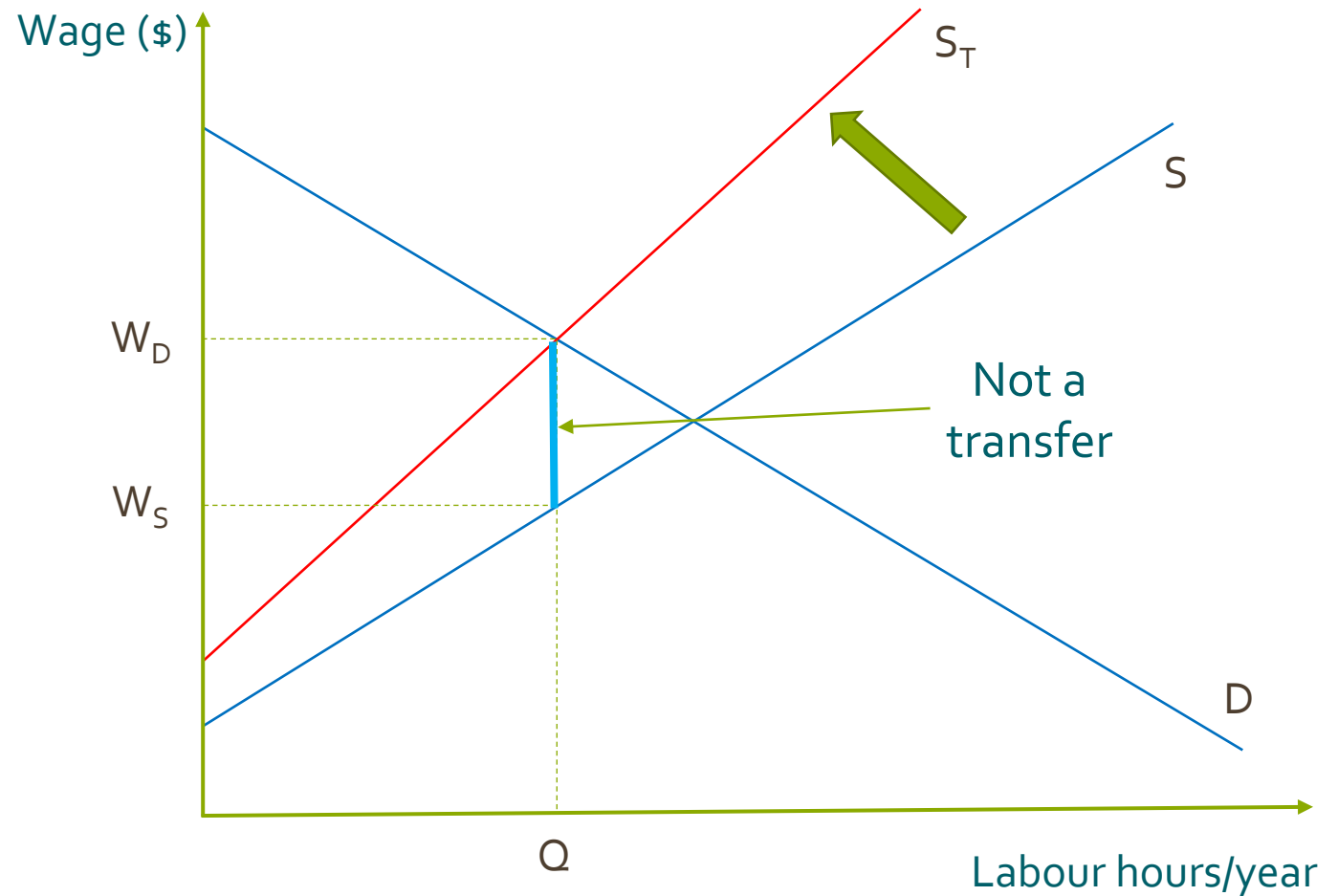
Unskilled labour subject to a minimum wage

- Suppose that unemployed labour receives an unemployment benefit (i.e. subsidy on leisure => reduce the level of unemployment).
- There are now two opportunity costs of otherwise unemployed labour: the OC to the economy, W_s ; and the OC to the unemployed person is the gross of subsidy, W_{ss} . The private OC is the value of leisure time, W_s , plus the unemployment benefit, which the person will forgo if they accept a job.
- The OC of labour to the project is still the value of output forgone W_s .



Skilled labour subject to an income tax

- Suppose an income tax t_L on the skilled labour income, then the market price of this labour is $W_m = W_d > W_s = W_m/(1 + t_L)$.
- If the labour employed on the project is sourced from an alternative market use, its OC is the value of its marginal product in its alternative job, which is the gross of tax wage, $W_m = W_D$.
- If we were to price labour at the after-tax wage (W_s), we would be understating its OC to the economy by the amount of the tax.



Land as a project's input

- The opportunity cost of land is the net value of the output which the land would have produced in its alternative use.
- If land use may be subject to taxes/subsidies, the value of foregone output is calculated according to the procedures discussed above.
- If land use is subject to regulations, the value of output in its constrained use measures the OC of the land.
 - Often the alternative use of land is in agriculture, in which case its OC is the time-stream of net returns – value of output less cost of inputs other than land – that would have been generated.
 - The annual OC of the land should take into account the soil degradation (i.e. declines over time).
 - Goods or services from vacant land are not traded in markets, we may apply the non-market valuation techniques to estimate the OC of vacant land.
- The OC of land is best entered as a series of annual values over the life of the project. In case in which the capital value of land is used to measure cost, the residual value of the land must be entered as a benefit at the end of the project's life.

Shadow pricing marketed outputs

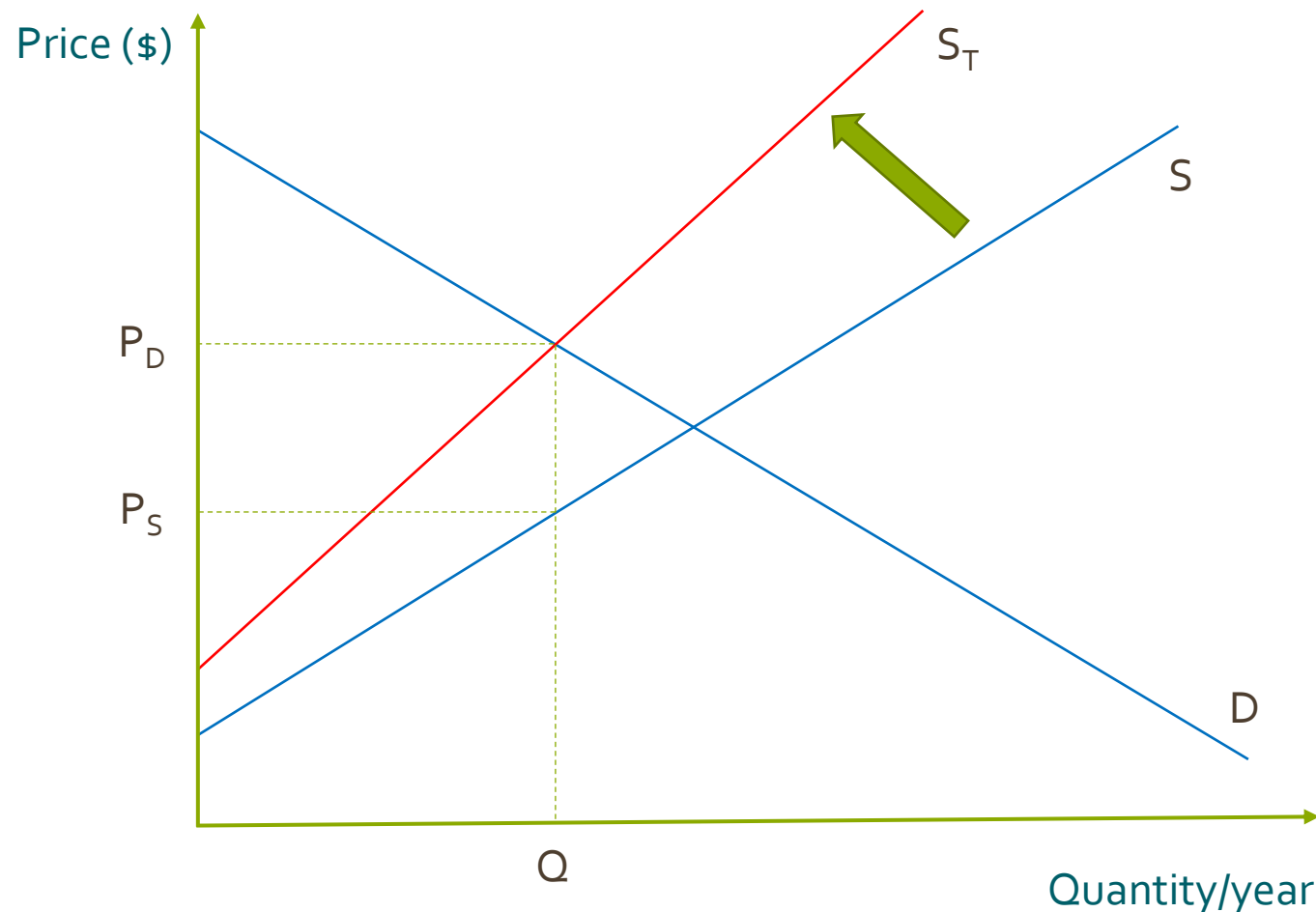
Distorted markets, small-scale projects



Indirect taxes on a project's output

- Suppose an indirect tax t_p on the producers, then the market price of the output is $P_m = P_d > P_s = P_m/(1 + t_p)$.
- If a small number of the output are produced by the project to meet additional demand, the shadow price is measured by P_d .
- If a small number of the output of other suppliers are replaced by the project output, the shadow price is measured by P_s , i.e. the opportunity cost of resources that would have been used by other suppliers to produce the replaced output (?).

Percentage tax levied at producers level



Indirect taxes on a project's output

(A small-scale project's output)

Financial benefit:

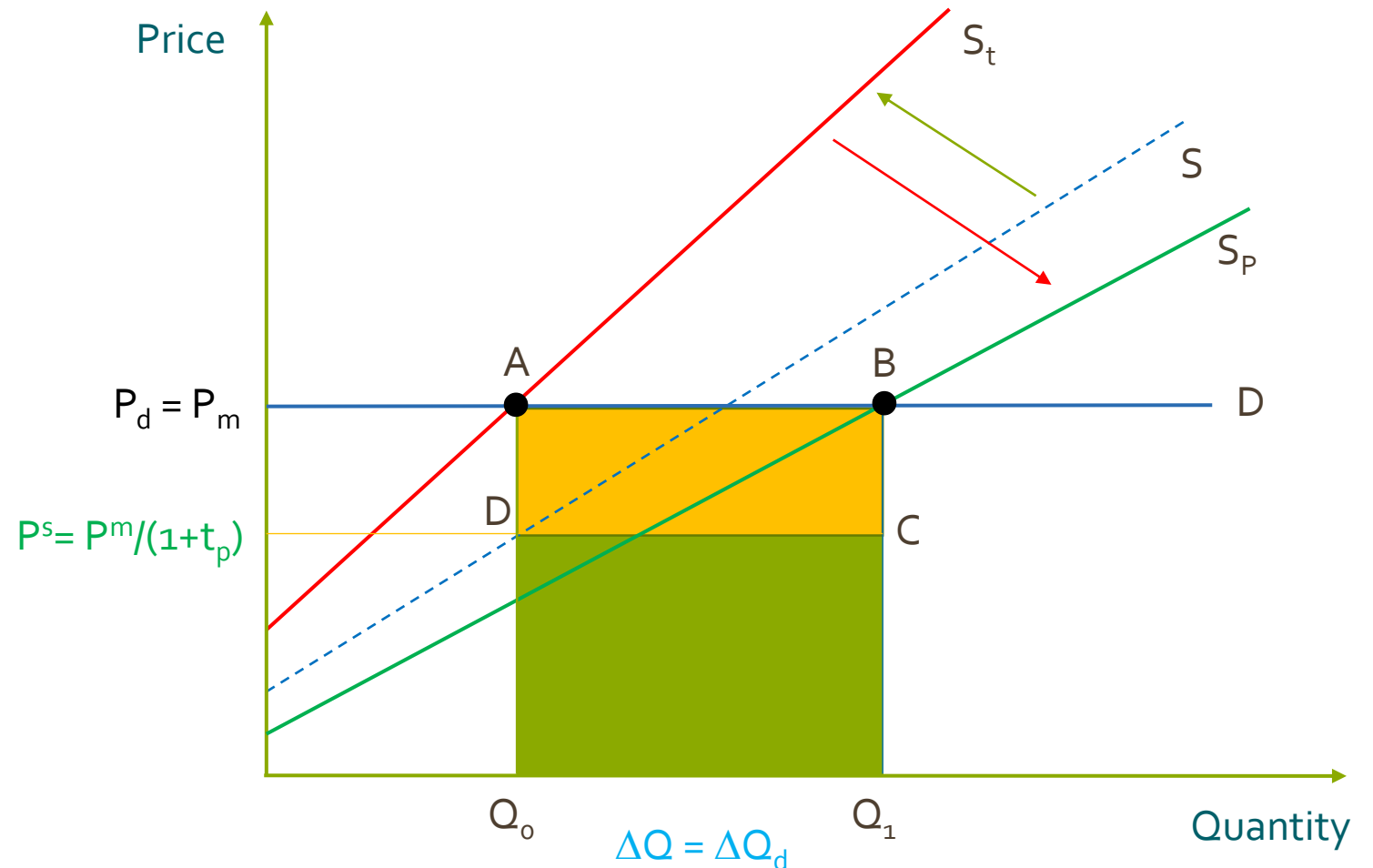
$$\begin{aligned}\Delta TR &= P_s(Q_1 - Q_0) = P_s \Delta Q \\ &= Q_0 DCQ_1\end{aligned}$$

Economic benefit:

$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P_s \Delta Q + 0 + 0 + ABCD \\ &= Q_0 ABQ_1\end{aligned}$$

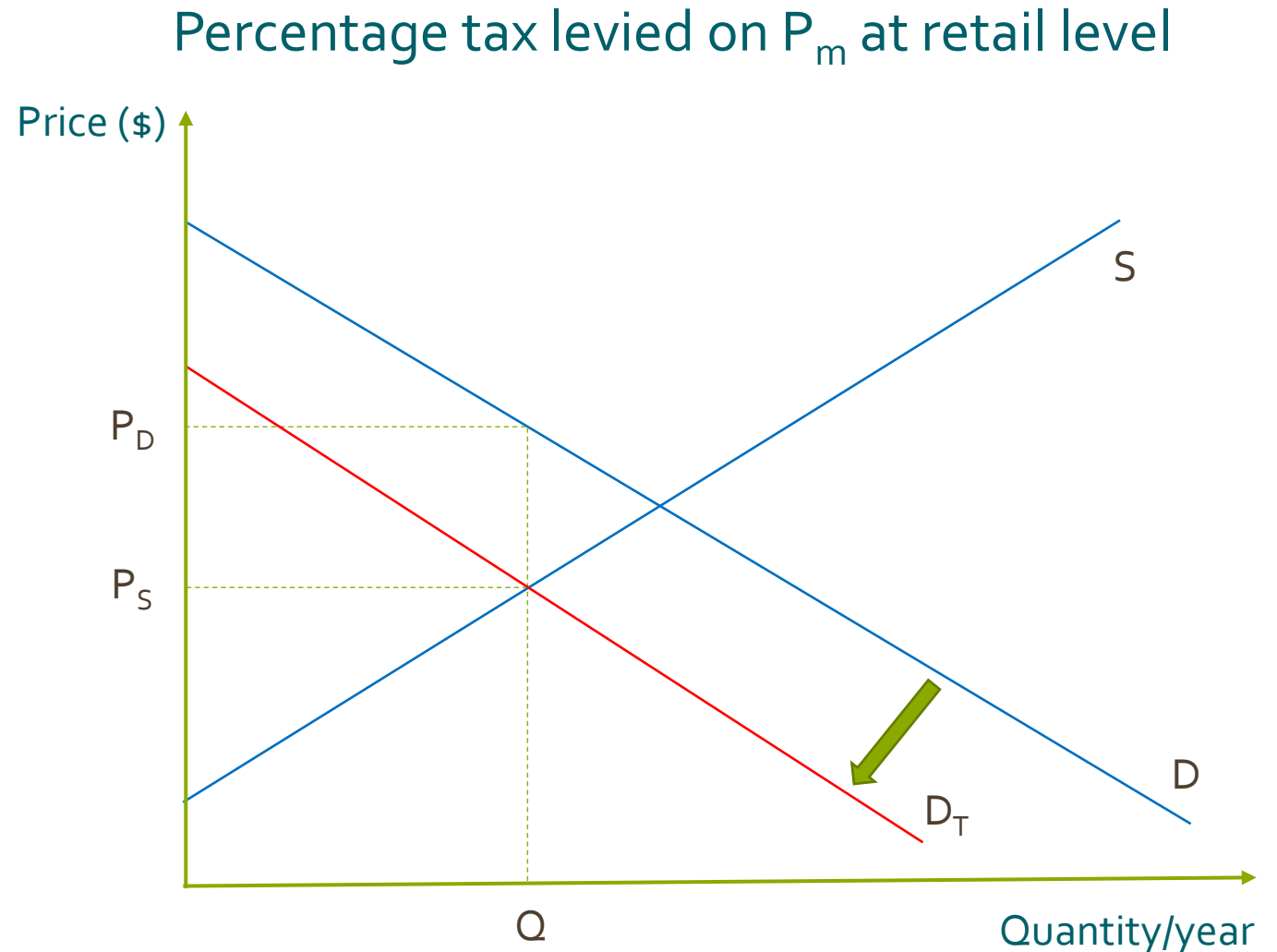
=> Economic benefit > Financial benefit

$$\Rightarrow P_s < P_e = P_d$$



Indirect taxes on a project's output

- Suppose an indirect tax t_c on the consumers, then the market price of this output is $P_m = P_s < P_d = P_m(1 + t_c)$.
- If a small number of the output are produced by the project to meet additional demand, the shadow price is measured by P_d .
- If a small number of the output of other suppliers are replaced by the project output, the shadow price is measured by P_s , i.e. the opportunity cost of resources that would have been used by other suppliers to produce the replaced output.



Indirect taxes on a project's output

(A small-scale project's output)

Financial benefit:

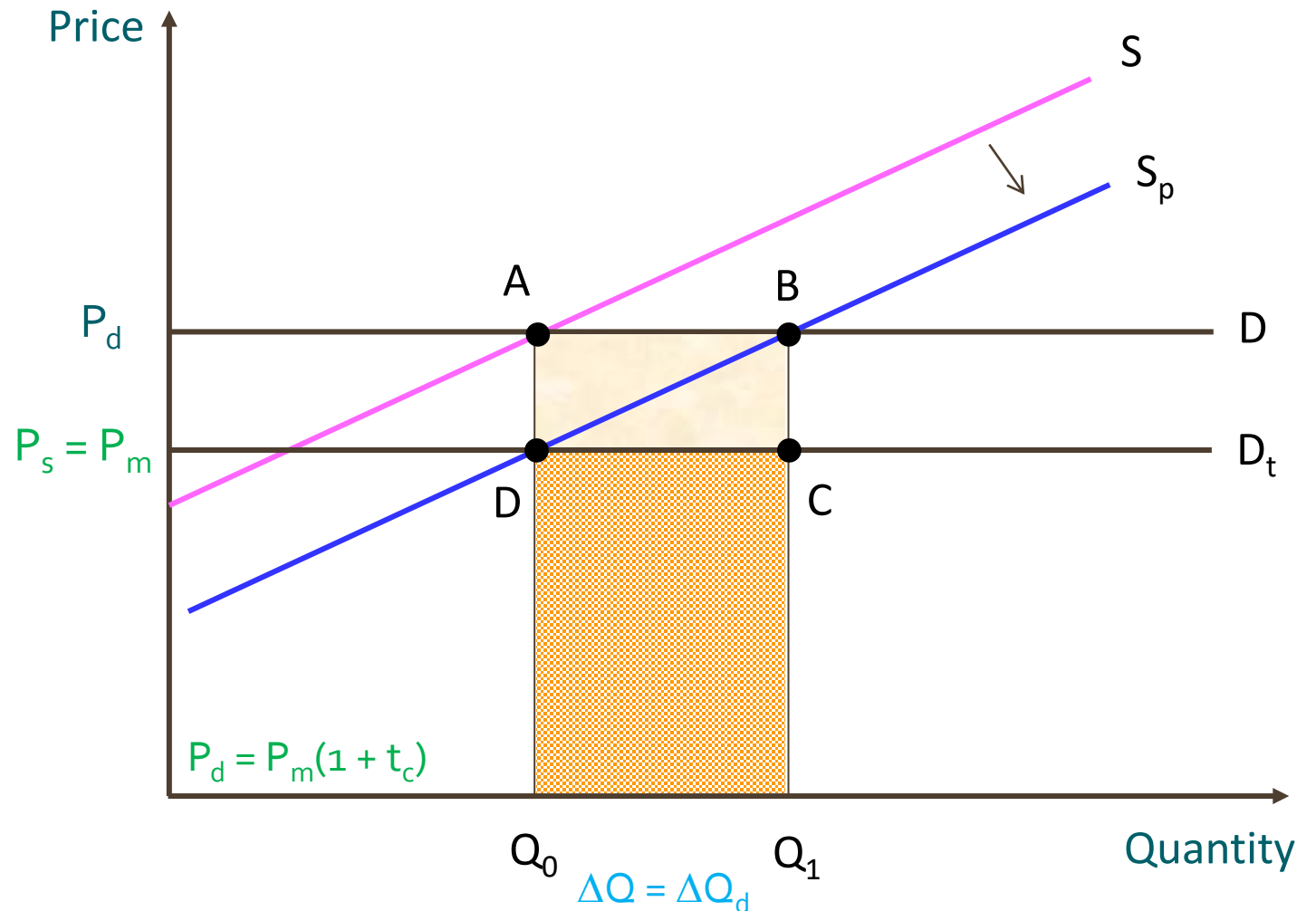
$$\begin{aligned}\Delta TR &= P_s(Q_1 - Q_0) = P_s\Delta Q \\ &= Q_0DCQ_1\end{aligned}$$

Economic benefit:

$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P_s\Delta Q + 0 + 0 + ABCD \\ &= Q_0ABQ_1\end{aligned}$$

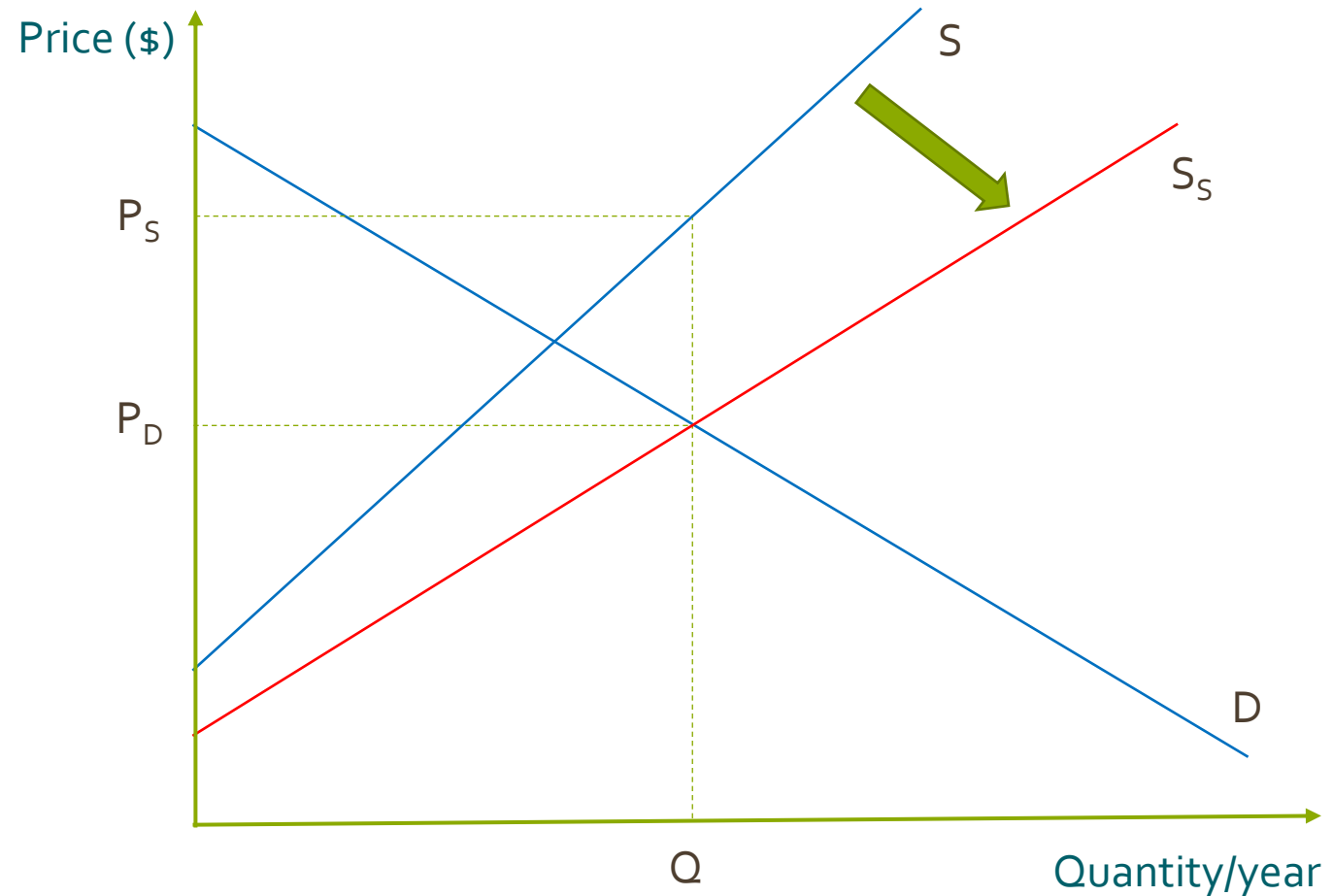
=> **Economic benefit > Financial benefit**

$$\Rightarrow P_e = P_d > P_s$$



A project's output subject to a subsidy

- Suppose an input of a project, such as agricultural products) is subject to an *ad valorem* subsidy s_p , the market price of this output is $P_m = P_d < P_s = P_m/(1 + s_p)$.
- If a small number of the output is produced by the project to satisfy additional demand, the shadow price is measured by P_d (i.e. subsidized price).
- If project output replaced a portion of current demand from an alternative unsubsidized source, such as imports, it would be valued at the unsubsidized price, P_s .



A project's output subject to a subsidy

(A small-scale project's output)

If the output of this project is not subject to a subsidy.

Financial benefit:

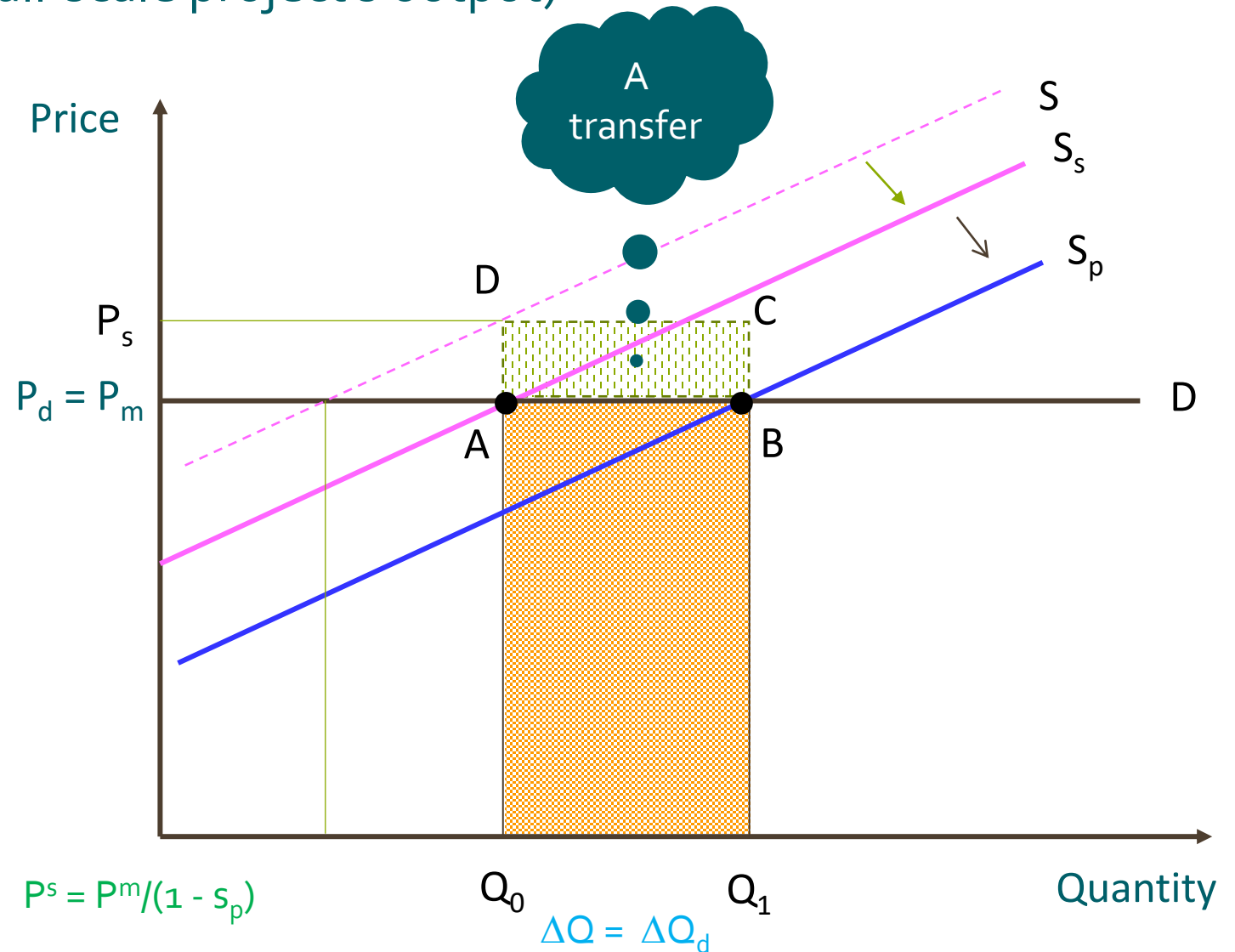
$$\begin{aligned}\Delta TR &= P_d(Q_1 - Q_0) = P_d\Delta Q \\ &= Q_0ABQ_1\end{aligned}$$

Economic benefit:

$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P_d\Delta Q + 0 + 0 + 0 \\ &= Q_0ABQ_1\end{aligned}$$

=> Economic benefit = Financial benefit

$$\Rightarrow P_e = P_d < P_s$$



A project's output subject to a subsidy

(A small-scale project's output)

If the output of this project is also subject to a subsidy.

Financial benefit:

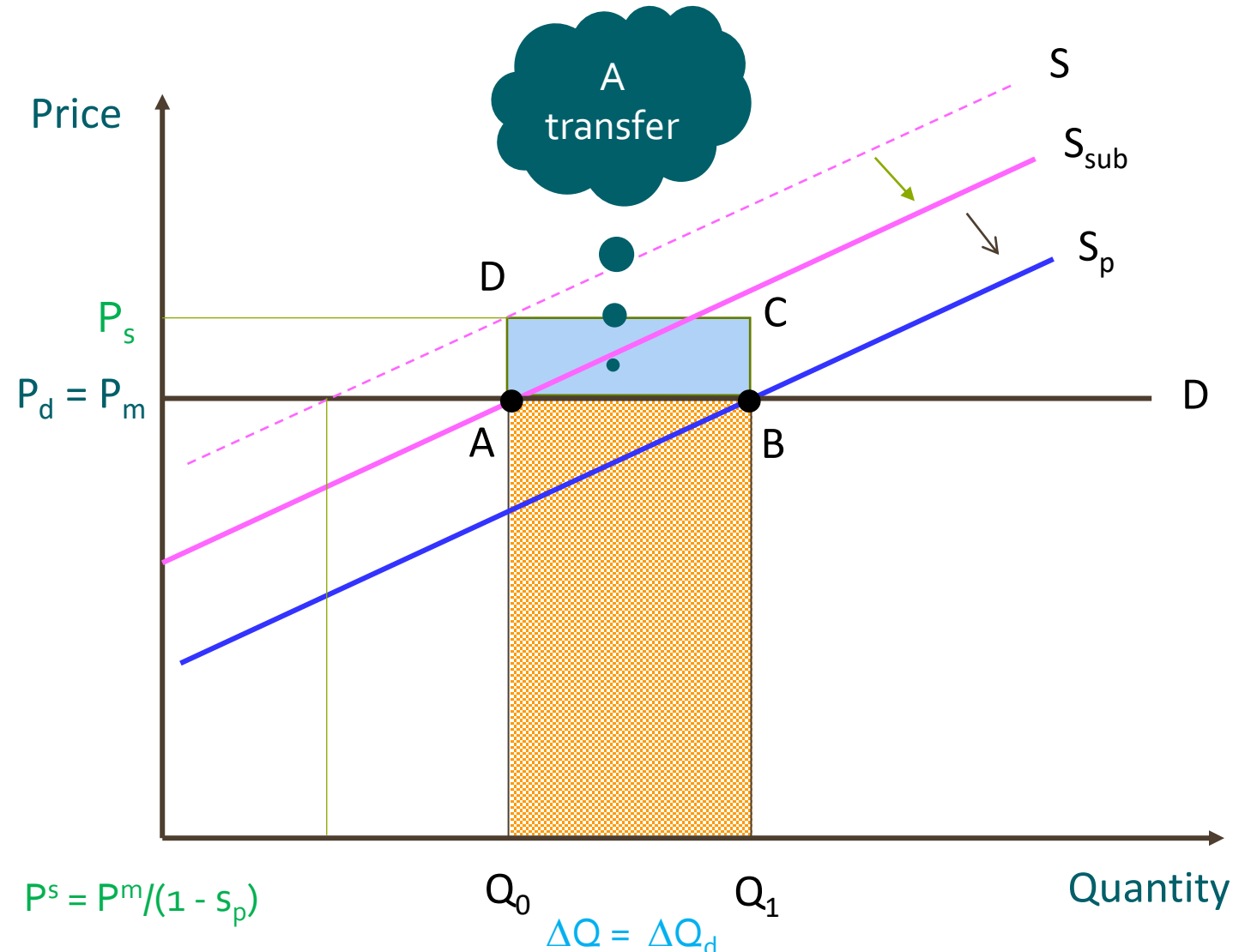
$$\begin{aligned}\Delta TR &= P_s(Q_1 - Q_0) = P_s \Delta Q \\ &= Q_0 D C Q_1\end{aligned}$$

Economic benefit:

$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P_s \Delta Q + 0 + 0 - ABCD \\ &= Q_0 A B Q_1\end{aligned}$$

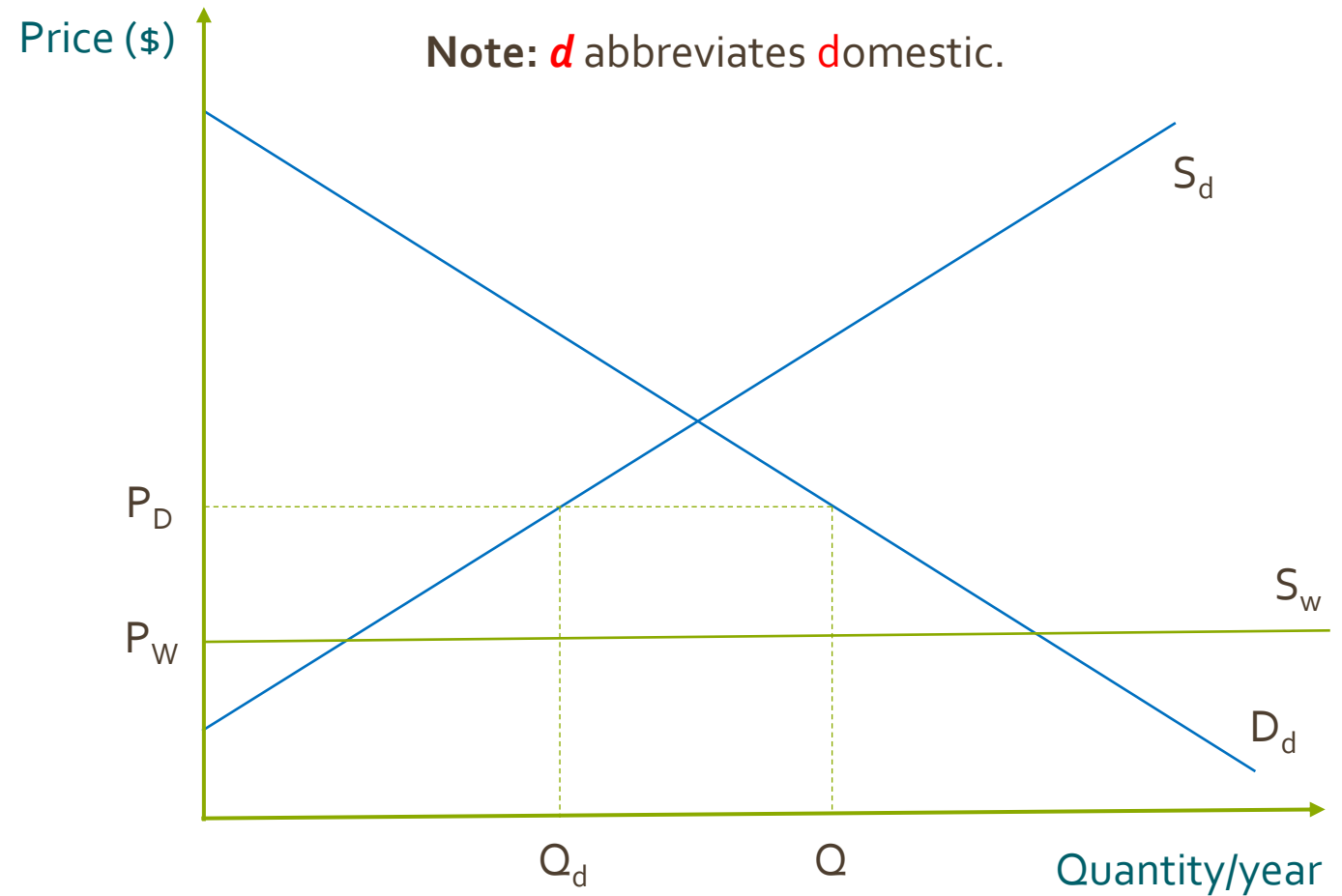
=> Economic benefit < Financial benefit

$$\Rightarrow P_e = P_d < P_s$$



An import-replacing output of a project

- If domestic production of an output is facing the imported product with a tariff t_m , the market price of this output is $P_m = P_d = P_w(1 + t_m)$.
- Consider a project produces an import-replacing output to satisfy an additional portion of existing demand from domestic production.
- Assume consumers remain at the same price-quantity equilibrium (P_d, Q); they are indifferent between domestically produced and imported product, and unaffected by the project.
- The value of additional output consists of the avoided cost of the imports displaced, and hence the shadow price is P_w .



An import-replacing output of a project

(A small- or large-scale project's output)

Financial benefit:

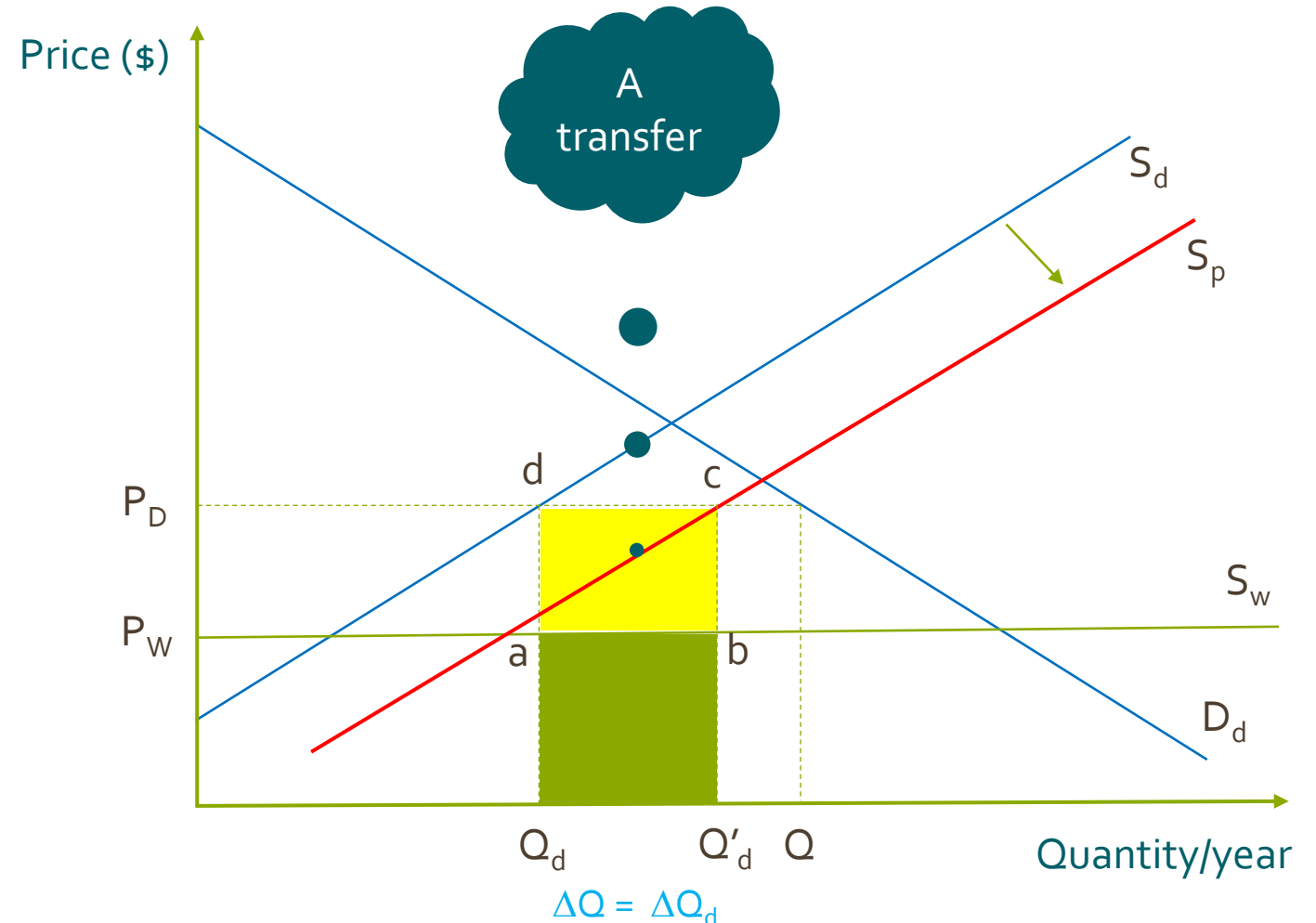
$$\begin{aligned}\Delta TR &= P_d(Q'_d - Q_d) = P_d \Delta Q \\ &= Q_d dc Q'_d\end{aligned}$$

Economic benefit:

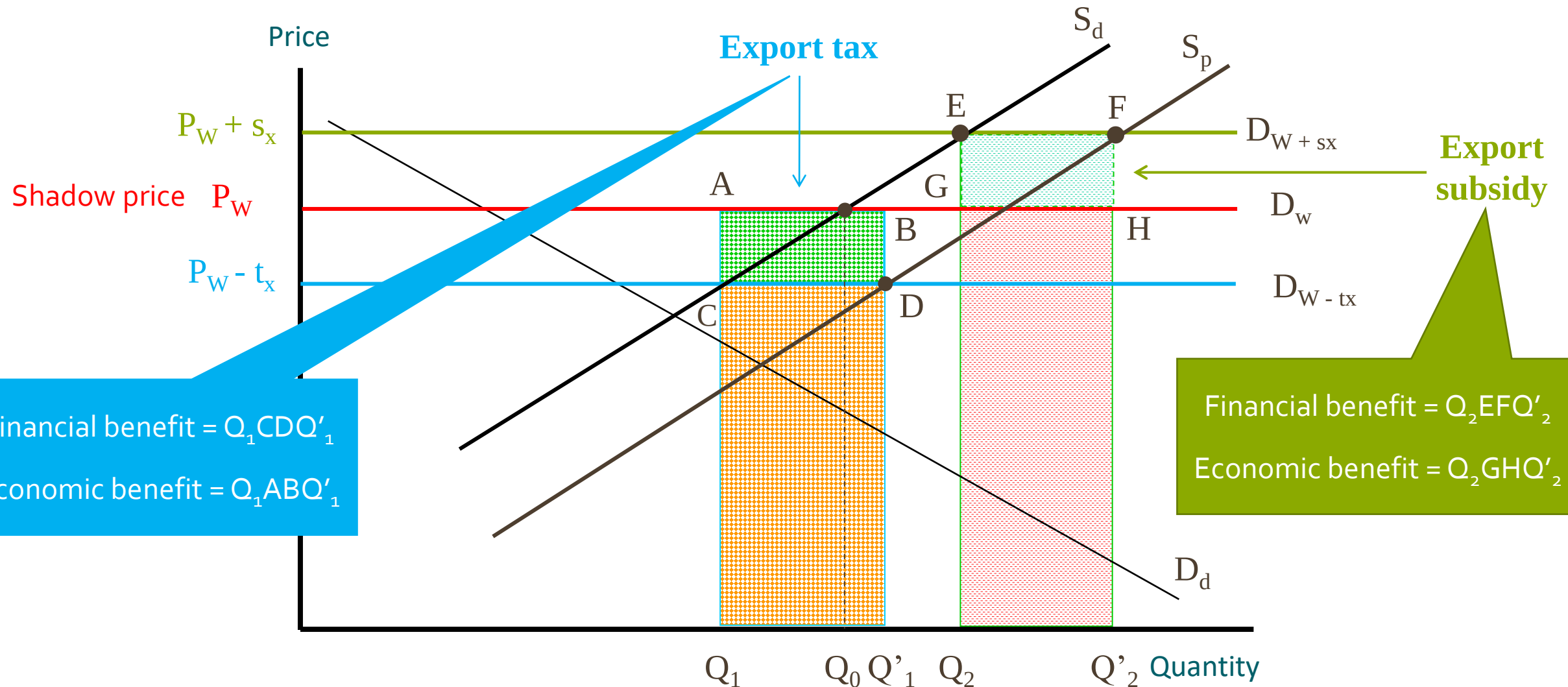
$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P_d \Delta Q + 0 + 0 - abcd \\ &= Q_d ab Q'_d\end{aligned}$$

=> Economic benefit < Financial benefit

$$\Rightarrow P_e = P_w < P_d$$

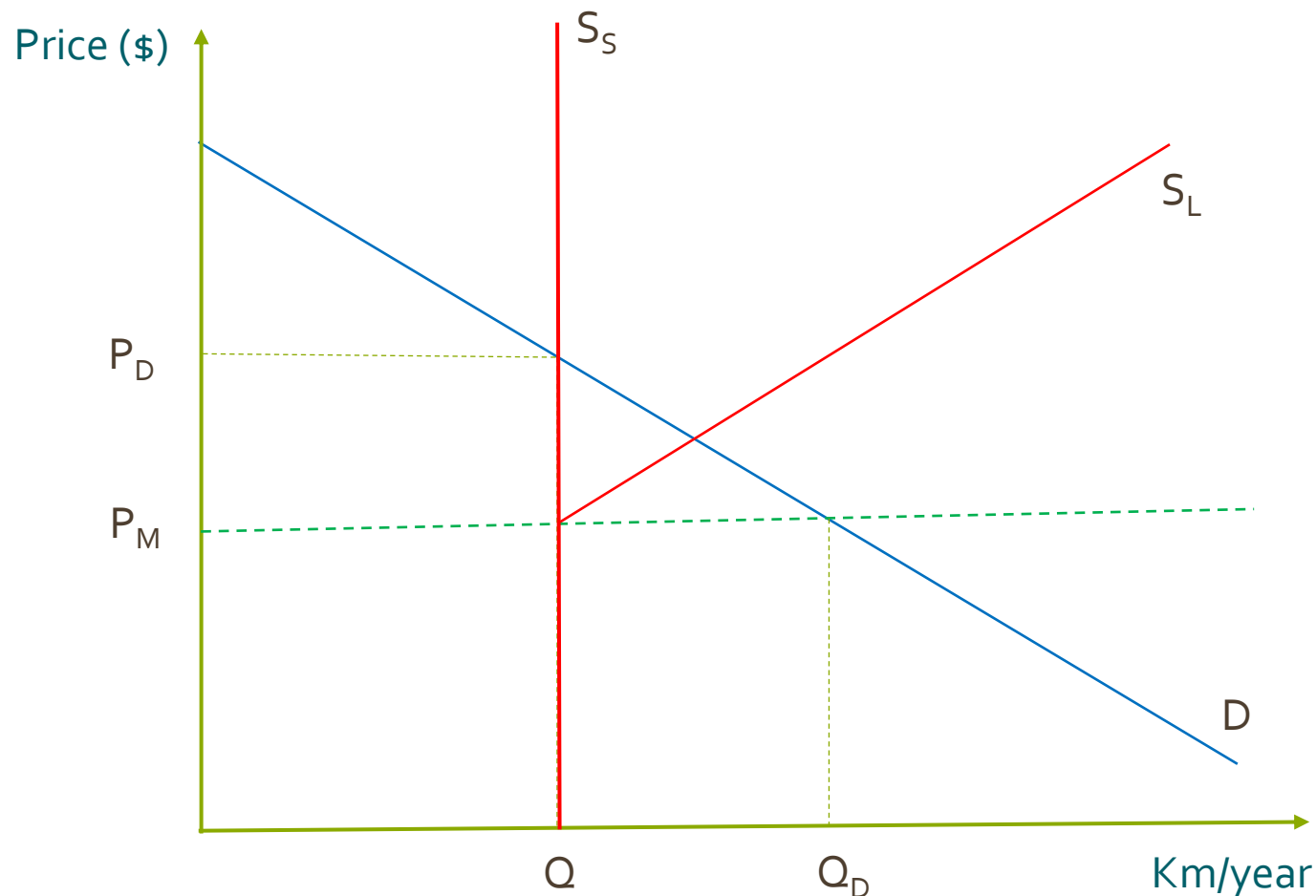


An exporting output of a project



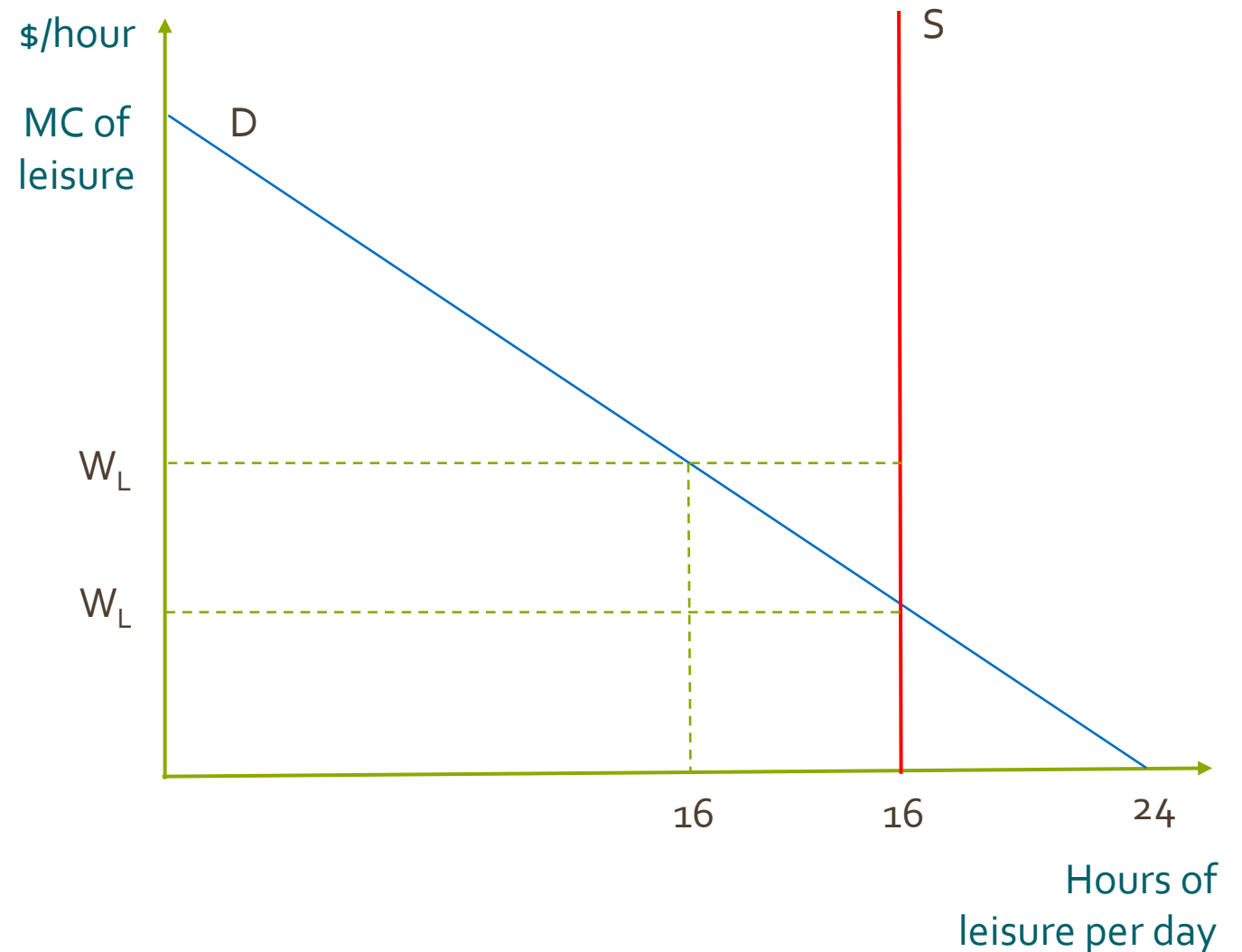
A project's output subject to a maximum price

- The market for public transport with price control at P_m . The short-run supply curve, S_s , is vertical at the current quantity supplied, Q . The long-run supply curve, S_L , offers the possibility of adding to the current public transport service to meet demand.
- If the project supplies additional buses at P_m for local residents who are willing to pay at P_d . Hence, P_d is the relevant shadow price of the output in an Efficiency Analysis.



Time saved by a project

- Many transportation investment have reductions in travel times as a principal output.
- Suppose a public project is proposed which will reduce the time taken to travel to work.
- At the wage W the individual would prefer 14 hours of leisure, with the remaining 10 hours in the day spent at work. However, he is constrained to accept 16 hours of leisure.
- Therefore, the wage W_L is the appropriate shadow price.



Shadow pricing marketed inputs

Distorted markets, large-scale projects



Indirect taxes on a project's input

Financial cost:

$$\begin{aligned}\Delta TC &= -P^d_1(Q^s_1 - Q^d_1) = -P^d_1\Delta Q \\ &= -Q^d_1\Delta EQ^s_1\end{aligned}$$

Economic cost:

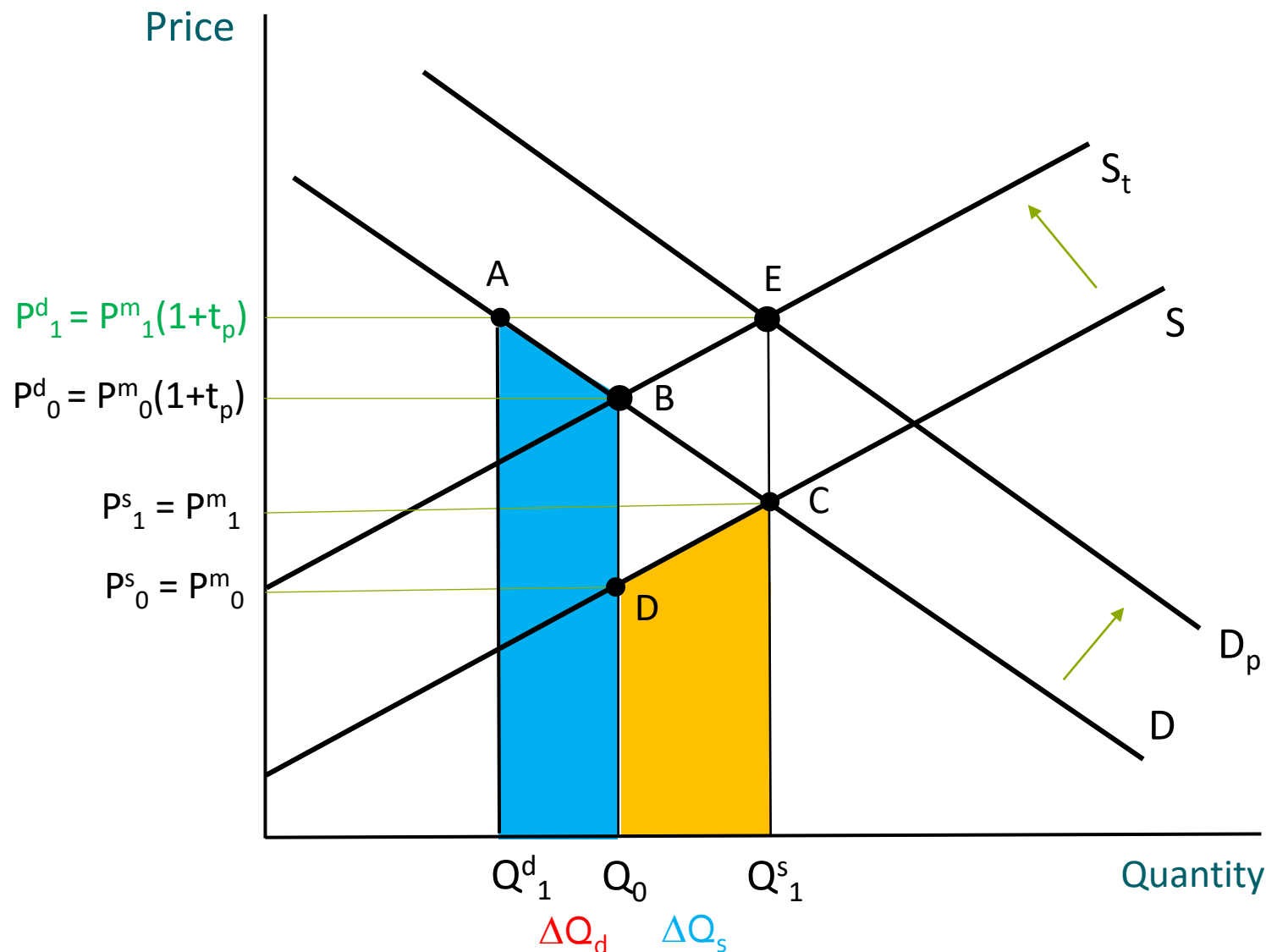
$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P^d_1\Delta Q - P^d_0BAP^d_1 + P^s_0DCP^s_1 + DCEB \\ &= -Q^d_1ABDCQ^s_1\end{aligned}$$

=> **Economic cost < Financial cost**

$$\Rightarrow P_e = (W_s \bar{P}_s + W_d \bar{P}_d) < P^d_1$$

$$\square W_s + W_d = 1$$

$$\square \bar{P}_s = (P^s_0 + P^s_1)/2; \bar{P}_d = (P^d_0 + P^d_1)/2$$



Indirect taxes on a project's input

Financial cost:

$$\begin{aligned}\Delta TC &= -P_1^d(Q_1^s - Q_1^d) = -P_1^d \Delta Q \\ &= -Q_1^d AA'Q_1^s\end{aligned}$$

Economic cost:

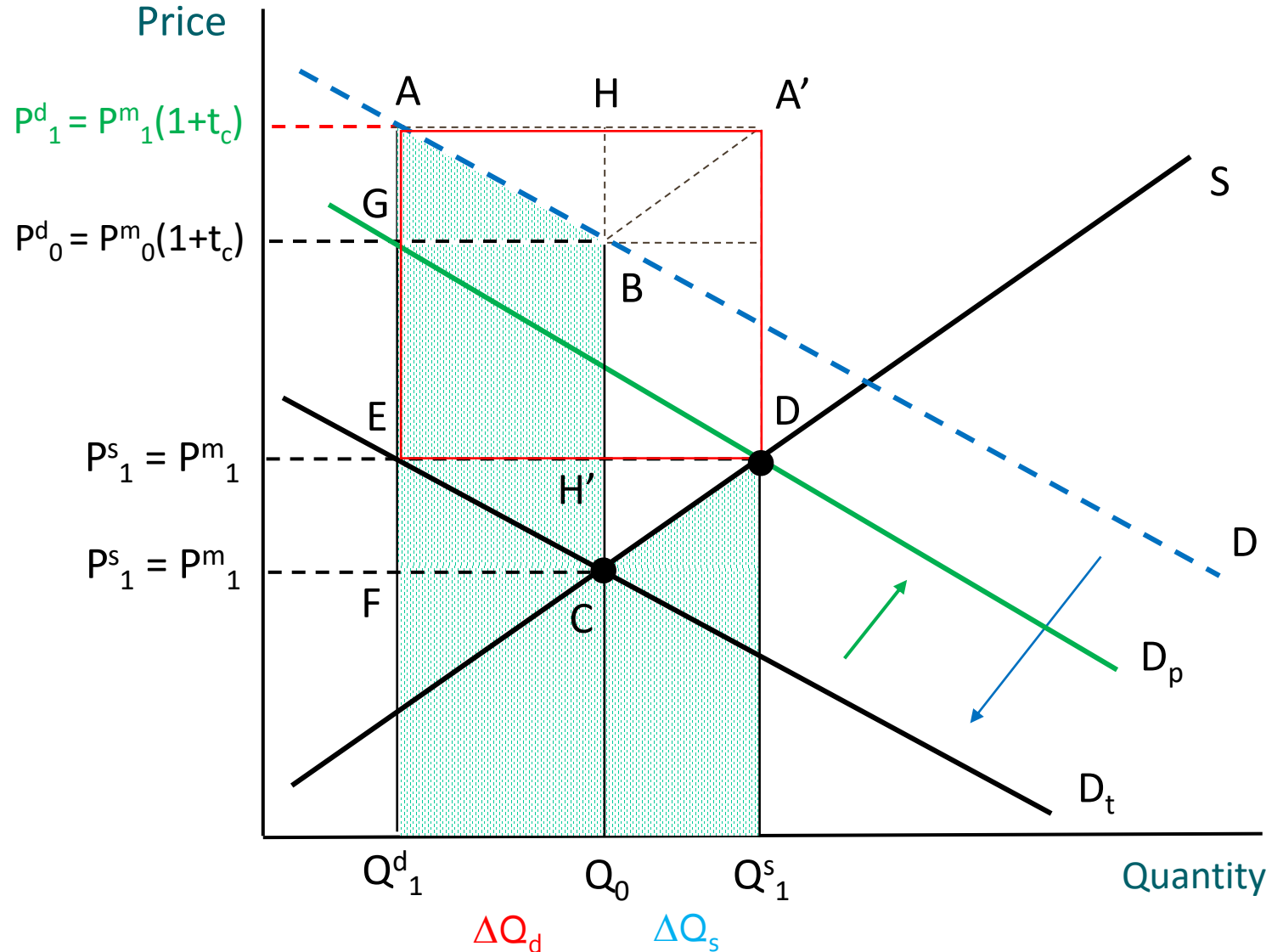
$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P_1^d \Delta Q - P_0^d BAP_1^d + P_0^s CDP_1^s + H'HA'D \\ &= -Q_1^d ABCDQ_1^s\end{aligned}$$

=> **Economic cost < Financial cost**

$$\Rightarrow P_1^m = P_1^s < P_e = (W_s \bar{P}_s + W_d \bar{P}_d) < P_1^d$$

$$\square W_s + W_d = 1$$

$$\square \bar{P}_s = (P_0^s + P_1^s)/2; \bar{P}_d = (P_0^d + P_1^d)/2$$



A project's input subject to a subsidy

Financial cost:

$$\begin{aligned}\Delta TC &= -P^d_1(Q_{1s} - Q_{1d}) = -P^d_1\Delta Q \\ &= -Q^d_1\Delta EQ^s_1\end{aligned}$$

Economic cost:

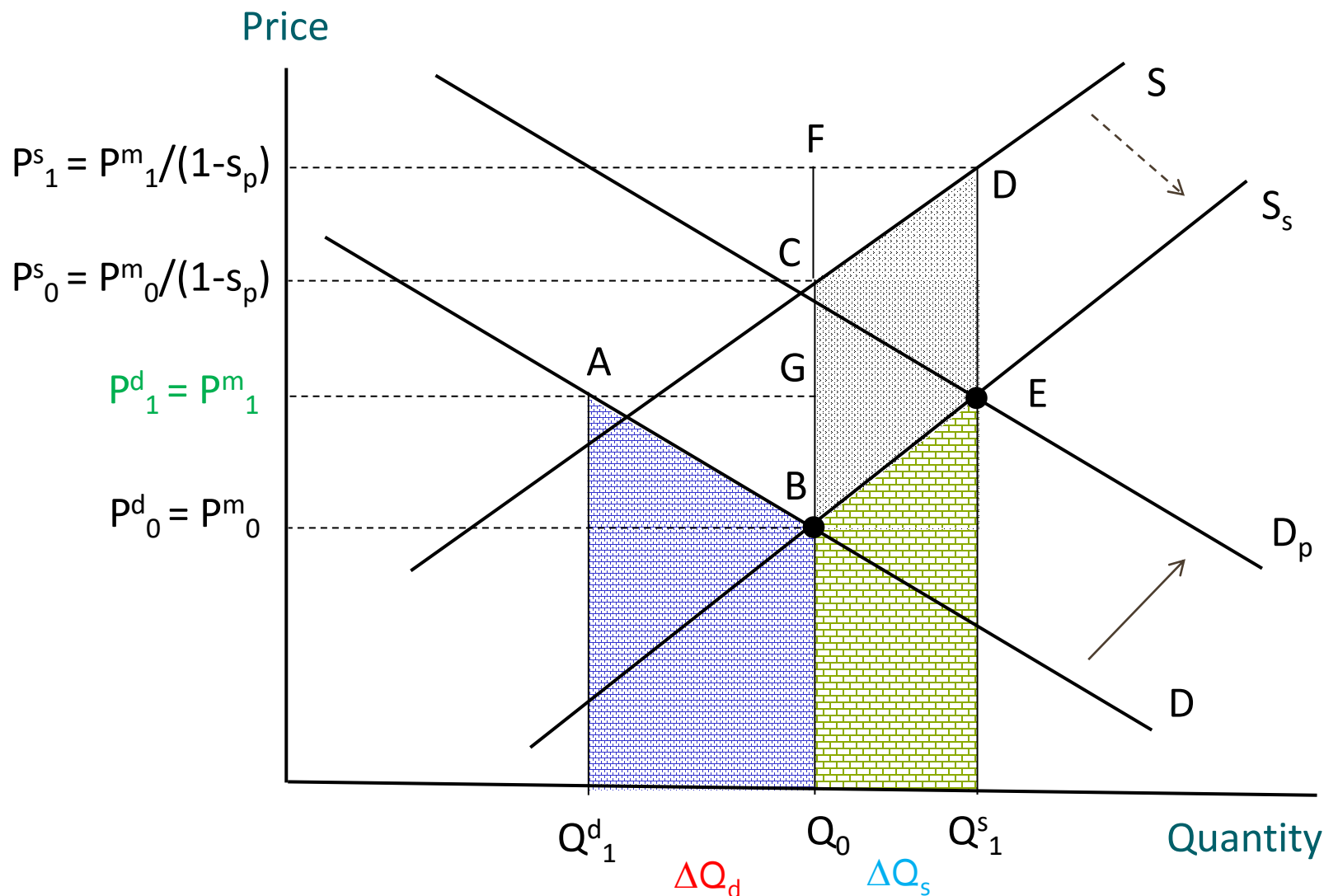
$$\begin{aligned}\Delta EV &= \Delta TC + \Delta CS + \Delta PS + \Delta GS \\ &= -P^d_1\Delta Q - P^d_0BAP^d_1 + P^s_0CDP^s_1 - BCDE \\ &= -Q^d_1ABCDQ^s_1\end{aligned}$$

=> **Economic cost > Financial cost**

$$\Rightarrow P_e = (W_s \bar{P}_s + W_d \bar{P}_d) > P^d_1 = P^m_1$$

$$\square W_s + W_d = 1$$

$$\square \bar{P}_s = (P^s_0 + P^s_1)/2; \bar{P}_d = (P^d_0 + P^d_1)/2$$



Shadow pricing marketed outputs

Distorted markets, large-scale projects



Indirect taxes on a project's output

Financial benefit:

$$\begin{aligned}\Delta TR &= P_1^s(Q_1^d - Q_1^s) = P_1^s \Delta Q \\ &= Q_1^s ABQ_1^d\end{aligned}$$

Economic benefit:

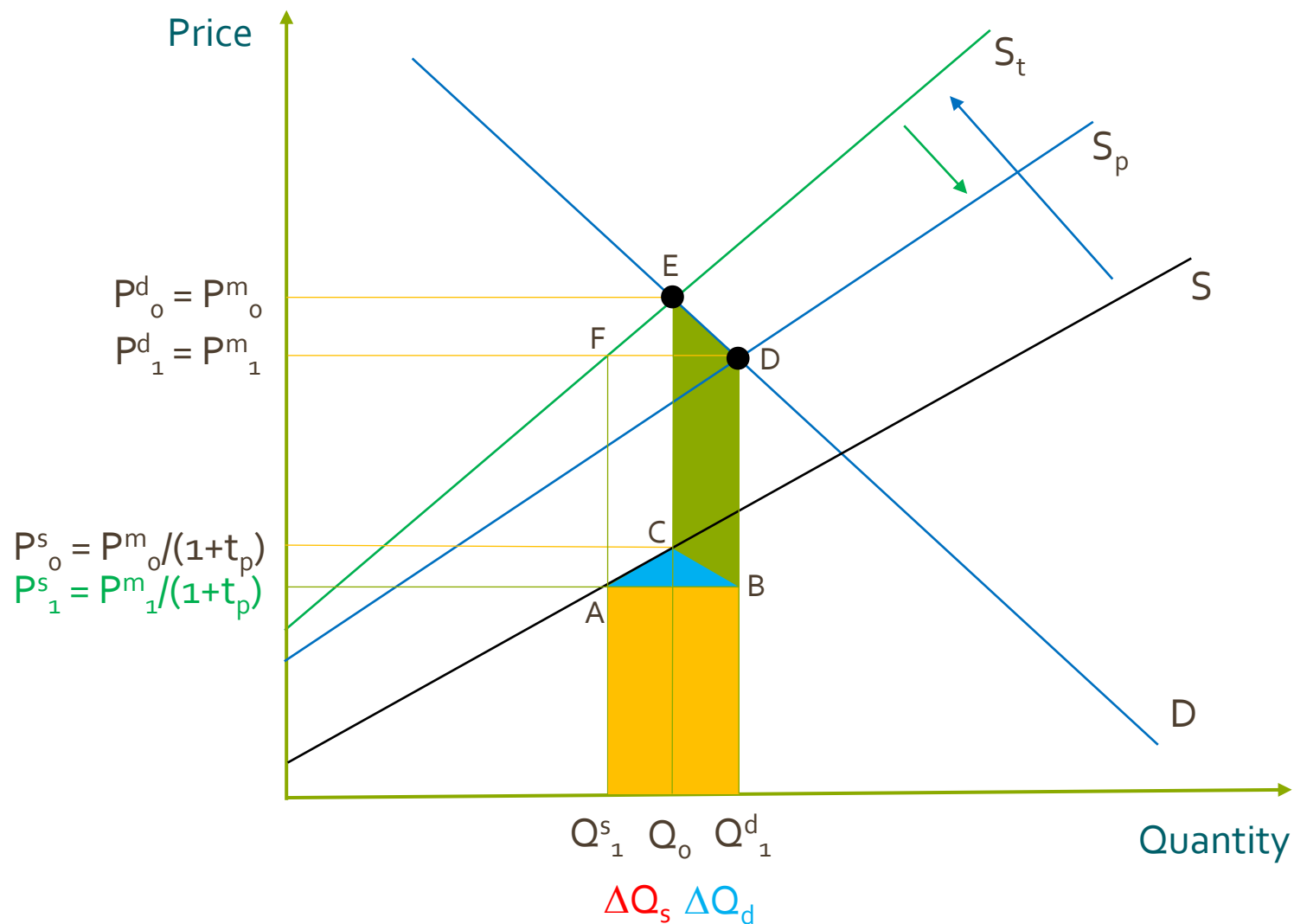
$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P_1^s \Delta Q + P_0^d EDP_1^d - P_0^s CAP_1^s + BCED \\ &= Q_1^s ACEDQ_1^d\end{aligned}$$

=> **Economic benefit > Financial benefit**

$$\Rightarrow P_1^s < P_e = (W_s \bar{P}_s + W_d \bar{P}_d) < P_1^d = P_1^m$$

$$\square W_s + W_d = 1$$

$$\square \bar{P}_s = (P_0^s + P_1^s)/2; \bar{P}_d = (P_0^d + P_1^d)/2$$



Indirect taxes on a project's output

Financial benefit:

$$\begin{aligned}\Delta TR &= P^s_1(Q^d_1 - Q^s_1) = P^s_1\Delta Q \\ &= Q^s_1 B E Q^d_1\end{aligned}$$

Economic benefit:

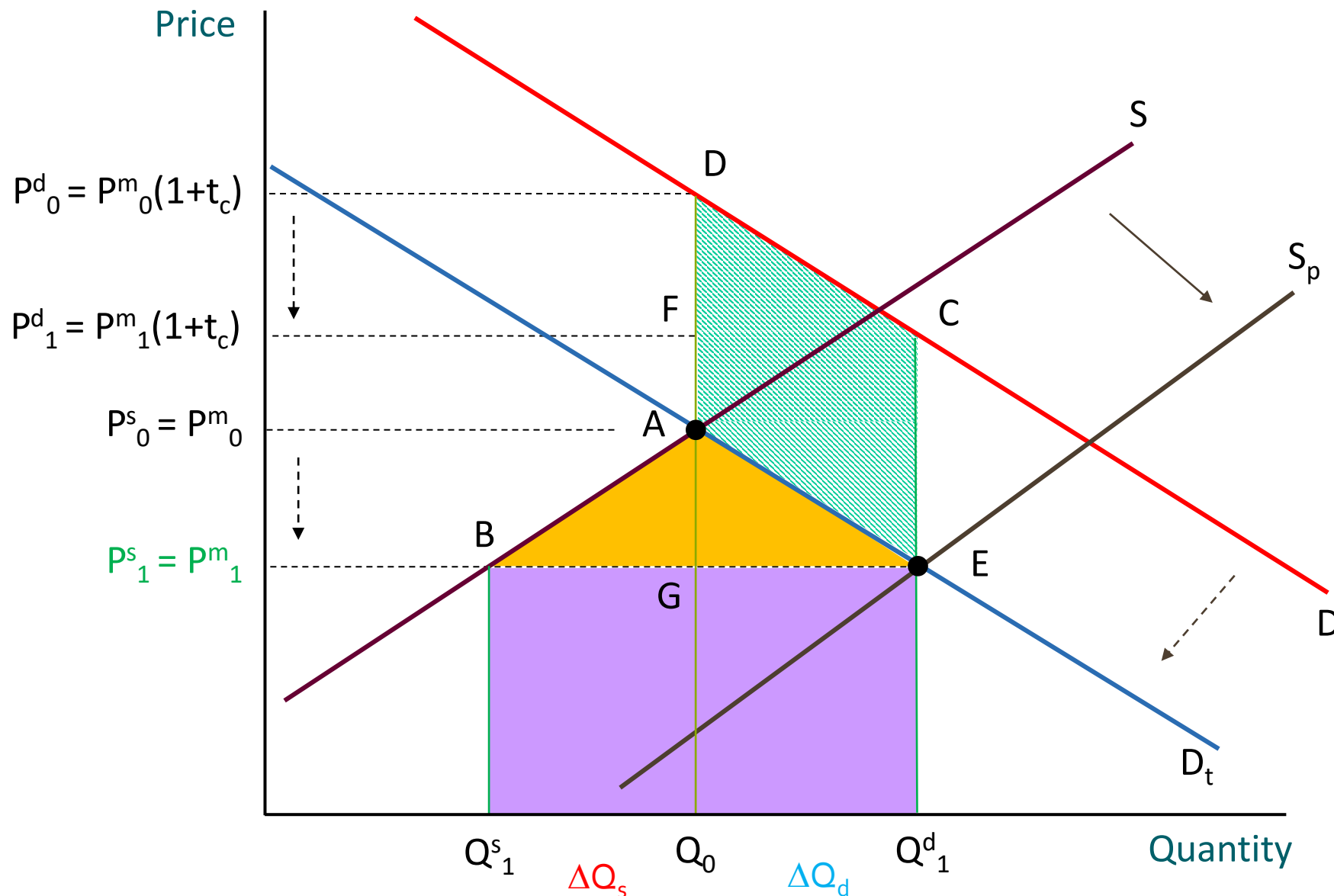
$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P^s_1\Delta Q + P^d_0 D C P^d_1 - P^s_0 A B P^s_1 + G F C E \\ &= Q^s_1 B A D C Q^d_1\end{aligned}$$

=> Economic benefit > Financial benefit

$$\Rightarrow P_e = (W_s \bar{P}_s + W_d \bar{P}_d) > P^s_1 = P^m_1$$

$$\square W_s + W_d = 1$$

$$\square \bar{P}_s = (P^s_0 + P^s_1)/2; \bar{P}_d = (P^d_0 + P^d_1)/2$$



A project's output subject to a subsidy

Financial benefit:

$$\begin{aligned}\Delta TR &= P_1^d(Q_{1d} - Q_{1s}) = P_1^d \Delta Q \\ &= Q_1^s EDQ_1^d\end{aligned}$$

Economic benefit:

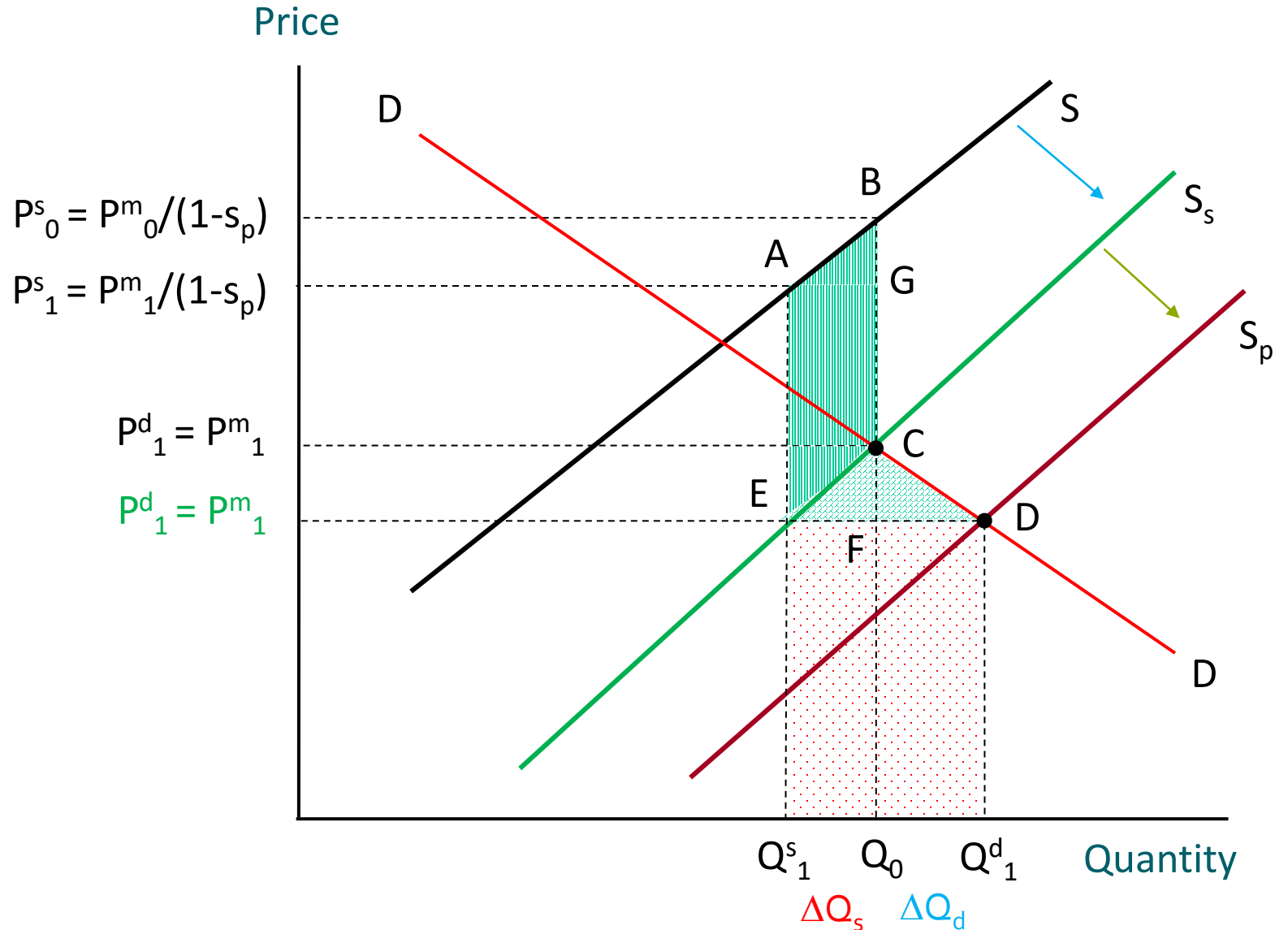
$$\begin{aligned}\Delta EV &= \Delta TR + \Delta CS + \Delta PS + \Delta GS \\ &= P_1^d \Delta Q + P_0^d CDP_1^d - P_0^s BAP_1^s + EABC \\ &= Q_1^s ABCDQ_1^d\end{aligned}$$

=> Economic benefit > Financial benefit

$$\Rightarrow P_e = (W_s \bar{P}_s + W_d \bar{P}_d) > P^d_1 = P^m_1$$

- $W_s + W_d = 1$

□ $\bar{P}_s = (P^s_0 + P^s_1)/2$; $\bar{P}_d = (P^d_0 + P^d_1)/2$



The efficiency pricing rules

Distorting taxes and subsidies



The efficiency pricing rules

Item to be valued	Valued at equilibrium point on a:	
	Demand curve (buyer's price)	Supply curve (seller's price)
Output	Satisfies additional demand	Satisfies existing demand from alternative source
Input	Sourced from an alternative market use	Sourced from additional supply

Rules for shadow-pricing marketed inputs

- If the project input is sourced from an increase in supply, its marginal cost is the supply price (P_s).
 - In the case of an indirect tax on the input, the supply price is the before-tax price.
 - In the case of a subsidy, the supply price is the unsubsidized price.
- If the project input is diverted from employment elsewhere in the economy, its marginal cost is the foregone benefit in its alternative use, which is measured by the demand price (P_d).
 - In case of an indirect tax, the demand price is the after-tax price;
 - In case of a subsidy, the demand price is the subsidized price which buyers actually pay.

Rules for shadow-pricing marketed outputs

- If the project output satisfies additional demand, it is valued at the demand price (P_d).
 - In the case of an indirect tax on the output, the demand price is the after-tax price.
 - In the case of a subsidy, the supply price is the subsidized price.
- If the project output satisfies existing demand from an alternative source, it is valued at the current supply price (P_s).
 - In case of a product subject to an indirect tax, the supply price is the before-tax price;
 - In case of a subsidy, the supply price is the unsubsidized price.

Corrective taxation

- Indirect taxes is not just to raise government revenue, or subsidies to assist producers.
- Another possible role for indirect taxes or subsidies is to discourage or encourage consumption of certain goods and services, or use of certain inputs in production.
- In other words, they are tools to internalize external effects.
- What are the appropriate shadow prices for inputs and outputs of a project?

Corrective taxation

- When a project output satisfies additional demand, the seller's price – the net of tax price (gross of subsidy price) – is used to shadow-price the output.
- When an input is in addition to existing supply, the buyer's price – the gross of tax price (net of subsidy price) – is used to shadow-price the input.
- There is no change in the rule in the case of an output which replaces existing supply or an input which is sourced from an alternative use because this involves no change in the level of the externality.

Conversion factors

Derivation of shadow prices from market prices



Conversion factors for project inputs

- Conversion factors (k_i) are defined as the ratio between shadow prices ($P_{e,i}$) and market prices ($P_{m,i}$). Formally:

$$k_i = \frac{P_{e_i}}{P_{m_i}} \leftrightarrow P_{e_i} = k_i \cdot P_{m_i}$$

- If $k_i > 1$, the observed price is lower than the shadow price, meaning that the opportunity cost of that good is higher than that captured by the market;
- If $k_i < 1$, the observed price is higher than the shadow price, due to taxes or other market distortions with add to the marginal social value of a good and determine a higher market price.
- In principle, conversion factors should be made available by a planning office and not calculated on a project-by-project basis.

Conversion factors for project inputs

- Minor non-tradable items are inputs that do not represent a significant share of the project total costs and for which no sufficient information on their cost structure is generally available, thus impeding an estimate of their long-term marginal cost.
- An adjustment method to estimate shadow prices of these items consists of computing the so-called standard conversion factor (SCF).
- The SCF is a proxy of the average distance between world prices and domestic prices, under the assumption that the world prices reflect the opportunity cost of the good.

Conversion factors for project inputs

- The full formula generally used to estimate the SCF is:

$$SCF = \frac{(M + X)}{(M + T_M - S_M) + (X - T_X + S_X)}$$

- M is the total value of import at shadow prices, i.e. CIF prices.
 - X is the total value of export at shadow prices, i.e. free on board (FOB) prices.
 - T_M and T_X are the value of duties on import and export respectively.
 - S_M and S_X are the value of subsidies on import and export respectively.
- The lower the distortion of the domestic market, the closer the SCF is to unity.

Shadow-pricing marketed inputs and outputs

Q&A section

